

Clinical Practice Guideline

2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults

A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines

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Preamble

Since 1980, the American College of Cardiology (ACC) and American Heart Association (AHA) have translated scientific evidence into clinical practice guidelines (guidelines) with recommendations to improve cardiovascular health. In 2013, the National Heart, Lung, and Blood Institute (NHLBI) Advisory Council recommended that the NHLBI focus specifically on reviewing the highest-quality evidence and partner with other organizations to develop recommendations.^{P-1,P-2} Accordingly, the ACC and AHA collaborated with the NHLBI and stakeholder and professional organizations to complete and publish 4 guidelines (on assessment of cardiovascular risk, lifestyle modifications to reduce cardiovascular risk, management of blood cholesterol in adults, and management of overweight and obesity in adults) to make them available to the widest possible constituency. In 2014, the ACC and AHA, in partnership with several other professional societies, initiated a guideline on the prevention, detection, evaluation, and management of high blood pressure (BP) in adults. Under the management of the ACC/AHA Task Force, a Prevention Subcommittee was appointed to help guide development of the suite of guidelines on prevention of cardiovascular disease (CVD). These guidelines, which are based on systematic methods to evaluate and classify evidence, provide a cornerstone for quality cardiovascular care. The ACC and AHA sponsor the development and publication of guidelines without commercial support, and members of each organization volunteer their time to the writing and review efforts. Guidelines are official policy of the ACC and AHA.

Intended Use

Practice guidelines provide recommendations applicable to patients with or at risk of developing CVD. The focus is on medical practice in the United States, but guidelines developed in collaboration with other organizations can have a global impact. Although guidelines may be used to inform regulatory or payer decisions, they are intended to improve patients' quality of care and align with patients' interests. Guidelines are intended to define practices meeting the needs of patients in most, but not all, circumstances and should not replace clinical judgment.

Clinical Implementation

Management in accordance with guideline recommendations is effective only when followed by both practitioners and patients. Adherence to recommendations can be enhanced by shared decision making between clinicians and patients, with patient engagement in selecting interventions on the basis of individual values, preferences, and associated conditions and comorbidities.

Methodology and Modernization

The ACC/AHA Task Force on Clinical Practice Guidelines (Task Force) continuously reviews, updates, and modifies guideline methodology on the basis of published standards from organizations, including the Institute of Medicine,^{P-3,P-4} and on the basis of internal reevaluation. Similarly, the presentation and delivery of guidelines are reevaluated and modified on the basis of evolving technologies and other factors to facilitate optimal dissemination of information to healthcare professionals at the point of care.

Toward this goal, this guideline continues the introduction of an evolved format of presenting guideline recommendations and associated text called the "modular knowledge chunk format." Each modular "chunk" includes a table of related recommendations, a brief synopsis, recommendation-specific supportive text, and when appropriate, flow diagrams or additional tables. References are provided within the modular chunk itself to facilitate quick review. Additionally, this format will facilitate seamless updating of guidelines with focused updates as new evidence is published, as well as content tagging for rapid electronic retrieval of related recommendations on a topic of interest. This evolved approach format was instituted when this guideline was near completion; therefore, the present document represents a transitional format that best suits the text as written. Future guidelines will fully implement this format, including provisions for limiting the amount of text in a guideline.

Recognizing the importance of cost-value considerations in certain guidelines, when appropriate and feasible, an analysis of the value of a drug, device, or intervention may be performed in accordance with the ACC/AHA methodology.^{P-5}

To ensure that guideline recommendations remain current, new data are reviewed on an ongoing basis, with full guideline revisions commissioned in approximately 6-year cycles. Publication of new, potentially practice-changing study results that are relevant to an existing or new drug, device, or management strategy will prompt evaluation by the Task Force, in consultation with the relevant guideline writing committee, to determine whether a focused update should be commissioned. For additional information and policies regarding guideline development, we encourage readers to consult the ACC/AHA guideline methodology manual^{P-6} and other methodology articles.^{P-7-P-10}

Selection of Writing Committee Members

The Task Force strives to avoid bias by selecting experts from a broad array of backgrounds. Writing committee members represent different geographic regions, sexes, ethnicities, races, intellectual perspectives/biases, and scopes

of clinical practice. The Task Force may also invite organizations and professional societies with related interests and expertise to participate as partners, collaborators, or endorers.

Relationships With Industry and Other Entities

The ACC and AHA have rigorous policies and methods to ensure that guidelines are developed without bias or improper influence. The complete relationships with industry and other entities (RWI) policy can be found [online](#). Appendix 1 of the present document lists writing committee members' relevant RWI. For the purposes of full transparency, writing committee members' comprehensive disclosure information is available [online](#). Comprehensive disclosure information for the Task Force is available [online](#).

Evidence Review and Evidence Review Committees

In developing recommendations, the writing committee uses evidence-based methodologies that are based on all available data.^{P-6-P-9} Literature searches focus on randomized controlled trials (RCTs) but also include registries, nonrandomized comparative and descriptive studies, case series, cohort studies, systematic reviews, and expert opinion. Only key references are cited.

An independent evidence review committee (ERC) is commissioned when there are 1 or more questions deemed of utmost clinical importance that merit formal systematic review. The systematic review will determine which patients are most likely to benefit from a drug, device, or treatment strategy and to what degree. Criteria for commissioning an ERC and formal systematic review include: a) the absence of a current authoritative systematic review, b) the feasibility of defining the benefit and risk in a time frame consistent with the writing of a guideline, c) the relevance to a substantial number of patients, and d) the likelihood that the findings can be translated into actionable recommendations. ERC members may include methodologists, epidemiologists, healthcare providers, and biostatisticians. The recommendations developed by the writing committee on the basis of the systematic review are marked with "SR."

Guideline-Directed Management and Therapy

The term guideline-directed management and therapy (GDMT) encompasses clinical evaluation, diagnostic testing, and pharmacological and procedural treatments. For these and all recommended drug treatment regimens, the reader should confirm the dosage by reviewing product insert material and evaluate the treatment regimen for contraindications and interactions. The recommendations are limited to drugs, devices, and treatments approved for clinical use in the United States.

Class of Recommendation and Level of Evidence

The Class of Recommendation (COR) indicates the strength of the recommendation, encompassing the estimated magnitude and certainty of benefit in proportion to risk. The Level of Evidence (LOE) rates the quality of scientific evidence that supports the intervention on the basis of the type, quantity,

and consistency of data from clinical trials and other sources (Table 1).^{P-6-P-8}

Glenn N. Levine, MD, FACC, FAHA
Chair, ACC/AHA Task Force on Clinical Practice Guidelines

1. Introduction

As early as the 1920s, and subsequently in the 1959 Build and Blood Pressure Study^{S1.5-1} of almost 5 million adults insured between 1934 and 1954, a strong direct relationship was noted between level of BP and risk of clinical complications and death. In the 1960s, these findings were confirmed in a series of reports from the Framingham Heart Study.^{S1.5-2} The 1967 and 1970 Veterans Administration Cooperative Study Group reports ushered in the era of effective treatment for high BP.^{S1.5-3,S1.5-4} The first comprehensive guideline for detection, evaluation, and management of high BP was published in 1977, under the sponsorship of the NHLBI.^{S1.5-5} In subsequent years, a series of Joint National Committee (JNC) BP guidelines were published to assist the practice community and improve prevention, awareness, treatment, and control of high BP.^{S1.5-5-S1.5-7} The present guideline updates prior JNC reports.

1.1. Methodology and Evidence Review

An extensive evidence review, which included literature derived from research involving human subjects, published in English, and indexed in MEDLINE (through PubMed), EMBASE, the Cochrane Library, the Agency for Healthcare Research and Quality, and other selected databases relevant to this guideline, was conducted between February and August 2015. Key search words included but were not limited to the following: *adherence; aerobic; alcohol intake; ambulatory care; antihypertensive; agents, drug, medication, therapy; beta adrenergic blockers; blood pressure: arterial, control, determination, devices, goal, high, improve, measurement, monitoring, ambulatory; calcium channel blockers; diet; diuretic agent; drug therapy; heart failure: diastolic, systolic; hypertension: white coat, masked, ambulatory, isolated ambulatory, isolated clinic, diagnosis, reverse white coat, prevention, therapy, treatment, control; intervention; lifestyle: measures, modification; office visits; patient outcome; performance measures; physical activity; potassium intake; protein intake; renin inhibitor; risk reduction: behavior, counseling; screening; sphygmomanometers; spironolactone; therapy; treatment: adherence, compliance, efficacy, outcome, protocol, regimen; weight*. Additional relevant studies published through June 2016, during the guideline writing process, were also considered by the writing committee and added to the evidence tables when appropriate. The final evidence tables included in the [Online Data Supplement](#) summarize the evidence used by the writing committee to formulate recommendations.

As noted in the preamble, an independent ERC was commissioned to perform a formal systematic review of 4 critical clinical questions related to hypertension (Table 2), the results of which were considered by the writing committee for incorporation into this guideline. Concurrent with this process, writing committee members evaluated other published data relevant to the guideline. The findings of the ERC and the writing committee members were formally presented and

Table 1. Applying Class of Recommendation and Level of Evidence to Clinical Strategies, Interventions, Treatments, or Diagnostic Testing in Patient Care* (Updated August 2015)

CLASS (STRENGTH) OF RECOMMENDATION	LEVEL (QUALITY) OF EVIDENCE†
CLASS I (STRONG) Benefit >>> Risk Suggested phrases for writing recommendations: - Is recommended - Is indicated/useful/effective/beneficial - Should be performed/administered/other - Comparative-Effectiveness Phrases‡: - Treatment/strategy A is recommended/indicated in preference to treatment B - Treatment A should be chosen over treatment B	LEVEL A - High-quality evidence‡ from more than 1 RCT - Meta-analyses of high-quality RCTs - One or more RCTs corroborated by high-quality registry studies
CLASS IIa (MODERATE) Benefit >> Risk Suggested phrases for writing recommendations: - Is reasonable - Can be useful/effective/beneficial - Comparative-Effectiveness Phrases‡: - Treatment/strategy A is probably recommended/indicated in preference to treatment B - It is reasonable to choose treatment A over treatment B	LEVEL B-R (Randomized) - Moderate-quality evidence‡ from 1 or more RCTs - Meta-analyses of moderate-quality RCTs
CLASS IIb (WEAK) Benefit ≥ Risk Suggested phrases for writing recommendations: - May/might be reasonable - May/might be considered - Usefulness/effectiveness is unknown/unclear/uncertain or not well established	LEVEL B-NR (Nonrandomized) - Moderate-quality evidence‡ from 1 or more well-designed, well-executed nonrandomized studies, observational studies, or registry studies - Meta-analyses of such studies
CLASS III: No Benefit (MODERATE) Benefit = Risk <i>(Generally, LOE A or B use only)</i> Suggested phrases for writing recommendations: - Is not recommended - Is not indicated/useful/effective/beneficial - Should not be performed/administered/other	LEVEL C-LD (Limited Data) - Randomized or nonrandomized observational or registry studies with limitations of design or execution - Meta-analyses of such studies - Physiological or mechanistic studies in human subjects
CLASS III: Harm (STRONG) Risk > Benefit Suggested phrases for writing recommendations: - Potentially harmful - Causes harm - Associated with excess morbidity/mortality - Should not be performed/administered/other	LEVEL C-EO (Expert Opinion) Consensus of expert opinion based on clinical experience

COR and LOE are determined independently (any COR may be paired with any LOE).

A recommendation with LOE C does not imply that the recommendation is weak. Many important clinical questions addressed in guidelines do not lend themselves to clinical trials. Although RCTs are unavailable, there may be a very clear clinical consensus that a particular test or therapy is useful or effective.

* The outcome or result of the intervention should be specified (an improved clinical outcome or increased diagnostic accuracy or incremental prognostic information).

† For comparative-effectiveness recommendations (COR I and IIa; LOE A and B only), studies that support the use of comparator verbs should involve direct comparisons of the treatments or strategies being evaluated.

‡ The method of assessing quality is evolving, including the application of standardized, widely used, and preferably validated evidence grading tools; and for systematic reviews, the incorporation of an Evidence Review Committee.

COR indicates Class of Recommendation; EO, expert opinion; LD, limited data; LOE, Level of Evidence; NR, nonrandomized; R, randomized; and RCT, randomized controlled trial.

discussed, and then guideline recommendations were developed. The systematic review report, "Systematic Review for the 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults," is published in conjunction with this guideline,^{S1,5-8} and its respective data supplements are available [online](#). No writing committee member reported a RWI. Drs. Whelton, Wright, and Williamson had leadership roles in SPRINT

(Systolic Blood Pressure Intervention Trial). Dr. Carey chaired committee discussions in which the SPRINT results were considered.

1.2. Organization of the Writing Committee

The writing committee consisted of clinicians, cardiologists, epidemiologists, internists, an endocrinologist, a geriatrician, a nephrologist, a neurologist, a nurse, a pharmacist, a physician assistant, and 2 lay/patient representatives. It

Table 2. Systematic Review Questions on High BP in Adults

Question Number	Question	Section Number
1	Is there evidence that self-directed monitoring of BP and/or ambulatory BP monitoring are superior to office-based measurement of BP by a healthcare worker for 1) preventing adverse outcomes for which high BP is a risk factor and 2) achieving better BP control?	4.2
2	What is the optimal target for BP lowering during antihypertensive therapy in adults?	8.1.5 9.3 9.6
3	In adults with hypertension, do various antihypertensive drug classes differ in their comparative benefits and harms?	8.1.6 8.2
4	In adults with hypertension, does initiating treatment with antihypertensive pharmacological monotherapy versus initiating treatment with 2 drugs (including fixed-dose combination therapy), either of which may be followed by the addition of sequential drugs, differ in comparative benefits and/or harms on specific health outcomes?	8.1.6.1

BP indicates blood pressure.

included representatives from the ACC, AHA, American Academy of Physician Assistants (AAPA), Association of Black Cardiologists (ABC), American College of Preventive Medicine (ACPM), American Geriatrics Society (AGS), American Pharmacists Association (APhA), American Society of Hypertension (ASH), American Society for Preventive Cardiology (ASPC), National Medical Association (NMA), and Preventive Cardiovascular Nurses Association (PCNA).

1.3. Document Review and Approval

This document was reviewed by 2 official reviewers nominated by the ACC and AHA; 1 reviewer each from the AAPA, ABC, ACPM, AGS, APhA, ASH, ASPC, NMA, and PCNA; and 38 individual content reviewers. Reviewers' RWI information was distributed to the writing committee and is published in this document (Appendix 2).

This document was approved for publication by the governing bodies of the ACC, AHA, AAPA, ABC, ACPM, AGS, APhA, ASH, ASPC, NMA, and PCNA.

1.4. Scope of the Guideline

The present guideline is intended to be a resource for the clinical and public health practice communities. It is designed to be comprehensive but succinct and practical in providing guidance for prevention, detection, evaluation, and management of high BP. It is an update of the NHLBI publication, "The Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation and Treatment of High Blood Pressure" (JNC 7).^{S1.5-7} It incorporates new information from studies of office-based BP-related risk of CVD, ambulatory blood pressure monitoring (ABPM), home blood pressure monitoring

(HBPM), telemedicine, and various other areas. This guideline does not address the use of BP-lowering medications for the purposes of prevention of recurrent CVD events in patients with stable ischemic heart disease (SIHD) or chronic heart failure (HF) in the absence of hypertension; these topics are the focus of other ACC/AHA guidelines.^{S1.5-9,S1.5-10} In developing the present guideline, the writing committee reviewed prior published guidelines, evidence reviews, and related statements. Table 3 contains a list of publications and statements deemed pertinent to this writing effort and is intended for use as a resource, thus obviating the need to repeat existing guideline recommendations.

1.5. Abbreviations and Acronyms

Abbreviation/Acronym	Meaning/Phrase
ABPM	ambulatory blood pressure monitoring
ACE	angiotensin-converting enzyme
AF	atrial fibrillation
ARB	angiotensin receptor blocker
BP	blood pressure
CCB	calcium channel blocker
CHD	coronary heart disease
CKD	chronic kidney disease
CPAP	continuous positive airway pressure
CVD	cardiovascular disease
DBP	diastolic blood pressure
DM	diabetes mellitus
ECG	electrocardiogram
ESRD	end-stage renal disease
GDMT	guideline-directed management and therapy
GFR	glomerular filtration rate
HBPM	home blood pressure monitoring
EHR	electronic health record
HF	heart failure
HFpEF	heart failure with preserved ejection fraction
HF/rEF	heart failure with reduced ejection fraction
ICH	intracerebral hemorrhage
JNC	Joint National Commission
LV	left ventricular
LVH	left ventricular hypertrophy
MI	myocardial infarction
MRI	magnetic resonance imaging
PAD	peripheral artery disease
RAS	renin-angiotensin system
RCT	randomized controlled trial
SBP	systolic blood pressure
SIHD	stable ischemic heart disease
TIA	transient ischemic attack

Table 3. Associated Guidelines and Statements

Title	Organization	Publication Year
Guidelines		
Lower-extremity peripheral artery disease	AHA/ACC	2016 ^{S1.5-11}
Management of primary aldosteronism: case detection, diagnosis, and treatment	Endocrine Society	2016 ^{S1.5-12}
Stable ischemic heart disease	ACC/AHA/AATS/PCNA/SCAI/STS	2014 ^{S1.5-13*} 2012 ^{S1.5-9}
Pheochromocytoma and paraganglioma	Endocrine Society	2014 ^{S1.5-14}
Atrial fibrillation	AHA/ACC/HRS	2014 ^{S1.5-15}
Valvular heart disease	ACC/AHA	2017 ^{S1.5-16}
Assessment of cardiovascular risk	ACC/AHA	2013 ^{S1.5-17}
Hypertension in pregnancy	ACOG	2013 ^{S1.5-18}
Heart failure	ACC/AHA	2017 ^{S1.5-19} 2013 ^{S1.5-10}
Lifestyle management to reduce cardiovascular risk	AHA/ACC	2013 ^{S1.5-20}
Management of arterial hypertension	ESH/ESC	2013 ^{S1.5-21}
Management of overweight and obesity in adults	AHA/ACC/TOS	2013 ^{S1.5-22}
ST-elevation myocardial infarction	ACC/AHA	2013 ^{S1.5-23}
Treatment of blood cholesterol to reduce atherosclerotic cardiovascular risk in adults	ACC/AHA	2013 ^{S1.5-24}
Cardiovascular diseases during pregnancy	ESC	2011 ^{S1.5-25}
Effectiveness-based guidelines for the prevention of cardiovascular disease in women	AHA/ACC	2011 ^{S1.5-26}
Secondary prevention and risk-reduction therapy for patients with coronary and other atherosclerotic vascular disease	AHA/ACC	2011 ^{S1.5-27}
Assessment of cardiovascular risk in asymptomatic adults	ACC/AHA	2010 ^{S1.5-28}
Thoracic aortic disease	ACC/AHA/AATS/ACR/ASA/SCA/SCAI/SIR/STS/SVM	2010 ^{S1.5-29}
Diagnosis, evaluation, and treatment of high blood pressure in children and adolescents	NHLBI	2004 ^{S1.5-30}
Statements		
Salt sensitivity of blood pressure	AHA	2016 ^{S1.5-31}
Cardiovascular team-based care and the role of advanced practice providers	ACC	2015 ^{S1.5-32}
Treatment of hypertension in patients with coronary artery disease	AHA/ACC/ASH	2015 ^{S1.5-33}
Ambulatory blood pressure monitoring in children and adolescents	AHA	2014 ^{S1.5-34}
An effective approach to high blood pressure control	AHA/ACC/CDC	2014 ^{S1.5-35}
Ambulatory blood pressure monitoring	ESH	2013 ^{S1.5-36}
Performance measures for adults with coronary artery disease and hypertension	ACC/AHA/AMA-PCPI	2011 ^{S1.5-37}
Interventions to promote physical activity and dietary lifestyle changes for cardiovascular risk factor reduction in adults	AHA	2010 ^{S1.5-38}
Resistant hypertension: diagnosis, evaluation, and treatment	AHA	2008 ^{S1.5-39}

*The full-text SIHD guideline is from 2012.^{S1.5-9} A focused update was published in 2014.^{S1.5-13}

AATS indicates American Association for Thoracic Surgery; ACC, American College of Cardiology; ACOG, American College of Obstetricians and Gynecologists; ACR, American College of Radiology; AHA, American Heart Association; AMA, American Medical Association; ASA, American Stroke Association; ASH, American Society of Hypertension; CDC, Centers for Disease Control and Prevention; ESC, European Society of Cardiology; ESH, European Society of Hypertension; HRS, Heart Rhythm Society; NHLBI, National Heart, Lung, and Blood Institute; PCNA, Preventive Cardiovascular Nurses Association; PCPI, Physician Consortium for Performance Improvement; SCA, Society of Cardiovascular Anesthesiologists; SCAI, Society for Cardiovascular Angiography and Interventions; SIHD, stable ischemic heart disease; SIR, Society of Interventional Radiology; STS, Society of Thoracic Surgeons; SVM, Society for Vascular Medicine; and TOS, The Obesity Society.

2. BP and CVD Risk

2.1. Observational Relationship

Observational studies have demonstrated graded associations between higher systolic blood pressure (SBP) and diastolic

blood pressure (DBP) and increased CVD risk.^{S2.1-1,S2.1-2} In a meta-analysis of 61 prospective studies, the risk of CVD increased in a log-linear fashion from SBP levels <115 mm Hg to >180 mm Hg and from DBP levels <75 mm Hg to >105 mm Hg.^{S2.1-1} In that analysis, 20 mm Hg higher SBP and

10 mm Hg higher DBP were each associated with a doubling in the risk of death from stroke, heart disease, or other vascular disease. In a separate observational study including >1 million adult patients ≥ 30 years of age, higher SBP and DBP were associated with increased risk of CVD incidence and angina, myocardial infarction (MI), HF, stroke, peripheral artery disease (PAD), and abdominal aortic aneurysm, each evaluated separately.^{S2.1-2} An increased risk of CVD associated with higher SBP and DBP has been reported across a broad age spectrum, from 30 years to >80 years of age. Although the relative risk of incident CVD associated with higher SBP and DBP is smaller at older ages, the corresponding high BP-related increase in absolute risk is larger in older persons (≥ 65 years) given the higher absolute risk of CVD at an older age.^{S2.1-1}

2.2. BP Components

Epidemiological studies have evaluated associations of SBP and DBP, as well as derived components of BP measurements (including pulse pressure, mean BP, and mid-BP), with CVD outcomes (Table 4). When considered separately, higher levels of both SBP and DBP have been associated with increased CVD risk.^{S2.2-1,S2.2-2} Higher SBP has consistently been associated with increased CVD risk after adjustment for, or within strata of, DBP.^{S2.2-3–S2.2-5} In contrast, after consideration of SBP through adjustment or stratification, DBP has not been consistently associated with CVD risk.^{S2.2-6,S2.2-7} Although pulse pressure and mid-BP have been associated with increased CVD risk independent of SBP and DBP in some studies, SBP (especially) and DBP are prioritized in the present document because of the robust evidence base for these measures in both observational studies and clinical trials and because of their ease of measurement in practice settings.^{S2.2-8–S2.2-11}

2.3. Population Risk

In 2010, high BP was the leading cause of death and disability-adjusted life years worldwide.^{S2.3-1,S2.3-2} In the United States, hypertension (see Section 3.1 for definition) accounted for more CVD deaths than any other modifiable CVD risk factor and was second only to cigarette smoking as a preventable cause of death for any reason.^{S2.3-3} In a follow-up study of 23 272 US NHANES (National Health and Nutrition Examination Survey) participants, >50% of deaths from coronary heart disease (CHD) and stroke occurred

among individuals with hypertension.^{S2.3-4} Because of the high prevalence of hypertension and its associated increased risk of CHD, stroke, and end-stage renal disease (ESRD), the population-attributable risk of these outcomes associated with hypertension is high.^{S2.3-4,S2.3-5} In the population-based ARIC (Atherosclerosis Risk in Communities) study, 25% of the cardiovascular events (CHD, coronary revascularization, stroke, or HF) were attributable to hypertension. In the Northern Manhattan study, the percentage of events attributable to hypertension was higher in women (32%) than in men (19%) and higher in blacks (36%) than in whites (21%).^{S2.3-6} In 2012, hypertension was the second leading assigned cause of ESRD, behind diabetes mellitus (DM), and accounted for 34% of incident ESRD cases in the US population.^{S2.3-7}

2.4. Coexistence of Hypertension and Related Chronic Conditions

Recommendation for Coexistence of Hypertension and Related Chronic Conditions

References that support the recommendation are summarized in Online Data Supplement 1.

COR	LOE	Recommendation
I	B-NR	1. Screening for and management of other modifiable CVD risk factors are recommended in adults with hypertension. ^{S2.4-1,S2.4-2}

Synopsis

Many adult patients with hypertension have other CVD risk factors; a list of such modifiable and relatively fixed risk factors is provided in Table 5. Among US adults with hypertension between 2009 and 2012, 15.5% were current smokers, 49.5% were obese, 63.2% had hypercholesterolemia, 27.2% had DM, and 15.8% had chronic kidney disease (CKD; defined as estimated glomerular filtration rate [eGFR] <60 mL/min/1.73 m² and/or urine albumin:creatinine ≥ 300 mg/g).^{S2.4-3}

Not only are CVD risk factors common among adults with hypertension, a higher percentage of adults with CVD risk

Table 4. BP Measurement Definitions

BP Measurement	Definition
SBP	First Korotkoff sound*
DBP	Fifth Korotkoff sound*
Pulse pressure	SBP minus DBP
Mean arterial pressure	DBP plus one third pulse pressure†
Mid-BP	Sum of SBP and DBP, divided by 2

*See Section 4 for a description of Korotkoff sounds.

†Calculation assumes normal heart rate.

BP indicates blood pressure; DBP, diastolic blood pressure; and SBP, systolic blood pressure.

Table 5. CVD Risk Factors Common in Patients With Hypertension

Modifiable Risk Factors†	Relatively Fixed Risk Factors†
Current cigarette smoking, secondhand smoking	CKD
	Family history
Diabetes mellitus	Increased age
Dyslipidemia/hypercholesterolemia	Low socioeconomic/educational status
Overweight/obesity	Male sex
Physical inactivity/low fitness	Obstructive sleep apnea
Unhealthy diet	Psychosocial stress

*Factors that can be changed and, if changed, may reduce CVD risk.

†Factors that are difficult to change (CKD, low socioeconomic/educational status, obstructive sleep apnea^{S2.4-12}), cannot be changed (family history, increased age, male sex), or, if changed through the use of current intervention techniques, may not reduce CVD risk (psychosocial stress).

CKD indicates chronic kidney disease; and CVD, cardiovascular disease.

factors have hypertension. For example, 71% of US adults with diagnosed DM have hypertension.^{S2.4.4} In the Chronic Renal Insufficiency Cohort (CRIC), 86% of the participants had hypertension.^{S2.4.5} Also, 28.1% of adults with hypertension and CKD in the population-based REGARDS (Reasons for Geographic and Racial Differences in Stroke) study had apparent resistant hypertension.^{S2.4.6} In NHANES 1999–2010, 35.7% of obese individuals had hypertension.^{S2.4.7} The presence of multiple CVD risk factors in individuals with hypertension results in high absolute risks for CHD and stroke in this population. For example, among US adults with hypertension between 2009 and 2012, 41.7% had a 10-year CHD risk >20%, 40.9% had a risk of 10% to 20%, and only 18.4% had a risk <10%.^{S2.4.3}

Modifiable risk factors for CVD that are common among adults with hypertension include cigarette smoking/tobacco smoke exposure, DM, dyslipidemia (including high levels of low-density lipoprotein cholesterol or hypercholesterolemia, high levels of triglycerides, and low levels of high-density lipoprotein cholesterol), overweight/obesity, physical inactivity/low fitness level, and unhealthy diet.^{S2.4.8} The relationship between hypertension and other modifiable risk factors is complex and interdependent, with several sharing mechanisms of action and pathophysiology. CVD risk factors affect BP through over activation of the renin-angiotensin-aldosterone system, activation of the sympathetic nervous system, inhibition of the cardiac natriuretic peptide system, endothelial dysfunction, and other mechanisms.^{S2.4.9–S2.4.11} Treating some of the other modifiable risk factors may reduce BP through modification of shared pathology, and CVD risk may be reduced by treating global risk factor burden.

Recommendation-Specific Supportive Text

1. Observational studies have demonstrated that CVD risk factors frequently occur in combination, with ≥3 risk factors present in 17% of patients.^{S2.4.1} A meta-analysis from 18 cohort studies involving 257 384 patients identified a lifetime risk of CVD death, nonfatal MI, and fatal or nonfatal stroke that was substantially higher in adults with ≥2 CVD risk factors than in those with only 1 risk factor.^{S2.4.1,S2.4.2}

3. Classification of BP

3.1. Definition of High BP

Recommendation for Definition of High BP		
References that support the recommendation are summarized in Online Data Supplement 2 .		
COR	LOE	Recommendation
I	B-NR	1. BP should be categorized as normal, elevated, or stage 1 or 2 hypertension to prevent and treat high BP (Table 6). ^{S3.1.1–S3.1.20}

Synopsis

Although a continuous association exists between higher BP and increased CVD risk (see Section 2.1), it is useful to categorize BP levels for clinical and public health decision making. In the present document, BP is categorized into 4

Table 6. Categories of BP in Adults*

BP Category	SBP		DBP
Normal	<120 mm Hg	and	<80 mm Hg
Elevated	120–129 mm Hg	and	<80 mm Hg
Hypertension			
Stage 1	130–139 mm Hg	or	80–89 mm Hg
Stage 2	≥140 mm Hg	or	≥90 mm Hg

*Individuals with SBP and DBP in 2 categories should be designated to the higher BP category.

BP indicates blood pressure (based on an average of ≥2 careful readings obtained on ≥2 occasions, as detailed in Section 4); DBP, diastolic blood pressure; and SBP, systolic blood pressure.

levels on the basis of average BP measured in a healthcare setting (office pressures): normal, elevated, and stage 1 or 2 hypertension (Table 6). [Online Data Supplement C](#) illustrates schematically the SBP and DBP categories defining normal BP, elevated BP, and stages 1 and 2 hypertension. This categorization differs from that previously recommended in the JNC 7 report, with stage 1 hypertension now defined as an SBP of 130–139 or a DBP of 80–89 mm Hg, and with stage 2 hypertension in the present document corresponding to stages 1 and 2 in the JNC 7 report.^{S3.1.21} The rationale for this categorization is based on observational data related to the association between SBP/DBP and CVD risk, RCTs of lifestyle modification to lower BP, and RCTs of treatment with antihypertensive medication to prevent CVD. The increased risk of CVD among adults with stage 2 hypertension is well established. An increasing number of individual studies and meta-analyses of observational data have reported a gradient of progressively higher CVD risk going from normal BP to elevated BP and stage 1 hypertension.^{S3.1.4–S3.1.10,S3.1.12,S3.1.13,S3.1.16} In many of these meta-analyses, the hazard ratios for CHD and stroke were between 1.1 and 1.5 for the comparison of SBP/DBP of 120–129/80–84 mm Hg versus <120/80 mm Hg and between 1.5 and 2.0 for the comparison of SBP/DBP of 130–139/85–89 mm Hg versus <120/80 mm Hg. This risk gradient was consistent across subgroups defined by sex and race/ethnicity. The relative increase in CVD risk associated with higher BP was attenuated but still present among older adults.^{S3.1.1} The prevalence of severe hypertension has been declining over time, but approximately 12.3% of US adults with hypertension have an average SBP ≥160 mm Hg or average DBP ≥100 mm Hg.^{S3.1.22} Lifestyle modification and pharmacological antihypertensive treatment recommendations for individuals with elevated BP and stages 1 and 2 hypertension are provided in Sections 6 and 8, respectively. The relationship of this classification schema with measurements obtained by ambulatory BP recording and home BP measurements is discussed in Section 4.2.

Recommendation-Specific Supportive Text

1. As was the case in previous BP classification systems, the choice and the naming of the categories were based on a pragmatic interpretation of BP-related CVD risk and benefit of BP reduction in clinical trials. Meta-analyses of observational studies have demonstrated that elevated

BP and hypertension are associated with increased risk of CVD, ESRD, subclinical atherosclerosis, and all-cause death.^{S3.1-1-S3.1-17} The recommended BP classification system is most valuable in untreated adults as an aid in decisions about prevention or treatment of high BP. However, it is also useful in assessing the success of interventions to reduce BP.

3.2. Lifetime Risk of Hypertension

Observational studies have documented a relatively high incidence of hypertension over periods of 5 to 10 years of follow-up.^{S3.2-1,S3.2-2} Thus, there is a much higher long-term population burden of hypertension as BP progressively increases with age. Several studies have estimated the long-term cumulative incidence of developing hypertension.^{S3.2-3,S3.2-4} In an analysis of 1132 white male medical students (mean age: approximately 23 years at baseline) in the Johns Hopkins Precursors study, 0.3%, 6.5%, and 37% developed hypertension at age 25, 45, and 65 years, respectively.^{S3.2-5} In MESA (Multi-Ethnic Study of Atherosclerosis), the percentage of the population developing hypertension over their lifetimes was higher for African Americans and Hispanics than for whites and Asians.^{S3.2-3} For adults 45 years of age without hypertension, the 40-year risk of developing hypertension was 93% for African-American, 92% for Hispanic, 86% for white, and 84% for Chinese adults.^{S3.2-3} In the Framingham Heart Study, approximately 90% of adults free of hypertension at age 55 or 65 years developed hypertension during their lifetimes.^{S3.2-4} All of these estimates were based on use of the 140/90-mm Hg cutpoint for recognition of hypertension and would have been higher had the 130/80-mm Hg cutpoint been used.

3.3. Prevalence of High BP

Prevalence estimates are greatly influenced by the choice of cutpoints to categorize high BP, the methods used to establish the diagnosis, and the population studied.^{S3.3-1,S3.3-2} Most general population prevalence estimates are derived from national surveys. Table 7 provides estimates for prevalence of hypertension in the US general adult population (≥20 years of age) that are based on the definitions of hypertension recommended in the present guideline and in the JNC 7 report. The prevalence of hypertension among US adults is substantially higher when the definition in the present guideline is used versus the JNC 7 definition (46% versus 32%). However, as described in Section 8.1, nonpharmacological treatment (not antihypertensive medication) is recommended for most US adults who have hypertension as defined in the present guideline but who would not meet the JNC 7 definition for hypertension. As a consequence, the new definition results in only a small increase in the percentage of US adults for whom antihypertensive medication is recommended in conjunction with lifestyle modification.

The prevalence of hypertension rises dramatically with increasing age and is higher in blacks than in whites, Asians, and Hispanic Americans. NHANES estimates of JNC 7–defined hypertension prevalence have remained fairly stable since the early 2000s.^{S3.3-1} Most contemporary population surveys, including NHANES, rely on an average of BP measurements obtained at a single visit,^{S3.3-2} which is likely to result

Table 7. Prevalence of Hypertension Based on 2 SBP/DBP Thresholds*†

	SBP/DBP ≥130/80 mm Hg or Self-Reported Antihypertensive Medication†		SBP/DBP ≥140/90 mm Hg or Self-Reported Antihypertensive Medication‡	
Overall, crude	46%		32%	
	Men (n=4717)	Women (n=4906)	Men (n=4717)	Women (n=4906)
Overall, age-sex adjusted	48%	43%	31%	32%
Age group, y				
20–44	30%	19%	11%	10%
45–54	50%	44%	33%	27%
55–64	70%	63%	53%	52%
65–74	77%	75%	64%	63%
75+	79%	85%	71%	78%
Race-ethnicity§				
Non-Hispanic white	47%	41%	31%	30%
Non-Hispanic black	59%	56%	42%	46%
Non-Hispanic Asian	45%	36%	29%	27%
Hispanic	44%	42%	27%	32%

The prevalence estimates have been rounded to the nearest full percentage.

*130/80 and 140/90 mm Hg in 9623 participants (≥20 years of age) in NHANES 2011–2014.

†BP cutpoints for definition of hypertension in the present guideline.

‡BP cutpoints for definition of hypertension in JNC 7.

§Adjusted to the 2010 age-sex distribution of the US adult population.

BP indicates blood pressure; DBP, diastolic blood pressure; NHANES, National Health and Nutrition Examination Survey; and SBP, systolic blood pressure.

in an overestimate of hypertension prevalence compared with what would be found by using an average of ≥2 readings taken on ≥2 visits,^{S3.3-1} as recommended in current and previous BP guidelines.^{S3.3-3-S3.3-5} The extent to which guideline recommendations for use of BP averages from ≥2 occasions is followed in practice is unclear. Adding self-report of previously diagnosed hypertension yields a 5% to 10% higher estimate of prevalence.^{S3.3-1,S3.3-6,S3.3-7} Most individuals who were added by use of this expanded definition have been diagnosed as having hypertension by a health professional on >1 occasion, and many have been advised to change their lifestyle.^{S3.3-2,S3.3-6}

3.4. Awareness, Treatment, and Control

Prevalence estimates for awareness, treatment, and control of hypertension are usually based on self-reports of the hypertension diagnosis (awareness), use of BP-lowering medications in those with hypertension (treatment), and achievement of a satisfactory SBP/DBP during treatment of hypertension (control). Before the present publication, awareness and treatment in adults were based on the SBP/DBP cutpoints of 140/90 mm Hg, and control was based on an SBP/DBP <140/90 mm Hg. In the US general adult population, hypertension awareness, treatment, and control

have been steadily improving since the 1960s,^{S3.4-1-S3.4-4} with NHANES 2009 to 2012 prevalence estimates for men and women, respectively, being 80.2% and 85.4% for awareness, 70.9% and 80.6% for treatment (88.4% and 94.4% in those who were aware), 69.5% and 68.5% for control in those being treated, and 49.3% and 55.2% for overall control in adults with hypertension.^{S3.4-5} The NHANES experience may underestimate awareness, treatment, and control of hypertension because it is based on BP estimates derived from an average of readings obtained at a single visit, whereas guidelines recommend use of BP averages of ≥ 2 readings obtained on ≥ 2 occasions. In addition, the current definition of control excludes the possibility of control resulting from life-style change or nonpharmacological interventions. NHANES hypertension control rates have been consistently higher in women than in men (55.3% versus 38.0% in 2009–2012); in whites than in blacks and Hispanics (41.3% versus 31.1% and 23.6%, respectively, in men, and 57.2% versus 43.2% and 52.9%, respectively, in women, for 2009–2012); and in older than in younger adults (50.5% in adults ≥ 60 years of age versus 34.4% in patients 18 to 39 years of age for 2011–2012) up to the seventh decade,^{S3.4-4,S3.4-5} although control rates are considerably lower for those ≥ 75 years (46%) and only 39.8% for adults ≥ 80 years.^{S3.4-6} In addition, control rates are higher for persons of higher socioeconomic status (43.2% for adults with an income $>400\%$ above the US government poverty line versus 30.2% for those below this line in 2003 to 2006).^{S3.4-5} Research studies have repeatedly demonstrated that structured, goal-oriented BP treatment initiatives with feedback and provision of free medication result in a substantial improvement in BP control.^{S3.4-7-S3.4-9} Control rates that are much higher than noted in the general population have been reported in care settings where a systems approach (detailed in Sections 12.2 and 12.3) has been implemented for insured adults.^{S3.4-10-S3.4-12}

4. Measurement of BP

4.1. Accurate Measurement of BP in the Office

Recommendation for Accurate Measurement of BP in the Office		
COR	LOE	Recommendation
I	C-E0	1. For diagnosis and management of high BP, proper methods are recommended for accurate measurement and documentation of BP (Table 8).

Synopsis

Although measurement of BP in office settings is relatively easy, errors are common and can result in a misleading estimation of an individual's true level of BP. There are various methods for measuring BP in the office. The clinical standard of auscultatory measures calibrated to a column of mercury has given way to oscillometric devices (in part because of toxicological issues with mercury). Oscillometric devices use a sensor that detects oscillations in pulsatile blood volume during cuff inflation and deflation. BP is indirectly calculated from maximum amplitude algorithms that involve population-based data. For this reason, only devices with a

Table 8. Checklist for Accurate Measurement of BP^{S4.1-3,S4.1-4}

Key Steps for Proper BP Measurements	Specific Instructions
Step 1: Properly prepare the patient	1. Have the patient relax, sitting in a chair (feet on floor, back supported) for >5 min. 2. The patient should avoid caffeine, exercise, and smoking for at least 30 min before measurement. 3. Ensure patient has emptied his/her bladder. 4. Neither the patient nor the observer should talk during the rest period or during the measurement. 5. Remove all clothing covering the location of cuff placement. 6. Measurements made while the patient is sitting or lying on an examining table do not fulfill these criteria.
Step 2: Use proper technique for BP measurements	1. Use a BP measurement device that has been validated, and ensure that the device is calibrated periodically.* 2. Support the patient's arm (eg, resting on a desk). 3. Position the middle of the cuff on the patient's upper arm at the level of the right atrium (the midpoint of the sternum). 4. Use the correct cuff size, such that the bladder encircles 80% of the arm, and note if a larger- or smaller-than-normal cuff size is used (Table 9). 5. Either the stethoscope diaphragm or bell may be used for auscultatory readings.^{S4.1-5,S4.1-6}
Step 3: Take the proper measurements needed for diagnosis and treatment of elevated BP/hypertension	1. At the first visit, record BP in both arms. Use the arm that gives the higher reading for subsequent readings. 2. Separate repeated measurements by 1–2 min. 3. For auscultatory determinations, use a palpated estimate of radial pulse obliteration pressure to estimate SBP. Inflate the cuff 20–30 mm Hg above this level for an auscultatory determination of the BP level. 4. For auscultatory readings, deflate the cuff pressure 2 mm Hg per second, and listen for Korotkoff sounds.
Step 4: Properly document accurate BP readings	1. Record SBP and DBP. If using the auscultatory technique, record SBP and DBP as onset of the first Korotkoff sound and disappearance of all Korotkoff sounds, respectively, using the nearest even number. 2. Note the time of most recent BP medication taken before measurements.
Step 5: Average the readings	Use an average of ≥ 2 readings obtained on ≥ 2 occasions to estimate the individual's level of BP.
Step 6: Provide BP readings to patient	Provide patients the SBP/DBP readings both verbally and in writing.

*See Section 4.2 for additional guidance.

Adapted with permission from Mancia et al^{S4.1-3} (Oxford University Press), Pickering et al^{S4.1-2} (American Heart Association, Inc.), and Weir et al^{S4.1-4} (American College of Physicians, Inc.).

BP indicates blood pressure; DBP, diastolic blood pressure; and SBP, systolic blood pressure.

validated measurement protocol can be recommended for use (see Section 4.2 for additional details). Many of the newer oscillometric devices automatically inflate multiple times (in 1- to 2-minute intervals), allowing patients to be alone and undisturbed during measurement. Although much of the available BP-related risk information and antihypertensive treatment trial experience have been generated by using “traditional” office methods of BP measurement, there is a growing evidence base supporting the use of automated office BP measurements.^{S4.1-1}

Recommendation-Specific Supportive Text

1. Accurate measurement and recording of BP are essential to categorize level of BP, ascertain BP-related CVD risk, and guide management of high BP. Most systematic errors in BP measurement can be avoided by following the suggestions provided in Table 8, including having the patient sit quietly for 5 minutes before a reading is taken, supporting the limb used to measure BP, ensuring the BP cuff is at heart level, using the correct cuff size (Table 9), and, for auscultatory readings, deflating the cuff slowly.^{S4.1-2} In those who are already taking medication that affects BP, the timing of BP measurements in relation to ingestion of the patient’s medication should be standardized. Because individual BP measurements tend to vary in an unpredictable or random fashion, a single reading is inadequate for clinical decision-making. An average of 2 to 3 BP measurements obtained on 2 to 3 separate occasions will minimize random error and provide a more accurate basis for estimation of BP. In addition to clinicians, other caregivers and patients who perform BP self-monitoring should be trained to follow the checklist in Table 8. Common errors in clinical practice that can lead to inaccurate estimation of BP include failure to allow for a rest period and/or talking with the patient during or immediately before the recording, improper patient positioning (eg, sitting or lying on an examination table), rapid cuff deflation (for auscultatory readings), and reliance on BPs measured at a single occasion.

4.2. Out-of-Office and Self-Monitoring of BP

Recommendation for Out-of-Office and Self-Monitoring of BP		
References that support the recommendation are summarized in Online Data Supplement 3 and Systematic Review Report.		
COR	LOE	Recommendation
I	ASR	1. Out-of-office BP measurements are recommended to confirm the diagnosis of hypertension (Table 11) and for titration of BP-lowering medication, in conjunction with telehealth counseling or clinical interventions. ^{S4.2-1–S4.2-4}

SR indicates systematic review.

Synopsis

Out-of-office measurement of BP can be helpful for confirmation and management of hypertension. Self-monitoring

Table 9. Selection Criteria for BP Cuff Size for Measurement of BP in Adults

Arm Circumference	Usual Cuff Size
22–26 cm	Small adult
27–34 cm	Adult
35–44 cm	Large adult
45–52 cm	Adult thigh

Adapted with permission from Pickering et al^{S4.1-2} (American Heart Association, Inc.).

BP indicates blood pressure.

of BP refers to the regular measurement of BP by an individual at home or elsewhere outside the clinic setting. Among individuals with hypertension, self-monitoring of BP, without other interventions, has shown limited evidence for treatment-related BP reduction and achievement of BP control.^{S4.2-1,S4.2-5,S4.2-6} However, with the increased recognition of inconsistencies between office and out-of-office BPs (see Section 4.4) and greater reduction in BP being recommended for hypertension control, increased attention is being paid to out-of-office BP readings. Although ABPM is generally accepted as the best out-of-office measurement method, HBPM is often a more practical approach in clinical practice. Recommended procedures for the collection of HBPM data are provided in Table 10. If self-monitoring is used, it is important to ensure that the BP measurement device used has been validated with an internationally accepted protocol and the results have been published in a peer-reviewed journal.^{S4.2-7} A guide to the relationship between HBPM BP readings and corresponding readings obtained in the office and by ABPM is presented in Table 11. The precise relationships between office readings, ABPM, and HBPM are unsettled, but there is general agreement that office BPs are often higher than ABPM or HBPM BPs, especially at higher BPs.

Recommendation-Specific Supportive Text

1. ABPM is used to obtain out-of-office BP readings at set intervals, usually over a period of 24 hours. HBPM is used to obtain a record of out-of-office BP readings taken by a patient. Both ABPM and HBPM typically provide BP estimates that are based on multiple measurements. A systematic review conducted by the US Preventive Services Task Force reported that ABPM provided a better method to predict long-term CVD outcomes than did office BPs. It incorporates new information from studies of HBPM, ABPM, the relationship of overall CVD risk to the effectiveness of blood pressure lowering, clinical outcomes related to different blood pressure goals, strategies to improve blood pressure control and various other areas. A small body of evidence suggested, but did not confirm, that HBPM could serve as a similar predictor of outcomes.^{S4.2-4} Meta-analyses of RCTs have identified clinically useful reductions in SBP and DBP and achievement of BP goals at 6 months and 1 year when self-monitoring of BP has been used in conjunction with other interventions, compared with usual care. Meta-analyses of RCTs have identified only small

Table 10. Procedures for Use of HBPM^{S4.2-5–S4.2-7}

Patient training should occur under medical supervision, including:
Information about hypertension
Selection of equipment
Acknowledgment that individual BP readings may vary substantially
Interpretation of results
Devices:
Verify use of automated validated devices. Use of auscultatory devices (mercury, aneroid, or other) is not generally useful for HBPM because patients rarely master the technique required for measurement of BP with auscultatory devices.
Monitors with provision for storage of readings in memory are preferred.
Verify use of appropriate cuff size to fit the arm (Table 9).
Verify that left/right inter-arm differences are insignificant. If differences are significant, instruct patient to measure BPs in the arm with higher readings.
Instructions on HBPM procedures:
Remain still:
Avoid smoking, caffeinated beverages, or exercise within 30 min before BP measurements.
Ensure ≥5 min of quiet rest before BP measurements.
Sit correctly:
Sit with back straight and supported (on a straight-backed dining chair, for example, rather than a sofa).
Sit with feet flat on the floor and legs uncrossed.
Keep arm supported on a flat surface (such as a table), with the upper arm at heart level.
Bottom of the cuff should be placed directly above the antecubital fossa (bend of the elbow).
Take multiple readings:
Take at least 2 readings 1 min apart in morning before taking medications and in evening before supper. Optimally, measure and record BP daily. Ideally, obtain weekly BP readings beginning 2 weeks after a change in the treatment regimen and during the week before a clinic visit.
Record all readings accurately:
Monitors with built-in memory should be brought to all clinic appointments.
BP should be based on an average of readings on ≥2 occasions for clinical decision making.
The information above may be reinforced with videos available online.

See Table 11 for HBPM targets.

BP indicates blood pressure; and HBPM, home blood pressure monitoring.

net reductions in SBP and DBP at 6 months and 1 year for use of self-monitoring of BP on its own, as compared with usual care.^{S4.2-1,S4.2-5,S4.2-6} See Section 4.4 for additional details of diagnostic classification and Section 12 for additional details of telehealth and out-of-office BP measurement for management of high BP.

4.3. Ambulatory BP Monitoring

All of the major RCTs have been based on use of clinic BP readings. However, ABPM is often used to supplement BP

Table 11. Corresponding Values of SBP/DBP for Clinic, HBPM, Daytime, Nighttime, and 24-Hour ABPM Measurements

Clinic	HBPM	Daytime ABPM	Nighttime ABPM	24-Hour ABPM
120/80	120/80	120/80	100/65	115/75
130/80	130/80	130/80	110/65	125/75
140/90	135/85	135/85	120/70	130/80
160/100	145/90	145/90	140/85	145/90

ABPM indicates ambulatory blood pressure monitoring; BP, blood pressure; DBP, diastolic blood pressure; HBPM, home blood pressure monitoring; and SBP, systolic blood pressure.

readings obtained in office settings.^{S4.3-1} The monitors are usually programmed to obtain readings every 15 to 30 minutes throughout the day and every 15 minutes to 1 hour during the night. ABPM is conducted while individuals go about their normal daily activities. ABPM can a) provide estimates of mean BP over the entire monitoring period and separately during nighttime and daytime, b) determine the daytime-to-nighttime BP ratio to identify the extent of nocturnal “dipping,” c) identify the early-morning BP surge pattern, d) estimate BP variability, and e) allow for recognition of symptomatic hypotension. The US Centers for Medicaid & Medicare Services and other agencies provide reimbursement for ABPM in patients with suspected white coat hypertension.^{S4.3-2} Medicare claims for ABPM between 2007 and 2010 were reimbursed at a median of \$52 and were submitted for <1% of beneficiaries.^{S4.3-3,S4.3-4} A list of devices validated for ABPM is available.^{S4.3-5,S4.3-6}

ABPM and HBPM definitions of high BP use different BP thresholds than those used by the previously mentioned office-based approach to categorize high BP identified in Section 3.1. Table 11 provides best estimates for corresponding home, daytime, nighttime, and 24-hour ambulatory levels of BP, including the values recommended for identification of hypertension with office measurements. Typically, a clinic BP of 140/90 mm Hg corresponds to home BP values of 135/85 mm Hg and to ABPM values defined as a daytime SBP/DBP of 135/85 mm Hg, a nighttime SBP/DBP of 120/70 mm Hg, and a 24-hour SBP/DBP of 130/80 mm Hg.^{S4.3-7,S4.3-8} These thresholds are based on data from European, Australian, and Asian populations, with few data available for establishing appropriate thresholds for US populations.^{S4.3-9–S4.3-13} They are provided as a guide but should be interpreted with caution. Higher daytime SBP measurements from ABPM can be associated with an increased risk of CVD and all-cause death independent of clinic-measured BP.^{S4.3-14} A meta-analysis of observational studies that included 13 844 individuals suggested nighttime BP is a stronger risk factor for CHD and stroke than either clinic or daytime BP.^{S4.3-15}

Methodological issues complicate the interpretation of data from studies that report office and out-of-office BP readings. Definitions and diagnostic methods for identifying white coat hypertension and masked hypertension (see Section 4.4) have not been standardized. The available studies have differed with regard to number of office readings obtained, use of 24-hour ABPM, use of daytime-only ABPM, inclusion of daytime and nighttime BP readings as separate categories, HBPM

for monitoring out-of-office BP levels, and even the BP thresholds used to define hypertension with ABPM or HBPM readings. In addition, there are few data that address reproducibility of these hypertension profiles over time, with several studies suggesting progression of white coat hypertension and especially of masked hypertension to sustained office-measured hypertension.^{S4.3-16–S4.3-22}

4.4. Masked and White Coat Hypertension

Recommendations for Masked and White Coat Hypertension		
References that support recommendations are summarized in Online Data Supplements 4, 5, and 6.		
COR	LOE	Recommendations
Ia	B-NR	1. In adults with an untreated SBP greater than 130 mm Hg but less than 160 mm Hg or DBP greater than 80 mm Hg but less than 100 mm Hg, it is reasonable to screen for the presence of white coat hypertension by using either daytime ABPM or HBPM before diagnosis of hypertension. ^{S4.4-1–S4.4-8}
Ia	C-LD	2. In adults with white coat hypertension, periodic monitoring with either ABPM or HBPM is reasonable to detect transition to sustained hypertension (S4.4-2, S4.4-5, S4.4-7).
Ia	C-LD	3. In adults being treated for hypertension with office BP readings not at goal and HBPM readings suggestive of a significant white coat effect, confirmation by ABPM can be useful (S4.4-9, S4.4-10).
Ia	B-NR	4. In adults with untreated office BPs that are consistently between 120 mm Hg and 129 mm Hg for SBP or between 75 mm Hg and 79 mm Hg for DBP, screening for masked hypertension with HBPM (or ABPM) is reasonable (S4.4-3, S4.4-4, S4.4-6, S4.4-8, S4.4-11).
IIb	C-LD	5. In adults on multiple-drug therapies for hypertension and office BPs within 10 mm Hg above goal, it may be reasonable to screen for white coat effect with HBPM (or ABPM) (S4.4-3, S4.4-7, S4.4-12).
IIb	C-EO	6. It may be reasonable to screen for masked uncontrolled hypertension with HBPM in adults being treated for hypertension and office readings at goal, in the presence of target organ damage or increased overall CVD risk.
IIb	C-EO	7. In adults being treated for hypertension with elevated HBPM readings suggestive of masked uncontrolled hypertension, confirmation of the diagnosis by ABPM might be reasonable before intensification of antihypertensive drug treatment.

Synopsis

The availability of noninvasive BP monitoring techniques has resulted in differentiation of hypertension into several clinically useful categories that are based on the place of BP

measurement (Table 12).^{S4.4-1, S4.4-13, S4.4-14} These include masked hypertension and white coat hypertension, in addition to sustained hypertension. White coat hypertension is characterized by elevated office BP but normal readings when measured outside the office with either ABPM or HBPM. In contrast, masked hypertension is characterized by office readings suggesting normal BP but out-of-office (ABPM/HBPM) readings that are consistently above normal.^{S4.4-15} In sustained hypertension, BP readings are elevated in both office and out-of-office settings.

In patients treated for hypertension, both “white coat effect” (higher office BPs than out-of-office BPs) and “masked uncontrolled hypertension” (controlled office BPs but uncontrolled BPs in out-of-office settings) categories have been reported.^{S4.4-5, S4.4-15, S4.4-16} The white coat effect (usually considered clinically significant when office SBP/DBPs are >20/10 mm Hg higher than home or ABPM SBP/DBPs) has been implicated in “pseudo-resistant hypertension” (see Section 11.1) and results in an underestimation of office BP control rates.^{S4.4-17, S4.4-18} The prevalence of masked hypertension varies from 10% to 26% (mean 13%) in population-based surveys and from 14% to 30% in normotensive clinic populations.^{S4.4-6, S4.4-16, S4.4-19–S4.4-21}

The risk of CVD and all-cause mortality in persons with masked hypertension is similar to that noted in those with sustained hypertension and about twice as high as the corresponding risk in their normotensive counterparts.^{S4.4-3, S4.4-4, S4.4-6, S4.4-8, S4.4-11} The prevalence of masked hypertension increases with higher office BP readings.^{S4.4-20, S4.4-22, S4.4-23}

The prevalence of white coat hypertension is higher with increasing age,^{S4.4-24} female versus male sex, nonsmoking versus current smoking status, and routine office measurement of BP by clinician observers versus unattended BP measurements. Many, but not all, studies^{S4.4-4, S4.4-6, S4.4-8, S4.4-25, S4.4-26} have identified a minimal increase in risk of CVD complications or all-cause mortality in patients who have white coat hypertension. This has resulted in a recommendation by some panels to screen for white coat hypertension with ABPM (or HBPM) to avoid initiating antihypertensive drug treatment in such individuals.^{S4.4-2, S4.4-5, S4.4-27} The white coat effect and masked uncontrolled hypertension appear to follow the risk profiles of their white coat hypertension and masked hypertension counterparts, respectively.^{S4.4-3, S4.4-12}

There are no data on the risks and benefits of treating white coat and masked hypertension. Despite these methodological differences, the data are consistent in indicating that

Table 12. BP Patterns Based on Office and Out-of-Office Measurements

	Office/Clinic/ Healthcare Setting	Home/Nonhealthcare/ ABPM Setting
Normotensive	No hypertension	No hypertension
Sustained hypertension	Hypertension	Hypertension
Masked hypertension	No hypertension	Hypertension
White coat hypertension	Hypertension	No hypertension

ABPM indicates ambulatory blood pressure monitoring; and BP, blood pressure.

masked hypertension and masked uncontrolled hypertension are associated with an increased prevalence of target organ damage and risk of CVD, stroke, and mortality compared with normotensive individuals and those with white coat hypertension.

Figure 1 is an algorithm on the detection of white coat hypertension or masked hypertension in patients not on drug therapy. Figure 2 is an algorithm on detection of white coat effect or masked uncontrolled hypertension in patients on drug therapy. Table 12 is a summary of BP patterns based on office and out-of-office measurements.

Recommendation-Specific Supportive Text

1. White coat hypertension prevalence averages approximately 13% and as high as 35% in some hypertensive populations,^{S4.4-1,S4.4-2} and ABPM and HBPM are better predictors of CVD risk due to elevated BP than are office BP measurements, with ABPM being the preferred measurement option. The major clinical relevance of white coat hypertension is that it has typically been associated with a minimal to only slightly increased risk of CVD and all-cause mortality risk.^{S4.4-3,S4.4-4,S4.4-6,S4.4-8,S4.4-11,S4.4-24} If ABPM resources are not readily available, HBPM provides a reasonable but less desirable alternative to screen for white coat hypertension, although the overlap with ABPM is only 60% to 70% for detection of white coat hypertension.^{S4.4-5,S4.4-9,S4.4-27-S4.4-30}
2. The incidence of white coat hypertension converting to sustained hypertension (justifying the addition of antihypertensive drug therapy to lifestyle modification) is 1% to 5% per year by ABPM or HBPM, with a higher incidence of conversion in those with elevated BP, older age, obesity, or black race.^{S4.4-2,S4.4-7}
3. The overlap between HBPM and both daytime and 24-hour ABPM in diagnosing white coat hypertension is only 60% to 70%, and the data for prediction of CVD risk are stronger with ABPM than with office measurements.^{S4.4-5,S4.4-9,S4.4-27-S4.4-30} Because a diagnosis of white coat hypertension may result in a decision not to treat or intensify treatment in patients with elevated office BP readings, confirmation of BP control by ABPM in addition to HBPM provides added support for this decision.
4. In contrast to white coat hypertension, masked hypertension is associated with a CVD and all-cause mortality risk

twice as high as that seen in normotensive individuals, with a risk range similar to that of patients with sustained hypertension.^{S4.4-3,S4.4-4,S4.4-6,S4.4-8,S4.4-11,S4.4-31} Therefore, out-of-office readings are reasonable to confirm BP control seen with office readings.

5. The white coat effect has been implicated in office-measured uncontrolled hypertension and pseudo-resistant hypertension, which may result in BP control being underestimated when subsequently assessed by ABPM.^{S4.4-17,S4.4-18} The risk of vascular complications in patients with office-measured uncontrolled hypertension with a white coat effect is similar to the risk in those with controlled hypertension.^{S4.4-3,S4.4-4,S4.4-7,S4.4-11,S4.4-12} White coat hypertension and white coat effect raise the concern that unnecessary antihypertensive drug therapy may be initiated or intensified. Because a diagnosis of white coat hypertension or white coat effect would result in a decision to not treat elevated office BP readings, confirmation of BP control by HBPM (or ABPM) provides more definitive support for the decision not to initiate antihypertensive drug therapy or accelerate treatment.
6. Analogous to masked hypertension in untreated patients, masked uncontrolled hypertension is defined in treated patients with hypertension by office readings suggesting adequate BP control but out-of-office readings (HBPM) that remain consistently above goal.^{S4.4-3,S4.4-15,S4.4-16,S4.4-32,S4.4-33} The CVD risk profile for masked uncontrolled hypertension appears to follow the risk profile for masked hypertension.^{S4.4-3,S4.4-12,S4.4-34} Although the evidence is consistent in identifying the increased risk of masked uncontrolled hypertension, evidence is lacking on whether the treatment of masked hypertension or masked uncontrolled hypertension reduces clinical outcomes. A suggestion for assessing CVD risk is provided in Section 8.
7. Although both ABPM and HBPM are better predictors of CVD risk than are office BP readings, ABPM confirmation of elevated BP by HBPM might be reasonable because of the more extensive documentation of CVD risk with ABPM. However, unlike the documentation of a significant white coat effect to justify the decision to not treat an elevated clinic BP, it is not mandatory to confirm masked uncontrolled hypertension determined by HBPM.

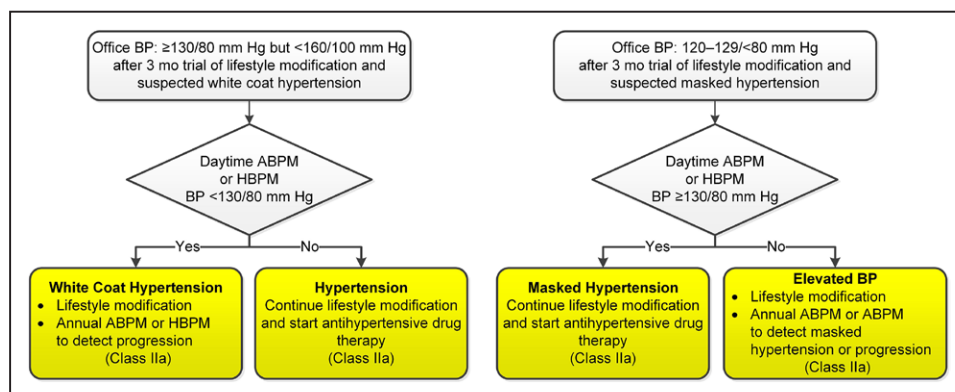


Figure 1. Detection of white coat hypertension or masked hypertension in patients not on drug therapy. Colors correspond to Class of Recommendation in Table 1. ABPM indicates ambulatory blood pressure monitoring; BP, blood pressure; and HBPM, home blood pressure monitoring.

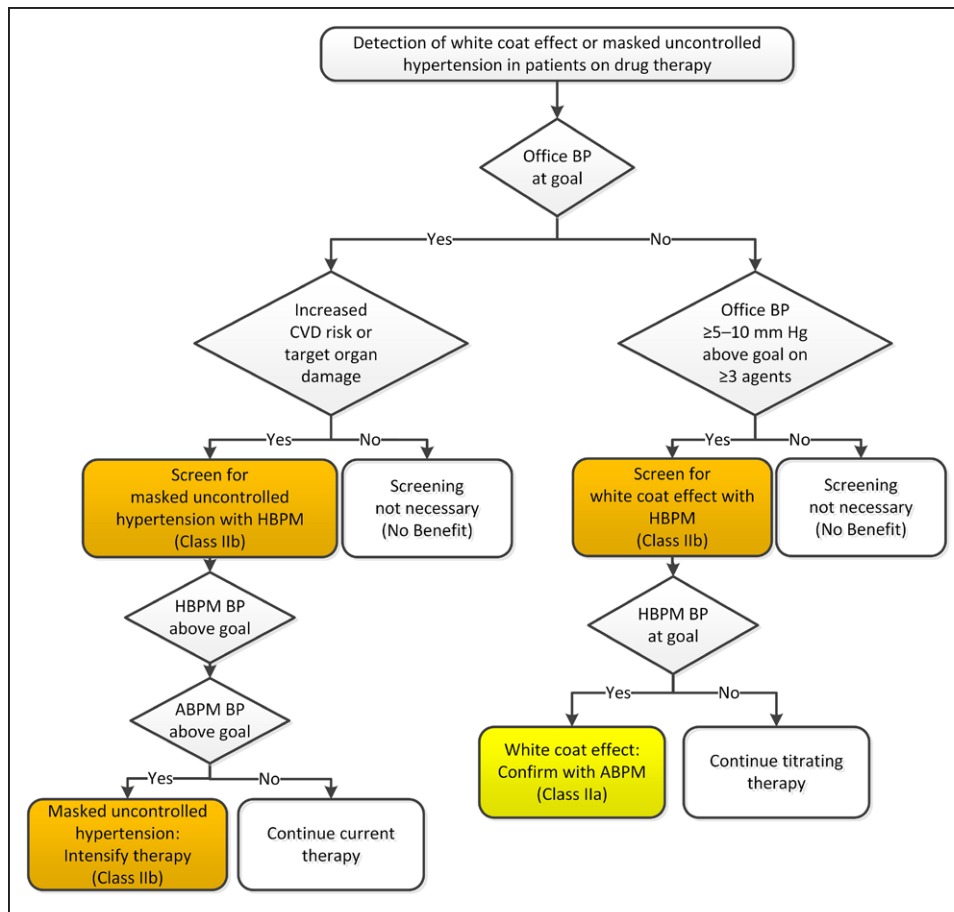


Figure 2. Detection of white coat effect or masked uncontrolled hypertension in patients on drug therapy. Colors correspond to Class of Recommendation in Table 1. See Section 8 for treatment options. ABPM indicates ambulatory blood pressure monitoring; BP, blood pressure; CVD, cardiovascular disease; and HBPM, home blood pressure monitoring.

5. Causes of Hypertension

5.1. Genetic Predisposition

Hypertension is a complex polygenic disorder in which many genes or gene combinations influence BP.^{S5.1-1,S5.1-2} Although several monogenic forms of hypertension have been identified, such as glucocorticoid-remediable aldosteronism, Liddle's syndrome, Gordon's syndrome, and others in which single-gene mutations fully explain the pathophysiology of hypertension, these disorders are rare.^{S5.1-3} The current tabulation of known genetic variants contributing to BP and hypertension includes more than 25 rare mutations and 120 single-nucleotide polymorphisms.^{S5.1-3,S5.1-4} However, even with the discovery of multiple single-nucleotide polymorphisms influencing control of BP since completion of the Human Genome Project in 2003, the associated variants have only small effects. Indeed, at present, the collective effect of all BP loci identified through genome-wide association studies accounts for only about 3.5% of BP variability.^{S5.1-4} The presence of a high number of small-effect alleles associated with higher BP results in a more rapid increase in BP with age.^{S5.1-5} Future studies will need to better elucidate genetic expression, epigenetic effects, transcriptomics, and proteomics that link genotypes with underlying pathophysiological mechanisms.

5.2. Environmental Risk Factors

Various environmental exposures, including components of diet, physical activity, and alcohol consumption, influence

BP. Many dietary components have been associated with high BP.^{S5.2-1,S5.2-2} Some of the diet-related factors associated with high BP include overweight and obesity, excess intake of sodium, and insufficient intake of potassium, calcium, magnesium, protein (especially from vegetables), fiber, and fish fats. Poor diet, physical inactivity, and excess intake of alcohol, alone or in combination, are the underlying cause of a large proportion of hypertension. Gut microbiota have also been linked to hypertension, especially in experimental animals.^{S5.2-3} Some of the best-proven environmental relationships with high BP are briefly reviewed below, and non-pharmacological interventions to lower BP are discussed in Section 6.2.

5.2.1. Overweight and Obesity

Insurance industry actuarial reports have identified a striking relationship between body weight and high BP^{S5.2.1-1} and a direct relationship between overweight/obesity and hypertension.^{S5.2.1-2} Epidemiological studies, including the Framingham Heart Study^{S5.2.1-3} and the Nurses' Health Study,^{S5.2.1-4} have consistently identified a direct relationship between body mass index and BP that is continuous and almost linear, with no evidence of a threshold.^{S5.2.1-5,S5.2.1-6} The relationship with BP is even stronger for waist-to-hip ratio and computed tomographic measures of central fat distribution.^{S5.2.1-7} Attributable risk estimates from the Nurses' Health Study suggest that obesity may be responsible for about 40% of hypertension, and in the Framingham Offspring Study, the corresponding estimates were even

higher (78% in men and 65% in women).^{S5.2.1-8,S5.2.1-9} The relationship between obesity at a young age and change in obesity status over time is strongly related to future risk of hypertension. In combined data from 4 longitudinal studies begun in adolescence with repeat examination in young adulthood to early middle age, being obese continuously or acquiring obesity was associated with a relative risk of 2.7 for developing hypertension. Becoming normal weight reduced the risk of developing hypertension to a level similar to those who had never been obese.^{S5.2.1-10}

5.2.2. Sodium Intake

Sodium intake is positively associated with BP in migrant,^{S5.2.2-11} cross-sectional,^{S5.2.2-12–S5.2.2-14} and prospective cohort studies^{S5.2.2-15} and accounts for much of the age-related increase in BP.^{S5.2.2-11,S5.2.2-16} In addition to the well-accepted and important relationship of dietary sodium with BP, excessive consumption of sodium is independently associated with an increased risk of stroke,^{S5.2.2-17,S5.2.2-18} CVD,^{S5.2.2-19} and other adverse outcomes.^{S5.2.2-20} Certain groups with various demographic, physiological, and genetic characteristics tend to be particularly sensitive to the effects of dietary sodium on BP.^{S5.2.2-21–S5.2.2-23} Salt sensitivity is a quantitative trait in which an increase in sodium load disproportionately increases BP.^{S5.2.2-21,S5.2.2-24} Salt sensitivity is especially common in blacks, older adults, and those with a higher level of BP or comorbidities such as CKD, DM, or the metabolic syndrome.^{S5.2.2-25} In aggregate, these groups constitute more than half of all US adults.^{S5.2.2-26} Salt sensitivity may be a marker for increased CVD and all-cause mortality risk independently of BP,^{S5.2.2-27,S5.2.2-28} and the trait has been demonstrated to be reproducible.^{S5.2.2-29} Current techniques for recognition of salt sensitivity are impractical in routine clinical practice, so salt sensitivity is best considered as a group characteristic.

5.2.3. Potassium

Potassium intake is inversely related to BP in migrant,^{S5.2.3-30} cross-sectional,^{S5.2.2-13,S5.2.2-16,S5.2.3-31,S5.2.3-32} and prospective cohort^{S5.2.3-33} studies. It is also inversely related to stroke.^{S5.2.3-34–S5.2.3-36} A higher level of potassium seems to blunt the effect of sodium on BP,^{S5.2.3-37} with a lower sodium–potassium ratio being associated with a lower level of BP than that noted for corresponding levels of sodium or potassium on their own.^{S5.2.3-38} Likewise, epidemiological studies suggest that a lower sodium–potassium ratio may result in a reduced risk of CVD as compared with the pattern for corresponding levels of either cation on its own.^{S5.2.3-39}

5.2.4. Physical Fitness

Epidemiological studies have demonstrated an inverse relationship between physical activity and physical fitness and level of BP and hypertension.^{S5.2.4-40} Even modest levels of physical activity have been associated with a decrease in the risk of incident hypertension.^{S5.2.4-41} In several observational studies, the relationship between physical activity and BP has been most apparent in white men.^{S5.2.4-40} With the advent of electronic activity trackers and ABPM, it has become increasingly feasible to conduct studies that relate physical activity and BP.^{S5.2.4-42} Physical fitness, measured objectively by graded exercise testing, attenuates the rise of BP with age and prevents the

development of hypertension. In the CARDIA (Coronary Artery Risk Development in Young Adults) study,^{S5.2.4-43} physical fitness measured at 18 to 30 years of age in the upper 2 deciles of an otherwise healthy population was associated with one third the risk of developing hypertension 15 years later, and one half the risk after adjustment for body mass index, as compared with the lowest quintile. Change in fitness assessed 7 years later further modified risk.^{S5.2.4-43} In a cohort of men 20 to 90 years of age who were followed longitudinally for 3 to 28 years, higher physical fitness decreased the rate of rise in SBP over time and delayed the time to onset of hypertension.^{S5.2.4-44}

5.2.5. Alcohol

The presence of a direct relationship between alcohol consumption and BP was first reported in 1915^{S5.2.5-45} and has been repeatedly identified in contemporary cross-sectional and prospective cohort studies.^{S5.2.5-46} Estimates of the contribution of alcohol consumption to population incidence and prevalence of hypertension vary according to level of intake. In the United States, it seems likely that alcohol may account for close to 10% of the population burden of hypertension (higher in men than in women). In contrast to its detrimental effect on BP, alcohol intake is associated with

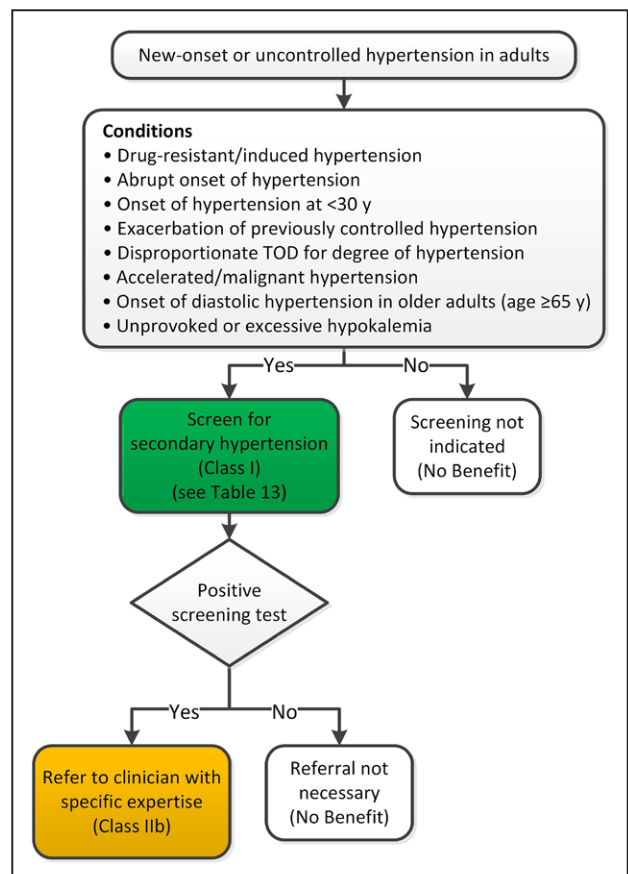


Figure 3. Screening for secondary hypertension. Colors correspond to Class of Recommendation in Table 1. TOD indicates target organ damage (eg, cerebrovascular disease, hypertensive retinopathy, left ventricular hypertrophy, left ventricular dysfunction, heart failure, coronary artery disease, chronic kidney disease, albuminuria, peripheral artery disease).

Table 13. Causes of Secondary Hypertension With Clinical Indications and Diagnostic Screening Tests

	Prevalence	Clinical Indications	Physical Examination	Screening Tests	Additional/Confirmatory Tests
Common causes					
Renal parenchymal disease ^{SS.4-1,SS.4-3}	1%–2%	Urinary tract infections; obstruction, hematuria; urinary frequency and nocturia; analgesic abuse; family history of polycystic kidney disease; elevated serum creatinine; abnormal urinalysis	Abdominal mass (polycystic kidney disease); skin pallor	Renal ultrasound	Tests to evaluate cause of renal disease
Renovascular disease ^{SS.4-4}	5%–34% [*]	Resistant hypertension; hypertension of abrupt onset or worsening or increasingly difficult to control; flash pulmonary edema (atherosclerotic); early-onset hypertension, especially in women (fibromuscular hyperplasia)	Abdominal systolic-diastolic bruit; bruits over other arteries (carotid – atherosclerotic or fibromuscular dysplasia), femoral	Renal Duplex Doppler ultrasound; MRA; abdominal CT	Bilateral selective renal intra-arterial angiography
Primary aldosteronism ^{SS.4-5,SS.4-6}	8%–20% [†]	Resistant hypertension; hypertension with hypokalemia (spontaneous or diuretic induced); hypertension and muscle cramps or weakness; hypertension and incidentally discovered adrenal mass; hypertension and obstructive sleep apnea; hypertension and family history of early-onset hypertension or stroke	Arrhythmias (with hypokalemia); especially atrial fibrillation	Plasma aldosterone/renin ratio under standardized conditions (correction of hypokalemia and withdrawal of aldosterone antagonists for 4–6 wk)	Oral sodium loading test (with 24-h urine aldosterone) or IV saline infusion test with plasma aldosterone at 4 h of infusion Adrenal CT scan, adrenal vein sampling.
Obstructive sleep apnea ^{SS.4-7‡}	25%–50%	Resistant hypertension; snoring; fitful sleep; breathing pauses during sleep; daytime sleepiness	Obesity, Mallampati class III–IV; loss of normal nocturnal BP fall	Berlin Questionnaire; ^{SS.4-8} Epworth Sleepiness Score; ^{SS.4-9} overnight oximetry	Polysomnography
Drug or alcohol induced ^{SS.4-10§}	2%–4%	Sodium-containing antacids; caffeine; nicotine (smoking); alcohol; NSAIDs; oral contraceptives; cyclosporine or tacrolimus; sympathomimetics (decongestants, anorectics); cocaine, amphetamines and other illicit drugs; neuropsychiatric agents; erythropoiesis-stimulating agents; clonidine withdrawal; herbal agents (Ma Huang, ephedra)	Fine tremor, tachycardia, sweating (cocaine, ephedrine, MAO inhibitors); acute abdominal pain (cocaine)	Urinary drug screen (illicit drugs)	Response to withdrawal of suspected agent
Uncommon causes					
Pheochromocytoma/paraganglioma ^{SS.4-11}	0.1%–0.6%	Resistant hypertension; paroxysmal hypertension or crisis superimposed on sustained hypertension; “spells,” BP lability, headache, sweating, palpitations, pallor; positive family history of pheochromocytoma/paraganglioma; adrenal incidentaloma	Skin stigmata of neurofibromatosis (café-au-lait spots; neurofibromas); Orthostatic hypotension	24-h urinary fractionated metanephrines or plasma metanephrines under standard conditions (supine position with indwelling IV cannula)	CT or MRI scan of abdomen/pelvis
Cushing’s syndrome ^{SS.4-12}	<0.1%	Rapid weight gain, especially with central distribution; proximal muscle weakness; depression; hyperglycemia	Central obesity, “moon” face, dorsal and supraclavicular fat pads, wide (1-cm) violaceous striae, hirsutism	Overnight 1-mg dexamethasone suppression test	24-h urinary free cortisol excretion (preferably multiple); midnight salivary cortisol

(Continued)

Table 13. Continued

	Prevalence	Clinical Indications	Physical Examination	Screening Tests	Additional/Confirmatory Tests
Uncommon causes (Continued)					
Hypothyroidism ^{S5.4-10}	<1%	Dry skin; cold intolerance; constipation; hoarseness; weight gain	Delayed ankle reflex; periorbital puffiness; coarse skin; cold skin; slow movement; goiter	Thyroid-stimulating hormone; free thyroxine	None
Hyperthyroidism ^{S5.4-10}	<1%	Warm, moist skin; heat intolerance; nervousness; tremulousness; insomnia; weight loss; diarrhea; proximal muscle weakness	Lid lag; fine tremor of the outstretched hands; warm, moist skin	Thyroid-stimulating hormone; free thyroxine	Radioactive iodine uptake and scan
Aortic coarctation (undiagnosed or repaired) ^{S5.4-13}	0.1%	Young patient with hypertension (<30 y of age)	BP higher in upper extremities than in lower extremities; absent femoral pulses; continuous murmur over patient's back, chest, or abdominal bruit; left thoracotomy scar (postoperative)	Echocardiogram	Thoracic and abdominal CT angiogram or MRA
Primary hyperparathyroidism ^{S5.4-14}	Rare	Hypercalcemia	Usually none	Serum calcium	Serum parathyroid hormone
Congenital adrenal hyperplasia ^{S5.4-15}	Rare	Hypertension and hypokalemia; virilization (11-beta-hydroxylase deficiency [11-beta-OH]); incomplete masculinization in males and primary amenorrhea in females (17-alpha-hydroxylase deficiency [17-alpha-OH])	Signs of virilization (11-beta-OH) or incomplete masculinization (17-alpha-OH)	Hypertension and hypokalemia with low or normal aldosterone and renin	11-beta-OH: elevated deoxycorticosterone (DOC), 11-deoxycortisol, and androgens; 17-alpha-OH: decreased androgens and estrogen; elevated deoxycorticosterone and corticosterone
Mineralocorticoid excess syndromes other than primary aldosteronism ^{S5.4-15}	Rare	Early-onset hypertension; resistant hypertension; hypokalemia or hyperkalemia	Arrhythmias (with hypokalemia)	Low aldosterone and renin	Urinary cortisol metabolites; genetic testing
Acromegaly ^{S5.4-16}	Rare	Acral features, enlarging shoe, glove, or hat size; headache, visual disturbances; diabetes mellitus	Acral features; large hands and feet; frontal bossing	Serum growth hormone ≥ 1 ng/mL during oral glucose load	Elevated age- and sex-matched IGF-1 level; MRI scan of the pituitary

*Depending on the clinical situation (hypertension alone, 5%; hypertension starting dialysis, 22%; hypertension and peripheral vascular disease, 28%; hypertension in the elderly with congestive heart failure, 34%).

†8% in general population with hypertension; up to 20% in patients with resistant hypertension.

‡Although obstructive sleep apnea is listed as a cause of secondary hypertension, RCTs on the effects of continuous positive airway pressure on lowering BP in patients with hypertension have produced mixed results (see Section 5.4.4 for details).

§For a list of frequently used drugs causing hypertension and accompanying evidence, see Table 14.

BP indicates blood pressure; CT, computed tomography; DOC, 11-deoxycorticosterone; IGF-1, insulin-like growth factor-1; IV, intravenous; MAO, monamine oxidase; MRI, magnetic resonance imaging; MRA, magnetic resonance arteriography; NSAIDs, nonsteroidal anti-inflammatory drugs; OH, hydroxylase; and RCT, randomized clinical trial.

a higher level of high-density lipoprotein cholesterol and, within modest ranges of intake, a lower level of CHD than that associated with abstinence.^{S5.2.3-35}

5.3. Childhood Risk Factors and BP Tracking

BP distribution in the general population increases with age. Multiple longitudinal studies have investigated the relationship of childhood BP to adult BP. A meta-analysis of 50 such

studies showed correlation coefficients of about 0.38 for SBP and 0.28 for DBP, with BPs in the upper range of the pediatric distribution (particularly BPs obtained in adolescence) predicting hypertension in adulthood.^{S5.3-1} Several factors, including genetic factors and development of obesity, increase the likelihood that a high childhood BP will lead to future hypertension.^{S5.3-2} Premature birth is associated with a 4-mm Hg higher SBP and a 3-mm Hg higher DBP in adulthood,

with somewhat larger effects in women than in men.^{S5.3-3} Low birth weight from other causes also contributes to higher BP in later life.^{S5.3-4}

5.4. Secondary Forms of Hypertension

Recommendations for Secondary Forms of Hypertension		
COR	LOE	Recommendations
I	C-EO	1. Screening for specific form(s) of secondary hypertension is recommended when the clinical indications and physical examination findings listed in Table 13 are present or in adults with resistant hypertension.
Iib	C-EO	2. If an adult with sustained hypertension screens positive for a form of secondary hypertension, referral to a physician with expertise in that form of hypertension may be reasonable for diagnostic confirmation and treatment.

Synopsis

A specific, remediable cause of hypertension can be identified in approximately 10% of adult patients with hypertension.^{S5.4-1} If a cause can be correctly diagnosed and treated, patients with secondary hypertension can achieve a cure or experience a marked improvement in BP control, with reduction in CVD risk. All new patients with hypertension should be screened with a history, physical examination, and laboratory investigations, as recommended in Section 7, before initiation of treatment.

Secondary hypertension can underlie severe elevation of BP, pharmacologically resistant hypertension, sudden onset of hypertension, increased BP in patients with hypertension previously controlled on drug therapy, onset of diastolic hypertension in older adults, and target organ damage disproportionate to the duration or severity of the hypertension. Although secondary hypertension should be suspected in younger patients (<30 years of age) with elevated BP, it is not uncommon for primary hypertension to manifest at a younger age, especially in blacks,^{S5.4-2} and some forms of secondary hypertension, such as renovascular disease, are more common at older age. Many of the causes of secondary hypertension are strongly associated with clinical findings or groups of findings that suggest a specific disorder.

Figure 3 is an algorithm on screening for secondary hypertension. Table 13 is a detailed list of clinical indications and diagnostic screening tests for secondary hypertension, and Table 14 is a list of drugs that can induce secondary hypertension.

Recommendation-Specific Supportive Text

1. The causes of secondary hypertension and recommended screening tests are provided in Table 13, and drugs that can induce secondary hypertension are provided in Table 14.
2. Diagnosis of many of these disorders requires a complex set of measurements, specialized technical expertise, and/or experience in data interpretation. Similarly, specific

treatment often requires a level of technical training and experience.

5.4.1. Drugs and Other Substances With Potential to Impair BP Control

Numerous substances, including prescription medications, over-the-counter medications, herbals, and food substances, may affect BP (Table 14).^{S5.4.1-1–S5.4.1-6} Changes in BP that occur because of drugs and other agents have been associated with the development of hypertension, worsening control in a patient who already has hypertension, or attenuation of the BP-lowering effects of antihypertensive therapy. A change in BP may also result from drug–drug or drug–food interactions.^{S5.4.1-2,S5.4.1-4} In the clinical assessment of hypertension, a careful history should be taken with regard to substances that may impair BP control, with close attention paid to not only prescription medications, but also over-the-counter substances, illicit drugs, and herbal products. When feasible, drugs associated with increased BP should be reduced or discontinued, and alternative agents should be used.

5.4.2. Primary Aldosteronism

Recommendations for Primary Aldosteronism		
COR	LOE	Recommendations
I	C-EO	1. In adults with hypertension, screening for primary aldosteronism is recommended in the presence of any of the following concurrent conditions: resistant hypertension, hypokalemia (spontaneous or substantial, if diuretic induced), incidentally discovered adrenal mass, family history of early-onset hypertension, or stroke at a young age (<40 years).
I	C-LD	2. Use of the plasma aldosterone: renin activity ratio is recommended when adults are screened for primary aldosteronism. ^{S5.4.2-1}
I	C-EO	3. In adults with hypertension and a positive screening test for primary aldosteronism, referral to a hypertension specialist or endocrinologist is recommended for further evaluation and treatment.

Synopsis

Primary aldosteronism is defined as a group of disorders in which aldosterone production is inappropriately high for sodium status, is relatively autonomous of the major regulators of secretion (angiotensin II and potassium), and cannot be suppressed with sodium loading.^{S5.4.2-2,S5.4.2-3} The increased production of aldosterone induces hypertension; cardiovascular and kidney damage; sodium retention; suppressed plasma renin activity; and increased potassium excretion, which, if prolonged and severe, may cause hypokalemia. However, hypokalemia is absent in the majority of cases and has a low negative predictive value for the diagnosis of primary aldosteronism.^{S5.4.2-4} In about 50% of the patients, primary aldosteronism is due to increased unilateral aldosterone production (usually aldosterone-producing adenoma or, rarely, unilateral adrenal hyperplasia); in the remaining 50%, primary aldosteronism is due to bilateral adrenal hyperplasia (idiopathic hyperaldosteronism).^{S5.4.2-2,S5.4.2-3}

Table 14. Frequently Used Medications and Other Substances That May Cause Elevated BP*

Agent	Possible Management Strategy
Alcohol	Limit alcohol to ≤1 drink daily for women and ≤2 drinks for men ^{S5.4.1-7}
Amphetamines (eg, amphetamine, methylphenidate, dextroamphetamine)	Discontinue or decrease dose ^{S5.4.1-8} Consider behavioral therapies for ADHD ^{S5.4.1-9}
Antidepressants (eg, MAOIs, SNRIs, TCAs)	Consider alternative agents (eg, SSRIs) depending on indication Avoid tyramine-containing foods with MAOIs
Atypical antipsychotics (eg, clozapine, olanzapine)	Discontinue or limit use when possible Consider behavior therapy where appropriate Recommend lifestyle modification (see Section 6.2) Consider alternative agents associated with lower risk of weight gain, diabetes mellitus, and dyslipidemia (eg, aripiprazole, ziprasidone) ^{S5.4.1-10, S5.4.1-11}
Caffeine	Generally limit caffeine intake to <300 mg/d Avoid use in patients with uncontrolled hypertension Coffee use in patients with hypertension is associated with acute increases in BP; long-term use is not associated with increased BP or CVD ^{S5.4.1-12}
Decongestants (eg, phenylephrine, pseudoephedrine)	Use for shortest duration possible, and avoid in severe or uncontrolled hypertension Consider alternative therapies (eg, nasal saline, intranasal corticosteroids, antihistamines) as appropriate
Herbal supplements (eg, Ma Huang [ephedra], St. John's wort [with MAO inhibitors, yohimbine])	Avoid use
Immunosuppressants (eg, cyclosporine)	Consider converting to tacrolimus, which may be associated with fewer effects on BP ^{S5.4.1-13–S5.4.1-15}
Oral contraceptives	Use low-dose (eg, 20–30 mcg ethinyl estradiol) agents ^{S5.4.1-16} or a progestin-only form of contraception, or consider alternative forms of birth control where appropriate (eg, barrier, abstinence, IUD) Avoid use in women with uncontrolled hypertension ^{S5.4.1-16}
NSAIDs	Avoid systemic NSAIDs when possible Consider alternative analgesics (eg, acetaminophen, tramadol, topical NSAIDs), depending on indication and risk
Recreational drugs (eg, “bath salts” [MDPV], cocaine, methamphetamine, etc.)	Discontinue or avoid use
Systemic corticosteroids (eg, dexamethasone, fludrocortisone, methylprednisolone, prednisone, prednisolone)	Avoid or limit use when possible Consider alternative modes of administration (eg, inhaled, topical) when feasible
Angiogenesis inhibitor (eg, bevacizumab) and tyrosine kinase inhibitors (eg, sunitinib, sorafenib)	Initiate or intensify antihypertensive therapy

*List is not all inclusive.

ADHD indicates attention-deficit/hyperactivity disorder; BP, blood pressure; CVD, cardiovascular disease; IUD, intra-uterine device; MAOI, monoamine-oxidase inhibitors; MDPV, methylenedioxypyrovalerone; NSAIDs, nonsteroidal anti-inflammatory drugs; SNRI, serotonin norepinephrine reuptake inhibitor; SSRI, selective serotonin reuptake inhibitor; and TCA, tricyclic antidepressant.

Recommendation-Specific Supportive Text

1. Primary aldosteronism is one of the most frequent disorders (occurring in 5% to 10% of patients with hypertension and 20% of patients with resistant hypertension) that causes secondary hypertension.^{S5.4.2-5, S5.4.2-6} The toxic tissue effects of aldosterone induce greater target organ damage in primary aldosteronism than in primary hypertension. Patients with primary aldosteronism have a 3.7-fold increase in HF, a 4.2-fold increase in stroke, a 6.5-fold increase in MI, a 12.1-fold increase in atrial fibrillation (AF), increased left ventricular hypertrophy (LVH) and diastolic dysfunction, increased stiffness of large arteries, widespread tissue fibrosis, increased remodeling of resistance vessels, and increased kidney damage as compared with patients with primary hypertension matched for BP level.^{S5.4.2-6–S5.4.2-8} Because the deleterious effects of

aldosterone overproduction are often reversible with unilateral laparoscopic adrenalectomy or treatment with mineralocorticoid receptor antagonists (ie, spironolactone or eplerenone), screening of patients with hypertension at increased risk of primary aldosteronism is beneficial.^{S5.4.2-2, S5.4.2-3} These include hypertensive patients with adrenal “incidentaloma,” an incidentally discovered adrenal lesion on a computed tomography or magnetic resonance imaging (MRI) scan performed for other purposes. Patients with hypertension and a history of early onset hypertension and/or cerebrovascular accident at a young age may have primary aldosteronism due to glucocorticoid-remediable aldosteronism (familial hyperaldosteronism type-1) and therefore warrant screening.^{S5.4.2-2, S5.4.2-3}

2. The aldosterone:renin activity ratio is currently the most accurate and reliable means of screening for primary

aldosteronism.^{S5.4.2-1} The most commonly used cutoff value is 30 when plasma aldosterone concentration is reported in nanograms per deciliter (ng/dL) and plasma renin activity in nanograms per milliliter per hour (ng/mL/h).^{S5.4.2-3} Because the aldosterone:renin activity ratio can be influenced by the presence of very low renin levels, the plasma aldosterone concentration should be at least 10 ng/dL to interpret the test as positive.^{S5.4.2-3} Patients should have unrestricted salt intake, serum potassium in the normal range, and mineralocorticoid receptor antagonists (eg, spironolactone or eplerenone) withdrawn for at least 4 weeks before testing.^{S5.4.2-2,S5.4.2-3}

3. The diagnosis of primary aldosteronism generally requires a confirmatory test (intravenous saline suppression test or oral salt-loading test).^{S5.4.2-2,S5.4.2-3} If the diagnosis of primary aldosteronism is confirmed (and the patient agrees that surgery would be desirable), the patient is referred for an adrenal venous sampling procedure to determine whether the increased aldosterone production is unilateral or bilateral in origin. If unilateral aldosterone production is documented on adrenal venous sampling, the patient is referred for unilateral laparoscopic adrenalectomy, which improves BP in virtually 100% of patients and results in a complete cure of hypertension in about 50%.^{S5.4.2-2,S5.4.2-3} If the patient has bilaterally increased aldosterone secretion on adrenal venous sampling or has a unilateral source of excess aldosterone production but cannot undergo surgery, the patient is treated with spironolactone or eplerenone as agent of choice.^{S5.4.2-2,S5.4.2-3} Both adrenalectomy and medical therapy are effective in lowering BP and reversing LVH. Treating primary aldosteronism, either by mineralocorticoid receptor antagonists or unilateral adrenalectomy (if indicated), resolves hypokalemia, lowers BP, reduces the number of antihypertensive medications required, and improves parameters of impaired cardiac and kidney function.^{S5.4.2-9,S5.4.2-10}

5.4.3. Renal Artery Stenosis

Recommendations for Renal Artery Stenosis		
References that support recommendations are summarized in Online Data Supplements 7 and 24.		
COR	LOE	Recommendations
I	A	1. Medical therapy is recommended for adults with atherosclerotic renal artery stenosis. ^{S5.4.3-1,S5.4.3-2}
IIb	C-E0	2. In adults with renal artery stenosis for whom medical management has failed (refractory hypertension, worsening renal function, and/or intractable HF) and those with nonatherosclerotic disease, including fibromuscular dysplasia, it may be reasonable to refer the patient for consideration of revascularization (percutaneous renal artery angioplasty and/or stent placement).

Synopsis

Renal artery stenosis refers to a narrowing of the renal artery that can result in a restriction of blood flow. Atherosclerotic disease (90%) is by far the most common cause of renal artery

stenosis, whereas nonatherosclerotic disease (of which fibromuscular dysplasia is the most common) is much less prevalent and tends to occur in younger, healthier patients.^{S5.4.3-3} Renal artery stenosis is a common form of secondary hypertension. Relieving ischemia and the ensuing postischemic release of renin by surgical renal artery reconstruction is an invasive strategy with a postoperative mortality as high as 13%.^{S5.4.3-4} With the advent of endovascular procedures to restore blood flow, several trials were designed to test the efficacy of these procedures against medical therapy, but they suggested no benefit over medical therapy alone.^{S5.4.3-1,S5.4.3-2}

Recommendation-Specific Supportive Text

1. Atherosclerotic disease in the renal arteries represents systemic disease and higher risk of both renal failure and cardiovascular morbidity and mortality. No RCT to date has demonstrated a clinical advantage of renal artery revascularization (with either angioplasty or stenting) over medical therapy.^{S5.4.3-2} On the basis of the CORAL (Cardiovascular Outcomes in Renal Atherosclerotic Lesions) trial, the recommended medical approach encompasses optimal management of hypertension with an antihypertensive regimen that includes a renin-angiotensin system (RAS) blocker, in addition to low-density lipoprotein cholesterol reduction with a high-intensity statin, smoking cessation, hemoglobin A1c reduction in patients with DM, and antiplatelet therapy.^{S5.4.3-1}
2. Revascularization may be considered for those who do not respond to medical therapy and for those who have nonatherosclerotic disease (eg, Takayasu arteritis in Asian populations, fibromuscular dysplasia in other populations). Fibromuscular dysplasia occurs over the lifespan of women (mean: 53 years of age) with almost equal frequency in the renal and carotid circulations.^{S5.4.3-3} Percutaneous transluminal angioplasty alone (without stenting) can improve BP control and even normalize BP, especially in patients with recent onset of hypertension or resistant hypertension.^{S5.4.3-5}

5.4.4. Obstructive Sleep Apnea

Recommendation for Obstructive Sleep Apnea		
References that support the recommendation are summarized in Online Data Supplement 8.		
COR	LOE	Recommendation
IIb	B-R	1. In adults with hypertension and obstructive sleep apnea, the effectiveness of continuous positive airway pressure (CPAP) to reduce BP is not well established. ^{S5.4.4-1-S5.4.4-5}

Synopsis

Obstructive sleep apnea is a common chronic condition characterized by recurrent collapse of upper airways during sleep, inducing intermittent episodes of apnea/hypopnea, hypoxemia, and sleep disruption.^{S5.4.4-6} Obstructive sleep apnea is a risk factor for several CVDs, including hypertension, coronary and cerebrovascular diseases, HF, and AF.^{S5.4.4-6-S5.4.4-9} Observational studies have shown that the presence of obstructive sleep apnea is associated with increased risk of incident hypertension.^{S5.4.4-10,S5.4.4-11} Obstructive sleep apnea is

highly prevalent in adults with resistant hypertension ($\geq 80\%$),^{S5.4.4-12,S5.4.4-13} and it has been hypothesized that treatment with CPAP may have more pronounced effects on BP reduction in resistant hypertension.^{S5.4.4-6}

Recommendation-Specific Supportive Text

1. CPAP is an efficacious treatment for improving obstructive sleep apnea. However, studies of the effects of CPAP on BP have demonstrated only small effects on BP (eg, 2- to 3-mm Hg reductions), with results dependent on patient compliance with CPAP use, severity of obstructive sleep apnea, and presence of daytime sleepiness in study participants.^{S5.4.4-1-S5.4.4-5} Although many RCTs have been reported that address the effects of CPAP on BP in obstructive sleep apnea, most of the patients studied did not have documented hypertension, and the studies were too small and the follow-up period too short to allow for adequate evaluation. In addition, a well-designed RCT demonstrated that CPAP plus usual care, compared with usual care alone, did not prevent cardiovascular events in patients with moderate-severe obstructive sleep apnea and established CVD.^{S5.4.4-14}

6. Nonpharmacological Interventions

Correcting the dietary aberrations, physical inactivity, and excessive consumption of alcohol that cause high BP is a fundamentally important approach to prevention and management of high BP, either on their own or in combination with pharmacological therapy. Prevention of hypertension and treatment of established hypertension are complementary approaches to reducing CVD risk in the population, but prevention of hypertension provides the optimal means of reducing risk and avoiding the harmful consequences of hypertension.^{S6.1-S6.3} Nonpharmacological therapy alone is especially useful for prevention of hypertension, including in adults with elevated BP, and for management of high BP in adults with milder forms of hypertension.^{S6.4,S6.5}

6.1. Strategies

Nonpharmacological interventions can be accomplished by means of behavioral strategies aimed at lifestyle change, prescription of dietary supplements, or implementation of kitchen-based interventions that directly modify elements of the diet. At a societal level, policy changes can enhance the availability of healthy foods and facilitate physical activity. The goal can be to modestly reduce BP in the general population or to undertake more intensive targeted lowering of BP in adults with hypertension or at high risk of developing hypertension.^{S6.1-1} The intent of the general population approach is to achieve a small downward shift in the general population distribution of BP, which would be expected to result in substantial health benefits.^{S6.1-2} The targeted approach focuses on BP reduction in adults at greatest risk of developing BP-related CVD, including individuals with hypertension, as well as those at increased risk of developing hypertension, especially blacks and adults who are overweight, consume excessive amounts of dietary sodium, have a high intake of alcohol, or are physically inactive. The targeted approach tends to be intensive, with a more ambitious goal for BP

reduction. Both approaches are complementary and mutually reinforcing, and modeling studies suggest they are likely to provide similar public health benefit.^{S6.1-3,S6.1-4} However, as the precision of risk prediction tools increases, targeted prevention strategies that focus on high-risk individuals seem to become more efficient than population-based strategies.^{S6.1-5}

6.2. Nonpharmacological Interventions

Recommendations for Nonpharmacological Interventions		
References that support recommendations are summarized in Online Data Supplements 9-21.		
COR	LOE	Recommendations
I	A	1. Weight loss is recommended to reduce BP in adults with elevated BP or hypertension who are overweight or obese. ^{S6.2-1-S6.2-4}
I	A	2. A heart-healthy diet, such as the DASH (Dietary Approaches to Stop Hypertension) diet, that facilitates achieving a desirable weight is recommended for adults with elevated BP or hypertension. ^{S6.2-5-S6.2-7}
I	A	3. Sodium reduction is recommended for adults with elevated BP or hypertension. ^{S6.2-8-S6.2-12}
I	A	4. Potassium supplementation, preferably in dietary modification, is recommended for adults with elevated BP or hypertension, unless contraindicated by the presence of CKD or use of drugs that reduce potassium excretion. ^{S6.2-13-S6.2-17}
I	A	5. Increased physical activity with a structured exercise program is recommended for adults with elevated BP or hypertension. ^{S6.2-3,S6.2-4,S6.2-12,S6.2-18-S6.2-22}
I	A	6. Adult men and women with elevated BP or hypertension who currently consume alcohol should be advised to drink no more than 2 and 1 standard drinks* per day, respectively. ^{S6.2-23-S6.2-28}

*In the United States, 1 "standard" drink contains roughly 14 g of pure alcohol, which is typically found in 12 oz of regular beer (usually about 5% alcohol), 5 oz of wine (usually about 12% alcohol), and 1.5 oz of distilled spirits (usually about 40% alcohol).^{S6.2-29}

Synopsis

Nonpharmacological interventions are effective in lowering BP, with the most important interventions being weight loss,^{S6.2-1} the DASH (Dietary Approaches to Stop Hypertension) diet,^{S6.2-5-S6.2-7,S6.2-30} sodium reduction,^{S6.2-8-S6.2-12} potassium supplementation,^{S6.2-13,S6.2-17} increased physical activity,^{S6.2-18-S6.2-22,S6.2-31} and a reduction in alcohol consumption.^{S6.2-23,S6.2-24} Various other nonpharmacological interventions have been reported to lower BP, but the extent and/or quality of the supporting clinical trial experience is less persuasive. Such interventions include consumption of probiotics,^{S6.2-32,S6.2-33,S6.2-34} increased intake of protein,^{S6.2-35-S6.2-37} fiber,^{S6.2-38,S6.2-39} flaxseed,^{S6.2-40} or fish oil;^{S6.2-41} supplementation with calcium^{S6.2-42,S6.2-43} or magnesium;^{S6.2-44,S6.2-45} and use of dietary patterns other than the DASH diet, including

low-carbohydrate, vegetarian, and Mediterranean diets.^{S6.2-46–S6.2-49} Stress reduction is intuitively attractive but insufficiently proved,^{S6.2-51} as are several other interventions, including consumption of garlic,^{S6.2-52} dark chocolate,^{S6.2-53,S6.2-54} tea,^{S6.2-55} or coffee.^{S6.2-56} Behavioral therapies, including guided breathing, yoga, transcendental meditation, and biofeedback, lack strong evidence for their long-term BP-lowering effect.^{S6.2-51,S6.2-57–S6.2-61} The best proven nonpharmacological measures to prevent and treat hypertension are summarized in Table 15.^{S6.2-62}

The nonpharmacological interventions presented in Table 15 may be sufficient to prevent hypertension and meet goal BP in managing patients with stage 1 hypertension, and they are an integral part of the management of persons with stage 2 hypertension. To a lesser extent, the Mediterranean diet^{S6.2-49,S6.2-63} (which incorporates the basics of healthy eating but emphasizes consumption of legumes and monounsaturated fat, avoidance of red meats, and moderate intake of wine) has been effective in reducing BP, as well as improving lipid profile.

Recommendation-Specific Supportive Text

1. Weight loss is a core recommendation and should be achieved through a combination of reduced calorie intake and increased physical activity.^{S6.2-1} The BP-lowering

effect of weight loss in patients with elevated BP is consistent with the corresponding effect in patients with established hypertension, with an apparent dose–response relationship of about 1 mm Hg per kilogram of weight loss. Achievement and maintenance of weight loss through behavior change are challenging^{S6.2-64–S6.2-66} but feasible over prolonged periods of follow-up.^{S6.2-64} For those who do not meet their weight loss goals with nonpharmacological interventions, pharmacotherapy or minimally invasive and bariatric surgical procedures can be considered.^{S6.2-67,S6.2-68} Surgical procedures tend to be more effective but are usually reserved for those with more severe and intractable obesity because of the frequency of complications.^{S6.2-69}

2. The DASH eating plan is the diet best demonstrated to be effective for lowering BP. Because the DASH diet is high in fruits, vegetables, and low-fat dairy products, it provides a means to enhance intake of potassium, calcium, magnesium, and fiber. In hypertensive and nonhypertensive adults, the DASH diet has produced overall reductions in SBP of approximately 11 mm Hg and 3 mm Hg, respectively,^{S6.2-7} and the diet was especially effective in blacks.^{S6.2-70} When combined with weight loss^{S6.2-6} or a reduction in sodium intake,^{S6.2-5,S6.2-30} the effect size was substantially increased. Most of the clinical trial

Table 15. Best Proven Nonpharmacological Interventions for Prevention and Treatment of Hypertension*

	Nonpharmacological Intervention	Dose	Approximate Impact on SBP		
			Hypertension	Normotension	Reference
Weight loss	Weight/body fat	Best goal is ideal body weight, but aim for at least a 1-kg reduction in body weight for most adults who are overweight. Expect about 1 mm Hg for every 1-kg reduction in body weight.	–5 mm Hg	–2/3 mm Hg	S6.2-1
Healthy diet	DASH dietary pattern	Consume a diet rich in fruits, vegetables, whole grains, and low-fat dairy products, with reduced content of saturated and total fat.	–11 mm Hg	–3 mm Hg	S6.2-6,S6.2-7
Reduced intake of dietary sodium	Dietary sodium	Optimal goal is <1500 mg/d, but aim for at least a 1000-mg/d reduction in most adults.	–5/6 mm Hg	–2/3 mm Hg	S6.2-9,S6.2-10
Enhanced intake of dietary potassium	Dietary potassium	Aim for 3500–5000 mg/d, preferably by consumption of a diet rich in potassium.	–4/5 mm Hg	–2 mm Hg	S6.2-13
Physical activity	Aerobic	90–150 min/wk 65%–75% heart rate reserve	–5/8 mm Hg	–2/4 mm Hg	S6.2-18,S6.2-22
	Dynamic resistance	90–150 min/wk 50%–80% 1 rep maximum 6 exercises, 3 sets/exercise, 10 repetitions/set	–4 mm Hg	–2 mm Hg	S6.2-18
	Isometric resistance	4 × 2 min (hand grip), 1 min rest between exercises, 30%–40% maximum voluntary contraction, 3 sessions/wk 8–10 wk	–5 mm Hg	–4 mm Hg	S6.2-19,S6.2-31
Moderation in alcohol intake	Alcohol consumption	In individuals who drink alcohol, reduce alcohol† to: Men: ≤2 drinks daily Women: ≤1 drink daily	–4 mm Hg	–3 mm Hg	S6.2-22–S6.2-24

Resources: Your Guide to Lowering Your Blood Pressure With DASH—How Do I Make the DASH? Available at: <https://www.nhlbi.nih.gov/health/resources/heart/hbp-dash-how-to>. Accessed September 15, 2017.^{S6.2-72}

Top 10 Dash Diet Tips. Available at: http://dashdiet.org/dash_diet_tips.asp. Accessed September 15, 2017.^{S6.2-73}

*Type, dose, and expected impact on BP in adults with a normal BP and with hypertension.

†In the United States, one “standard” drink contains roughly 14 g of pure alcohol, which is typically found in 12 oz of regular beer (usually about 5% alcohol), 5 oz of wine (usually about 12% alcohol), and 1.5 oz of distilled spirits (usually about 40% alcohol).^{S6.2-29}

DASH indicates Dietary Approaches to Stop Hypertension; and SBP, systolic blood pressure.

experience comes from short-term feeding studies,^{S6.2-7} but lifestyle change with the DASH diet has been successful in at least 2 trials that used a behavioral intervention over a 4-month^{S6.2-30} or 6-month^{S6.2-6} period of follow-up. Websites and books provide advice on implementation of the DASH diet.^{S6.2-13,S6.2-71-S6.2-74} Counseling by a knowledgeable nutritionist can be helpful. Several other diets, including diets that are low in calories from carbohydrates,^{S6.2-46} high-protein diets,^{S6.2-75} vegetarian diets,^{S6.2-48} and a Mediterranean dietary pattern,^{S6.2-49,S6.2-63} have been shown to lower BP.

3. Sodium reduction interventions prevent hypertension and lower BP in adults with hypertension, especially in those with higher levels of BP, blacks, older persons, and others who are particularly susceptible to the effects of sodium on BP.^{S6.2-8-S6.2-11} Sodium reduction interventions may prevent CVD.^{S6.2-76,S6.2-77} Lifestyle change (behavioral) interventions usually reduce sodium intake by about 25% (approximately 1000 mg per day) and result in an average of about a 2-mm Hg to 3-mm Hg reduction in SBP in nonhypertensive individuals, though the reduction can be more than double this in more susceptible individuals, those with hypertension, and those concurrently on the DASH diet^{S6.2-5} or receiving a weight loss intervention.^{S6.2-12} Sodium reduction in adults with hypertension who are already being treated with BP-lowering medications further reduces SBP by about 3 mm Hg and can facilitate discontinuation of medication, although this requires maintenance of the lifestyle change and warrants careful monitoring.^{S6.2-12} When combined with weight loss, the reduction in BP is almost doubled. A reduction in sodium intake may also lower SBP significantly in individuals with resistant hypertension who are taking multiple antihypertensive medications^{S6.2-78} (see Section 11.1). Reduced dietary sodium has been reported to augment the BP-lowering effects of RAS blocker therapy.^{S6.2-79} Maintenance of the lifestyle changes necessary to reduce sodium intake is challenging,^{S6.2-2-S6.2-4,S6.2-12} but even a small decrement in sodium consumption is likely to be safe^{S6.2-2,S6.2-4,S6.2-9,S6.2-12,S6.2-80} and beneficial,^{S6.2-8,S6.2-81} especially in those whose BP is salt sensitive.^{S6.2-82} In the United States, most dietary sodium comes from additions during food processing or during commercial food preparation at sit-down and fast-food restaurants.^{S6.2-83,S6.2-84} Person-specific and policy approaches can be used to reduce dietary sodium intake.^{S6.2-85,S6.2-86} Individuals can take action to reduce their dietary intake of sodium by choice of fresh foods, use of food labels to choose foods that are lower in sodium content, choice of foods with a “no added sodium” label, judicious use of condiments and sodium-infused foods, use of spices and low-sodium flavorings, careful ordering when eating out, control of food portion size, and avoiding or minimizing use of salt at the table. Dietary counseling by a nutritionist with expertise in behavior modification can be helpful. A reduction in the amount of sodium added during food processing, as well as fast food and restaurant food preparation, has the potential to substantially reduce sodium intake without the need for a conscious change in lifestyle.^{S6.2-81,S6.2-85,S6.2-87}

4. Dietary potassium is inversely related to BP and hypertension in migrant studies,^{S6.2-88} cross-sectional reports,^{S6.2-89-S6.2-91} and prospective cohort studies.^{S6.2-92} Likewise, dietary potassium^{S6.2-93-S6.2-96} and a high intake of fruits and vegetables are associated with a lower incidence of stroke.^{S6.2-97} Potassium interventions have been effective in lowering BP,^{S6.2-13,S6.2-14,S6.2-16,S6.2-81} especially in adult patients consuming an excess of sodium^{S6.2-13,S6.2-74,S6.2-98} and in blacks.^{S6.2-13} The typical BP-lowering effect of a 60-mmol (1380-mg) administration of potassium chloride has been about 2 mm Hg and 4 to 5 mm Hg in adults with normotension and hypertension, respectively, although the response is up to twice as much in persons consuming a high-sodium diet. A reduction in the sodium/potassium index may be more important than the corresponding changes in either electrolyte alone.^{S6.2-99} Some but not all studies suggest that the intervention effect may be restricted to adult patients with a low (1500-mg to 2000-mg) daily intake of potassium.^{S6.2-92,S6.2-100} Most of the intervention experience comes from trials of relatively short duration (median of 5 to 6 weeks),^{S6.2-13,S6.2-14} but the BP-lowering effect of potassium in adult patients consuming a high-sodium diet has been reproduced after an interval of 4.4 years.^{S6.2-98} In most trials, potassium supplementation was achieved by administration of potassium chloride pills, but the BP response pattern was similar when dietary modification was used.^{S6.2-13} Because potassium-rich diets tend to be heart healthy, they are preferred over use of pills for potassium supplementation. The 2015 Dietary Guidelines for Americans^{S6.2-101} encourage a diet rich in potassium and identify the adequate intake level for adult patients as 4700 mg/day.^{S6.2-102} The World Health Organization recommends a potassium intake of at least 90 mmol (3510 mg) per day from food for adult patients.^{S6.2-15} Good sources of dietary potassium include fruits and vegetables, as well as low-fat dairy products, selected fish and meats, nuts, and soy products. Four to five servings of fruits and vegetables will usually provide 1500 to >3000 mg of potassium. This can be achieved by a diet, such as the DASH diet, that is high in potassium content.^{S6.2-7}
5. A BP-lowering effect of increased physical activity has been repeatedly demonstrated in clinical trials, especially during dynamic aerobic exercise,^{S6.2-18,S6.2-20,S6.2-22} but also during dynamic resistance training^{S6.2-18,S6.2-21} and static isometric exercise.^{S6.2-18,S6.2-19,S6.2-31} The average reductions in SBP with aerobic exercise are approximately 2 to 4 mm Hg and 5 to 8 mm Hg in adult patients with normotension and hypertension, respectively.^{S6.2-18} Most trials have been of relatively short duration, but increased physical activity has been an intrinsic component of longer-term weight reduction interventions used to reduce BP and prevent hypertension.^{S6.2-3,S6.2-4,S6.2-12} BP-lowering effects have been reported with lower- and higher-intensity exercise and with continuous and interval exercise training.^{S6.2-18,S6.2-103} Meta-analyses suggest isometric exercise results in substantial lowering of BP.^{S6.2-18,S6.2-19,S6.2-31}
6. In observational studies, there is a strong, predictable direct relationship between alcohol consumption and BP,

especially above an intake of 3 standard drinks per day (approximately 36 ounces of regular beer, 15 ounces of wine, or 4.5 ounces of distilled spirits).^{S6.2-29,S6.2-104,S6.2-105}

Meta-analyses of RCTs that have studied the effect of reduced alcohol consumption on BP in adults have identified a significant reduction in SBP and DBP.^{S6.2-23,S6.2-24}

The benefit has seemed to be consistent across trials, but confined to those consuming ≥ 3 drinks/day, as well as dose dependent, with those consuming ≥ 6 drinks/day at baseline who reduce their alcohol intake by about 50%, experiencing an average reduction in SBP/DBP of approximately 5.5/4.0 mm Hg.^{S6.2-23,S6.2-24} Only limited information is available on the effect of alcohol reduction on BP in blacks.^{S6.2-23,S6.2-106} In contrast to its effect on BP, alcohol seems to have a beneficial effect on several biomarkers for CVD risk, including high-density lipoprotein cholesterol.^{S6.2-107,S6.2-108} Observational studies have shown a relatively consistent finding of an inverse relationship between alcohol intake and CHD,^{S6.2-109,S6.2-110} within a moderate range (approximately 12–14 and ≤ 9 standard drinks/week for men and women, respectively). On balance, it seems reasonable for those who are consuming moderate quantities of alcohol (≤ 2 drinks/day) to continue their moderate consumption of alcohol.

7. Patient Evaluation

The patient evaluation is designed to identify target organ damage and possible secondary causes of hypertension and to assist in planning an effective treatment regimen. Historical features are relevant to the evaluation of the patient (Table 16). The pattern of BP measurements and changes over time may differentiate primary from secondary causes of hypertension. A rise in BP associated with weight gain, lifestyle factors (such as a job change requiring travel and meals away from home), reduced frequency or intensity of physical activity, or advancing age in a patient with a strong family history of hypertension would suggest the diagnosis of primary hypertension. An evaluation of the patient's dietary habits, physical activity, alcohol consumption, and tobacco use should be performed, with recommendation of the nonpharmacological interventions detailed in Section 6.2 where appropriate. The history should also include inquiry into possible occurrence of symptoms to indicate a secondary cause (Tables 13 and 16). The patient's treatment goals and risk tolerance should also be elicited. This is especially true for older persons, for whom an assessment of multiple chronic conditions, frailty, and prognosis should be performed, including consideration of the time required to see benefit from intervention, which may not be realized for some individuals.

The physical examination should include accurate measurement of BP (Table 8). Automated oscillometric devices provide an opportunity to obtain repeated measurements without a provider present, thereby minimizing the potential for a white coat effect. Change in BP from seated to standing position should be measured to detect orthostatic hypotension (a decline >20 mm Hg in SBP or >10 mm Hg in DBP after 1 minute is abnormal). For adults ≤ 30 years of age with elevated brachial BP, a thigh BP measurement is indicated; if the thigh measurement is lower than arm pressures, a diagnosis of coarctation of the aorta should be considered. The physical examination should

Table 16. Historical Features Favoring Hypertension Cause

Primary Hypertension	Secondary Hypertension
Gradual increase in BP, with slow rate of rise in BP	BP lability, episodic pallor and dizziness (pheochromocytoma)
Lifestyle factors that favor higher BP (eg, weight gain, high-sodium diet, decreased physical activity, job change entailing increased travel, excessive consumption of alcohol)	Snoring, hypersomnolence (obstructive sleep apnea)
	Prostatism (chronic kidney disease due to post-renal urinary tract obstruction)
Family history of hypertension	Muscle cramps, weakness (hypokalemia from primary aldosteronism or secondary aldosteronism due to renovascular disease)
	Weight loss, palpitations, heat intolerance (hyperthyroidism)
	Edema, fatigue, frequent urination (kidney disease or failure)
	History of coarctation repair (residual hypertension associated with coarctation)
	Central obesity, facial rounding, easy bruisability (Cushing's syndrome)
	Medication or substance use (eg, alcohol, NSAIDs, cocaine, amphetamines)
	Absence of family history of hypertension

BP indicates blood pressure; and NSAIDs, nonsteroidal anti-inflammatory drugs.

include assessment of hypertension-related target organ damage. Attention should be paid to physical features that suggest secondary hypertension (Table 13).

7.1. Laboratory Tests and Other Diagnostic Procedures

Laboratory measurements should be obtained for all patients with a new diagnosis of hypertension to facilitate CVD risk factor profiling, establish a baseline for medication use, and screen for secondary causes of hypertension (Table 17). Optional tests may provide information on target organ damage. Monitoring of serum sodium and potassium levels is helpful during diuretic or RAS blocker titration, as are serum creatinine and urinary albumin as markers of CKD progression.^{S7.1-1} Measurement of thyroid-stimulating hormone is a simple test to easily detect hypothyroidism and hyperthyroidism, 2 remediable causes of hypertension. A decision to conduct additional laboratory testing would be appropriate in the context of increased hypertension severity, poor response to standard treatment approaches, a disproportionate severity of target organ damage for the level of BP, or historical or clinical clues that support a secondary cause.

7.2. Cardiovascular Target Organ Damage

Pulse-wave velocity, carotid intima-media thickness, and coronary artery calcium score provide noninvasive estimates

Table 17. Basic and Optional Laboratory Tests for Primary Hypertension

Basic testing	Fasting blood glucose*
	Complete blood count
	Lipid profile
	Serum creatinine with eGFR*
	Serum sodium, potassium, calcium*
	Thyroid-stimulating hormone
	Urinalysis
	Electrocardiogram
Optional testing	Echocardiogram
	Uric acid
	Urinary albumin to creatinine ratio

*May be included in a comprehensive metabolic panel.
eGFR indicates estimated glomerular filtration rate.

of vascular target organ injury and atherosclerosis.^{S7.2-1} High BP readings, especially when obtained several years before a noninvasive measurement, are associated with an increase in subclinical CVD risk.^{S7.2-2–S7.2-4} Although carotid intima-media thickness values and coronary artery calcium scores are associated with cardiovascular events, inadequate or absent information on the effect of improvement in these markers on cardiovascular events prevents their routine use as surrogate markers in the treatment of hypertension.

LVH is a secondary manifestation of hypertension and independently predicts future CVD events. LVH is commonly measured by electrocardiography, echocardiography, or MRI.^{S7.2-5,S7.2-6} Left ventricular (LV) mass is associated with body size (particularly lean body mass), tobacco use, heart rate (inverse), and long-standing DM.^{S7.2-7–S7.2-9} BP lowering leads to a reduction in LV mass. In TOMHS (Treatment of Mild Hypertension Study), the long-acting diuretic chlorthalidone was slightly more effective in reducing LVH than were a calcium channel blocker (CCB) (amlodipine), ACE inhibitor (enalapril), alpha-receptor blocker (doxazosin), or beta-receptor blocker (acebutolol).^{S7.2-10} Beta blockers are inferior to angiotensin receptor blockers (ARBs), angiotensin-converting enzyme (ACE) inhibitors, and CCBs in reducing LVH.^{S7.2-11}

Hypertension adversely impacts other echocardiographic markers of cardiac structure and function, including left atrial size (both diameter and area; left atrial size is also a precursor of AF); diastolic function (many parameters; a precursor of HF with preserved ejection fraction [HFpEF]); cardiac structure; and subclinical markers of LV systolic function, such as myocardial strain assessment with echocardiography and MRI.

Assessment of LVH by means of echocardiography or MRI is not universally recommended during evaluation and management of hypertension in adults because there are limited data on the cost and value of these measures for CVD risk reclassification and changes in type or intensity of treatment. Assessment of LVH is most useful in adults

who are young (≤ 18 years of age) or have evidence of secondary hypertension, chronic uncontrolled hypertension, or history of symptoms of HF. Electrocardiographic criteria for LVH correlate weakly with echocardiographic or MRI definitions of LVH and are less strongly related to CVD outcomes.^{S7.2-12–S7.2-15} Imprecision in lead placement accounts, in part, for the poor correlation of electrocardiographic measurements with direct imaging results. However, electrocardiographic LVH has been valuable in predicting CVD risk in some reports.^{S7.2-16,S7.2-17} Electrocardiography may also be useful in the assessment of comorbidities, such as rhythm disturbances and prior MI.

LVH, as assessed by electrocardiography, echocardiography, or MRI, is an independent predictor of CVD complications.^{S7.2-18,S7.2-19} Reduction in LVH can predict a reduction in CVD risk, independent of change in BP.^{S7.2-20} When used in CVD risk predictor models, echocardiographic LVH has a small but significant independent effect on CVD risk in younger patients. At older ages, LVH measured by electrocardiography or MRI provides no independent contribution to prediction of CVD risk.^{S7.2-21–S7.2-23} Patients can be classified into 4 groups on the basis of the presence or absence of LVH and a determination of whether the LVH has an eccentric (normal relative wall thickness) or concentric geometry.^{S7.2-6,S7.2-22}

8. Treatment of High BP

Clinicians managing adults with high BP should focus on overall patient health, with a particular emphasis on reducing the risk of future adverse CVD outcomes. All patient risk factors need to be managed in an integrated fashion with a comprehensive set of nonpharmacological (see Section 6) and pharmacological strategies. As patient BP and risk of future CVD events increase, BP management should be intensified.

8.1. Pharmacological Treatment

8.1.1. Initiation of Pharmacological BP Treatment in the Context of Overall CVD Risk

For any specific difference in BP, the relative risk of CVD is constant across groups that differ in absolute risk of atherosclerotic CVD.^{S8.1.1-1–S8.1.1-4} albeit with some evidence of lesser relative risk but greater excess risk in older than in younger adults.^{S8.1.1-5–S8.1.1-8} Thus, there are more potentially preventable CVD events attributable to elevated BP in individuals with higher than with lower risk of CVD and in older than in younger adults. The relative risk reduction for CVD prevention with use of BP-lowering medications is fairly constant for groups that differ in CVD risk across a wide range of estimated absolute risk^{S8.1.1-9,S8.1.1-10} and across groups defined by sex, age, body mass index, and the presence or absence of DM, AF, and CKD.^{S8.1.1-5,S8.1.1-11–S8.1.1-21} As a consequence, the absolute CVD risk reduction attributable to BP lowering is greater at greater absolute levels of CVD risk.^{S8.1.1-9,S8.1.1-10,S8.1.1-12,S8.1.1-15–S8.1.1-19,S8.1.1-22,S8.1.1-23} Put another way, for a given magnitude of BP reduction due to antihypertensive medications, fewer individuals at high CVD risk would need to be treated to prevent a CVD event (ie, lower number needed to treat) than those at low CVD risk.

8.1.2. BP Treatment Threshold and the Use of CVD Risk Estimation* to Guide Drug Treatment of Hypertension

Recommendations for BP Treatment Threshold and Use of Risk Estimation* to Guide Drug Treatment of Hypertension		
References that support recommendations are summarized in Online Data Supplement 23.		
COR	LOE	Recommendations
I	SBP: A	1. Use of BP-lowering medications is recommended for secondary prevention of recurrent CVD events in patients with clinical CVD and an average SBP of 130 mm Hg or higher or an average DBP of 80 mm Hg or higher, and for primary prevention in adults with an estimated 10-year atherosclerotic cardiovascular disease (ASCVD) risk of 10% or higher and an average SBP 130 mm Hg or higher or an average DBP 80 mm Hg or higher. ^{S8.1.2-1-S8.1.2-9}
	DBP: C-EO	
I	C-LD	2. Use of BP-lowering medication is recommended for primary prevention of CVD in adults with no history of CVD and with an estimated 10-year ASCVD risk <10% and an SBP of 140 mm Hg or higher or a DBP of 90 mm Hg or higher. ^{S8.1.2-3,S8.1.2-10-S8.1.2-13}

*ACC/AHA Pooled Cohort Equations (<http://tools.acc.org/ASCVD-Risk-Estimator/>)^{S8.1.2-13a} to estimate 10-year risk of atherosclerotic CVD. ASCVD was defined as a first CHD death, non-fatal MI or fatal or non-fatal stroke.

Synopsis

Whereas treatment of high BP with BP-lowering medications on the basis of BP level alone is considered cost effective,^{S8.1.2-14} use of a combination of absolute CVD risk and BP level to guide such treatment is more efficient and cost effective at reducing risk of CVD than is use of BP level alone.^{S8.1.2-15-S8.1.2-24} Practical approaches have been developed to translate evidence from RCTs into individual patient treatment recommendations that are based on absolute net benefit for CVD risk,^{S8.1.2-25} and several national and international guidelines recommend basing use of BP-lowering medications on a combination of absolute risk of CVD and level of BP instead of relying solely on level of BP.^{S8.1.2-26-S8.1.2-31}

Attempts to use absolute risk to guide implementation of pharmacological treatment to prevent CVD have had mixed results, with many reports of improvements in provider prescribing behaviors, patient adherence, and reductions in risk,^{S8.1.2-32-S8.1.2-38} but with others showing no impact on provider behaviors.^{S8.1.2-39,S8.1.2-40} Use of global CVD risk assessment is infrequent in routine clinical practice,^{S8.1.2-41-S8.1.2-46} which suggests that intensive efforts would be required to achieve universal implementation. The choice of specific risk calculators for estimation of risk and risk threshold has been an important source of variability, ambiguity, and controversy.^{S8.1.2-47-S8.1.2-54} In addition, implementation of a standard (worldwide) absolute CVD risk threshold for initiating use of BP-lowering medications would result in large variations in medication use at a given level of BP across countries.^{S8.1.2-48,S8.1.2-54,S8.1.2-55} Future research in this area should focus on issues related to implementation of a risk-based approach to CVD prevention,

including the use of BP-lowering medications. Although several CVD risk assessment tools are available, on the basis of current knowledge, we recommend use of the ACC/AHA Pooled Cohort Equations (<http://tools.acc.org/ASCVD-Risk-Estimator/>) to estimate 10-year risk of atherosclerotic CVD (ASCVD) to establish the BP threshold for treatment.^{S8.1.2-56,S8.1.2-57} It should be kept in mind that the ACC/AHA Pooled Cohort Equations are validated for US adults ages 40 to 79 years in the absence of concurrent statin therapy.^{S8.1.2-56} For those >79 years old, the 10-year ASCVD risk is generally >10%, and thus the SBP threshold for antihypertensive drug treatment for patients >79 years old is 130 mm Hg. Two recent reviews have highlighted the importance of using predicted CVD risk together with BP to guide antihypertensive drug therapy.^{S8.1.2-22,S8.1.2-23}

Figure 4 is an algorithm on BP thresholds and recommendations for treatment and follow-up.

Recommendation-Specific Supportive Text

1. For the purposes of secondary prevention, clinical CVD is defined as CHD, congestive HF, and stroke. Several meta-analyses of RCTs support the value of using BP-lowering medications, in addition to nonpharmacological treatment, in patients with established CVD in the absence of hypertension, defined previously by an SBP ≥140 mm Hg or a DBP ≥90 mm Hg.^{S8.1.2-1,S8.1.2-6,S8.1.2-7,S8.1.2-9} Many RCTs of BP lowering in adults without CVD have used inclusion criteria designed to increase the level of CVD risk in the study populations to increase trial efficiency by facilitating shorter duration and a smaller sample size. As a consequence, few relatively low-risk adults with hypertension have been included in the trials. Trial results provide evidence of CVD prevention from use of BP-lowering medications in adults with an average SBP ≥130 mm Hg or an average DBP ≥80 mm Hg and clinical CVD; 5-year risk of CVD (defined as stroke, CHD, HF, or other CVD death) of approximately 6% to 7%;^{S8.1.2-3,S8.1.2-5} an estimated 10-year CVD death rate of approximately 4.5%;^{S8.1.2-4} or an annual rate of major CVD events of approximately 0.9% per year.^{S8.1.2-7} In the absence of clinical CVD, these risk estimates are roughly equivalent to a 10-year risk of ASCVD exceeding 10% as per the ACC/AHA Pooled Cohort Equations.^{S8.1.2-56} SPRINT (Systolic Blood Pressure Intervention Trial) provides additional support for the use of BP-lowering medications in patients without CVD at SBP levels ≥130 mm Hg; however, it is important to note that few SPRINT participants had untreated SBP between 130 mm Hg and 139 mm Hg at baseline. Furthermore, SPRINT used a Framingham 10-year risk of general CVD exceeding 15% to identify increased CVD risk.^{S8.1.2-8} Although this level of risk is lower than the levels described previously, being roughly equivalent to a 6% to 7% 10-year ASCVD risk per the ACC/AHA Pooled Cohort Equations, most of the participants in SPRINT had a much higher level of CVD risk. This recommendation differs from JNC 7 in its use of CVD risk, rather than diabetes or CKD, to recognize patients, including older adults, with a SBP/DBP <140/90 mm Hg who are likely to benefit from BP lowering drug therapy in addition to nonpharmacological antihypertensive treatment. In JNC 7, the BP

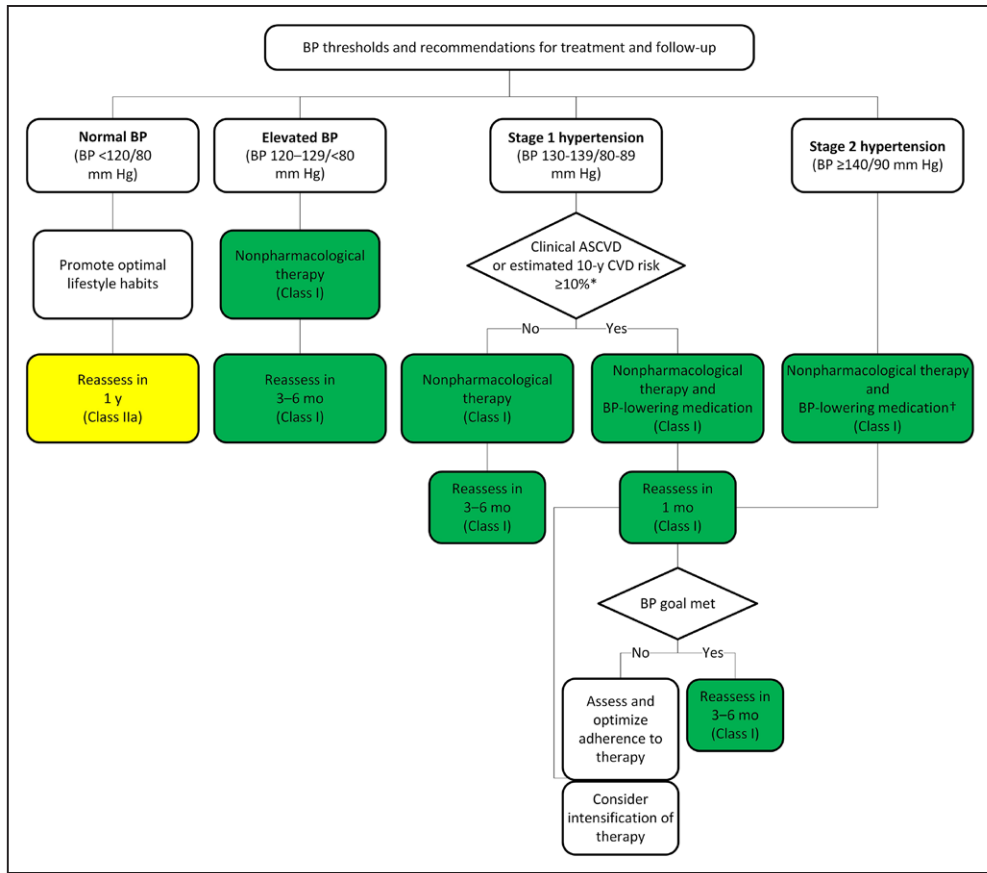


Figure 4. Blood pressure (BP) thresholds and recommendations for treatment and follow-up. Colors correspond to Class of Recommendation in Table 1. *Using the ACC/AHA Pooled Cohort Equations.^{S8.1.2-56,S8.1.2-57} Note that patients with DM or CKD are automatically placed in the high-risk category. For initiation of RAS inhibitor or diuretic therapy, assess blood tests for electrolytes and renal function 2 to 4 weeks after initiating therapy. †Consider initiation of pharmacological therapy for stage 2 hypertension with 2 antihypertensive agents of different classes. Patients with stage 2 hypertension and BP ≥160/100 mm Hg should be promptly treated, carefully monitored, and subject to upward medication dose adjustment as necessary to control BP. Reassessment includes BP measurement, detection of orthostatic hypotension in selected patients (eg, older or with postural symptoms), identification of white coat hypertension or a white coat effect, documentation of adherence, monitoring of the response to therapy, reinforcement of the importance of adherence, reinforcement of the importance of treatment, and assistance with treatment to achieve BP target. ACC indicates American College of Cardiology; AHA, American Heart Association; ASCVD, atherosclerotic cardiovascular disease; BP, blood pressure; CKD, chronic kidney disease; DM, diabetes mellitus; and RAS, renin-angiotensin system.

threshold for initiation of antihypertensive drug therapy was ≥140/90 mm Hg for the general adult population and ≥130/80 mm Hg for adults with diabetes or CKD. Since the publication of JNC 7 in 2003, we have gained additional experience with risk assessment and new data from randomized trials, observational studies and simulation analyses have demonstrated that antihypertensive drug treatment based on overall ASCVD risk assessment combined with BP levels may prevent more CVD events than treatment based on BP levels alone.^{S8.1.2-15–S8.1.2-24} According to an analysis of NHANES 2011–2014, the new definition results in only a small increase in the percentage of US adults for whom antihypertensive medication is recommended in conjunction with lifestyle modification. The previously cited meta-analyses are consistent with the conclusion that lowering of BP results in benefit in higher-risk individuals, regardless of their baseline treated or untreated BP ≥130/80 mm Hg and irrespective of the specific cause of their elevated risk. These analyses indicate that the benefit of treatment outweighs the potential harm at threshold BP ≥130/80 mm Hg.

2. This recommendation is consistent with prior guidelines, such as JNC 7. In addition, for those for whom nonpharmacological therapy has been ineffective, antihypertensive drug treatment should be added in patients with an SBP ≥140 mm Hg or a DBP ≥90 mm Hg, even in adults who are at lower risk than those included in RCTs. The rationale for drug treatment in patients with an SBP ≥140 mm Hg or a DBP ≥90 mm Hg and an estimated 10-year risk of CVD <10% is based on several lines of evidence. First, the relationship of SBP with risk of CVD is known to be continuous across levels of SBP and similar across groups that differ in level of absolute risk.^{S8.1.2-10} Second, the relative risk reduction attributable to BP-lowering medication therapy is consistent across the range of absolute risk observed in trials,^{S8.1.2-3,S8.1.2-11,S8.1.2-58} supporting the contention that the relative risk reduction may be similar at lower levels of absolute risk. This is the case even in a meta-analysis of trials in adults without clinical CVD and an average SBP/DBP of 146/84 mm Hg.^{S8.1.2-5} Finally, modeling studies support the effectiveness and cost-effectiveness of treatment of younger, lower-risk patients

over the course of their life spans.^{S8.1.2-12,S8.1.2-13} Although the numbers needed to treat with BP-lowering medications to prevent a CVD event in the short term are greater in younger, lower-risk individuals with hypertension than in older, higher-risk adults with hypertension, the estimated gains in life expectancy attributable to long-term use of BP-lowering medications are correspondingly greater in younger, lower-risk individuals than in older adults with a higher risk of CVD.^{S8.1.2-12,S8.1.2-13} Indirect support is also provided by evidence from trials using BP-lowering medications to reduce the risk of developing higher levels of BP^{S8.1.2-59-S8.1.2-61} and, in one case, to achieve a reduction in LV mass.^{S8.1.2-62} In the HOPE-3 (Heart Outcomes Prevention Evaluation-3) BP Trial, there was no evidence of short-term benefit during treatment of adults (average age 66 years) with a relatively low risk of CVD (3.8% CVD event rate during 5.6 years of follow-up). However, subgroup analysis suggested benefit in those with an average SBP approximately >140 mm Hg (and a CVD risk of 6.5% during the 5.6 years of follow-up).^{S8.1.2-63} We acknowledge the importance of excluding white coat hypertension before initiating pharmacological therapy in hypertensive patients with low ASCVD risk. This may be accomplished (as described in Section 4) by HBPM or ABPM as appropriate.

8.1.3. Follow-Up After Initial BP Evaluation

Recommendations for Follow-Up After Initial BP Evaluation		
References that support recommendations are summarized in Online Data Supplement 24.		
COR	LOE	Recommendations
I	B-R	1. Adults with an elevated BP or stage 1 hypertension who have an estimated 10-year ASCVD risk less than 10% should be managed with nonpharmacological therapy and have a repeat BP evaluation within 3 to 6 months. ^{S8.1.3-1,S8.1.3-2}
I	B-R	2. Adults with stage 1 hypertension who have an estimated 10-year ASCVD risk of 10% or higher should be managed initially with a combination of nonpharmacological and antihypertensive drug therapy and have a repeat BP evaluation in 1 month. ^{S8.1.3-1,S8.1.3-2}
I	B-R	3. Adults with stage 2 hypertension should be evaluated by or referred to a primary care provider within 1 month of the initial diagnosis, have a combination of nonpharmacological and antihypertensive drug therapy (with 2 agents of different classes) initiated, and have a repeat BP evaluation in 1 month. ^{S8.1.3-1,S8.1.3-2}
I	B-R	4. For adults with a very high average BP (eg, SBP ≥180 mm Hg or DBP ≥110 mm Hg), evaluation followed by prompt antihypertensive drug treatment is recommended. ^{S8.1.3-1,S8.1.3-2}
Ila	C-E0	5. For adults with a normal BP, repeat evaluation every year is reasonable.

Synopsis

An important component of BP management in hypertensive patients is follow-up. Different periods of time for follow-up

are recommended depending on the stage of hypertension, the presence or absence of target organ damage, treatment with antihypertensive medications, and the level of BP control. Recommendations for follow-up are summarized in Figure 4.

Recommendation-Specific Supportive Text

1. Nonpharmacological therapy (see Section 6.2) is the preferred therapy for adults with elevated BP and an appropriate first-line therapy for adults with stage 1 hypertension who have an estimated 10-year ASCVD risk of <10%. Adherence to and impact of nonpharmacological therapy should be assessed within 3 to 6 months.
2. Nonpharmacological therapy can help reduce BP in patients with stage 1 hypertension with an estimated 10-year ASCVD risk of ≥10% and should be used in addition to pharmacological therapy as first-line therapy in such patients (see Section 6.2).
3. Prompt evaluation and treatment of patients with stage 2 hypertension with a combination of drug and nonpharmacological therapy are important because of the elevated risk of CVD events in this subgroup, especially those with multiple ASCVD risk factors or target organ damage.^{S8.1.3-1,S8.1.3-2}
4. Prompt management of very high BP is important to reduce the risk of target organ damage (see Section 11.2). The rapidity of the treatment needed is dependent on the patient's clinical presentation (presence of new or worsening target organ damage) and presence or absence of CVD complications, but treatment should be initiated within at least 1 week.
5. Given that the lifetime risk of hypertension exceeds 80% in US adults,^{S8.1.3-3} it is likely that individuals with a normal BP will develop elevated BP in the future. BP may change over time because of changes in BP-related lifestyle factors, such as degree of sedentary lifestyle, dietary sodium intake, body weight, and alcohol intake. Less commonly, secondary causes of hypertension can occur over time and lead to an increase in BP. Periodic BP screening can identify individuals who develop elevated BP over time. More frequent BP screening may be particularly important for individuals with elevated ASCVD risk.

8.1.4. General Principles of Drug Therapy

Recommendation for General Principle of Drug Therapy		
References that support recommendations are summarized in Online Data Supplement 25.		
COR	LOE	Recommendation
III: Harm	A	1. Simultaneous use of an ACE inhibitor, ARB, and/or renin inhibitor is potentially harmful and is not recommended to treat adults with hypertension. ^{S8.1.4-1-S8.1.4-3}

Synopsis

Pharmacological agents, in addition to lifestyle modification (see Section 6.2), provide the primary basis for treatment of high BP. A large number of clinical trials have demonstrated that antihypertensive pharmacotherapy not only lowers BP but reduces the risk of CVD, cerebrovascular events, and death.^{S8.1.4-4-S8.1.4-7}

Numerous classes of antihypertensive agents are available to treat high BP (Table 18). Agents that have been shown to reduce clinical events should be used preferentially. Therefore, the primary agents used in the treatment of hypertension include thiazide diuretics, ACE inhibitors, ARBs, and CCBs^{S8.1.4-8–S8.1.4-11} (see Section 8.1.6). Although many other drugs and drug classes are available, either confirmation that these agents decrease clinical outcomes to an extent similar to that of the primary agents is lacking, or safety and tolerability may relegate their role to use as secondary agents. In particular, there is inadequate evidence to support the initial use of beta blockers for hypertension in the absence of specific cardiovascular comorbidities (see Section 9).

When the initial drug treatment of high BP is being considered, several different strategies may be contemplated. Many patients can be started on a single agent, but consideration should be given to starting with 2 drugs of different classes for those with stage 2 hypertension (see Section 8.1.6.1). In addition, other patient-specific factors, such as age, concurrent medications, drug adherence, drug interactions, the overall treatment regimen, out-of-pocket costs, and comorbidities, should be considered. From a societal perspective, total costs must be taken into account. Shared decision making, with the patient influenced by clinician judgment, should drive the ultimate choice of antihypertensive agent(s).

Many patients started on a single agent will subsequently require ≥ 2 drugs from different pharmacological classes to reach their BP goals.^{S8.1.4-12,S8.1.4-13,S8.1.4-14} Knowledge of the pharmacological mechanisms of action of each agent is important. Drug regimens with complementary activity, where a second antihypertensive agent is used to block compensatory responses to the initial agent or affect a different pressor mechanism, can result in additive lowering of BP. For example, thiazide diuretics may stimulate the renin-angiotensin-aldosterone system. By adding an ACE inhibitor or ARB to the thiazide, an additive BP-lowering effect may be obtained.^{S8.1.4-13} Use of combination therapy may also improve adherence. Several 2- and 3-fixed-dose drug combinations of antihypertensive drug therapy are available, with complementary mechanisms of action among the components (Online Data Supplement D). However, it should be noted that many triple-dose combinations may contain a lower-than-optimal dose of thiazide diuretic.

Table 18 is a summary of oral antihypertensive drugs.

Recommendation-Specific Supportive Text

1. Drug combinations that have similar mechanisms of action or clinical effects should be avoided. For example, 2 drugs from the same class should not be administered together (eg, 2 different beta blockers, ACE inhibitors, or nondihydropyridine CCBs). Likewise, 2 drugs from classes that target the same BP control system are less effective and potentially harmful when used together (eg, ACE inhibitors, ARBs). Exceptions to this rule include concomitant use of a thiazide diuretic, K-sparing diuretic, and/or loop diuretic in various combinations. Also, dihydropyridine and nondihydropyridine CCBs can be combined. High-quality RCT data demonstrate that simultaneous administration of RAS blockers (ie, ACE inhibitor with ARB; ACE inhibitor or ARB with renin

inhibitor aliskiren) increases cardiovascular and renal risk.^{S8.1.4-1–S8.1.4-3}

8.1.5. BP Goal for Patients With Hypertension

Recommendations for BP Goal for Patients With Hypertension		
References that support recommendations are summarized in Online Data Supplement 26 and Systematic Review Report.		
COR	LOE	Recommendations
I	SBP: B-R ^{SR}	1. For adults with confirmed hypertension and known CVD or 10-year ASCVD event risk of 10% or higher (see Section 8.1.2), a BP target of less than 130/80 mm Hg is recommended. ^{S8.1.5-1–S8.1.5-5}
	DBP: C-EO	
IIb	SBP: B-NR	2. For adults with confirmed hypertension, without additional markers of increased CVD risk, a BP target of less than 130/80 mm Hg may be reasonable. ^{S8.1.5-6–S8.1.5-9}
	DBP: C-EO	

SR indicates systematic review.

Synopsis

Refer to the “Systematic Review for the 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults” for the complete systematic evidence review for additional data and analyses.^{S8.1.5-10} Several trials have tested whether more intensive BP control improves major CVD outcomes. Meta-analyses and systematic reviews of these trials provide strong support for the more intensive approach, but the data are less clear in identification of a specific optimal BP target.^{S8.1.5-1–S8.1.5-5,S8.1.5-7,S8.1.5-11–S8.1.5-13} Recent trials that address optimal BP targets include SPRINT and ACCORD (Action to Control Cardiovascular Risk in Diabetes), with targets for more intensive (SBP <120 mm Hg) and standard (SBP <140 mm Hg) treatment,^{S8.1.5-14,S8.1.5-15} and SPS-3, with a more intensive target of <130/80 mm Hg.^{S8.1.5-16} These trials yielded mixed results in achieving their primary endpoints. SPRINT was stopped early, after a median follow-up of 3.26 years, when more intensive treatment resulted in a significant reduction in the primary outcome (a CVD composite) and in all-cause mortality rate. In ACCORD, more intensive BP treatment failed to demonstrate a significant reduction in the primary outcome (a CVD composite). However, the incidence of stroke, a component of the primary outcome, was significantly reduced. The standard glycemia subgroup did show significant benefit in ACCORD, and a meta-analysis of the only 2 trials (ACCORD and SPRINT) testing an SBP goal of <120 mm Hg showed significant reduction in CVD events.^{S8.1.5-17} SPS-3 failed to demonstrate benefit for the primary endpoint of recurrent stroke ($P=0.08$) but found a significant reduction in a subgroup with hemorrhagic stroke. Pooling of the experience from 19 trials (excluding SPRINT) that randomly assigned participants to different BP treatment targets identified a significant reduction in CVD events, MI, and stroke in those assigned to a lower (average achieved SBP/DBP was 133/76 mm Hg) versus a higher BP treatment target.^{S8.1.5-2} Similar patterns of benefit were reported in 3 other meta-analyses of trials in which participants were randomly

Table 18. Oral Antihypertensive Drugs

Class	Drug	Usual Dose, Range (mg/d)*	Daily Frequency	Comments
Primary agents				
Thiazide or thiazide-type diuretics	Chlorthalidone	12.5–25	1	Chlorthalidone is preferred on the basis of prolonged half-life and proven trial reduction of CVD. Monitor for hyponatremia and hypokalemia, uric acid and calcium levels. Use with caution in patients with history of acute gout unless patient is on uric acid–lowering therapy.
	Hydrochlorothiazide	25–50	1	
	Indapamide	1.25–2.5	1	
	Metolazone	2.5–5	1	
ACE inhibitors	Benazepril	10–40	1 or 2	Do not use in combination with ARBs or direct renin inhibitor. There is an increased risk of hyperkalemia, especially in patients with CKD or in those on K ⁺ supplements or K ⁺ -sparing drugs. There is a risk of acute renal failure in patients with severe bilateral renal artery stenosis. Do not use if patient has history of angioedema with ACE inhibitors. Avoid in pregnancy.
	Captopril	12.5–150	2 or 3	
	Enalapril	5–40	1 or 2	
	Fosinopril	10–40	1	
	Lisinopril	10–40	1	
	Moexipril	7.5–30	1 or 2	
	Perindopril	4–16	1	
	Quinapril	10–80	1 or 2	
	Ramipril	2.5–20	1 or 2	
	Trandolapril	1–4	1	
ARBs	Azilsartan	40–80	1	Do not use in combination with ACE inhibitors or direct renin inhibitor. There is an increased risk of hyperkalemia in CKD or in those on K ⁺ supplements or K ⁺ -sparing drugs. There is a risk of acute renal failure in patients with severe bilateral renal artery stenosis. Do not use if patient has history of angioedema with ARBs. Patients with a history of angioedema with an ACE inhibitor can receive an ARB beginning 6 weeks after ACE inhibitor is discontinued. Avoid in pregnancy.
	Candesartan	8–32	1	
	Eprosartan	600–800	1 or 2	
	Irbesartan	150–300	1	
	Losartan	50–100	1 or 2	
	Olmesartan	20–40	1	
	Telmisartan	20–80	1	
	Valsartan	80–320	1	
CCB—dihydropyridines	Amlodipine	2.5–10	1	Avoid use in patients with HF/EF; amlodipine or felodipine may be used if required. They are associated with dose-related pedal edema, which is more common in women than men.
	Felodipine	2.5–10	1	
	Isradipine	5–10	2	
	Nicardipine SR	60–120	2	
	Nifedipine LA	30–90	1	
	Nisoldipine	17–34	1	
CCB—nondihydropyridines	Diltiazem ER	120–360	1	Avoid routine use with beta blockers because of increased risk of bradycardia and heart block. Do not use in patients with HF/EF. There are drug interactions with diltiazem and verapamil (CYP3A4 major substrate and moderate inhibitor).
	Verapamil IR	120–360	3	
	Verapamil SR	120–360	1 or 2	
	Verapamil-delayed onset ER	100–300	1 (in the evening)	
Secondary agents				
Diuretics—loop	Bumetanide	0.5–2	2	These are preferred diuretics in patients with symptomatic HF. They are preferred over thiazides in patients with moderate-to-severe CKD (eg, GFR <30 mL/min).
	Furosemide	20–80	2	
	Torsemide	5–10	1	
Diuretics—potassium sparing	Amiloride	5–10	1 or 2	These are monotherapy agents and minimally effective antihypertensive agents. Combination therapy of potassium-sparing diuretic with a thiazide can be considered in patients with hypokalemia on thiazide monotherapy. Avoid in patients with significant CKD (eg, GFR <45 mL/min).
	Triamterene	50–100	1 or 2	

(Continued)

Table 18. Continued

Class	Drug	Usual Dose, Range (mg/d)*	Daily Frequency	Comments
Diuretics—aldosterone antagonists	Eplerenone	50–100	1 or 2	These are preferred agents in primary aldosteronism and resistant hypertension. Spironolactone is associated with greater risk of gynecomastia and impotence as compared with eplerenone. This is common add-on therapy in resistant hypertension. Avoid use with K ⁺ supplements, other K ⁺ -sparing diuretics, or significant renal dysfunction. Eplerenone often requires twice-daily dosing for adequate BP lowering.
	Spironolactone	25–100	1	
Beta blockers—cardioselective	Atenolol	25–100	2	Beta blockers are not recommended as first-line agents unless the patient has IHD or HF. These are preferred in patients with bronchospastic airway disease requiring a beta blocker. Bisoprolol and metoprolol succinate are preferred in patients with HF/EF. Avoid abrupt cessation.
	Betaxolol	5–20	1	
	Bisoprolol	2.5–10	1	
	Metoprolol tartrate	100–200	2	
	Metoprolol succinate	50–200	1	
Beta blockers—cardioselective and vasodilatory	Nebivolol	5–40	1	Nebivolol induces nitric oxide–induced vasodilation. Avoid abrupt cessation.
Beta blockers—noncardioselective	Nadolol	40–120	1	Avoid in patients with reactive airways disease. Avoid abrupt cessation.
	Propranolol IR	80–160	2	
	Propranolol LA	80–160	1	
Beta blockers—intrinsic sympathomimetic activity	Acebutolol	200–800	2	Generally avoid, especially in patients with IHD or HF. Avoid abrupt cessation.
	Penbutolol	10–40	1	
	Pindolol	10–60	2	
Beta blockers—combined alpha- and beta-receptor	Carvedilol	12.5–50	2	Carvedilol is preferred in patients with HF/EF. Avoid abrupt cessation.
	Carvedilol phosphate	20–80	1	
	Labetalol	200–800	2	
Direct renin inhibitor	Aliskiren	150–300	1	Do not use in combination with ACE inhibitors or ARBs. Aliskiren is very long acting. There is an increased risk of hyperkalemia in CKD or in those on K ⁺ supplements or K ⁺ -sparing drugs. Aliskiren may cause acute renal failure in patients with severe bilateral renal artery stenosis. Avoid in pregnancy.
Alpha-1 blockers	Doxazosin	1–16	1	These are associated with orthostatic hypotension, especially in older adults. They may be considered as second-line agent in patients with concomitant BPH.
	Prazosin	2–20	2 or 3	
	Terazosin	1–20	1 or 2	
Central alpha ₂ -agonist and other centrally acting drugs	Clonidine oral	0.1–0.8	2	These are generally reserved as last-line because of significant CNS adverse effects, especially in older adults. Avoid abrupt discontinuation of clonidine, which may induce hypertensive crisis; clonidine must be tapered to avoid rebound hypertension.
	Clonidine patch	0.1–0.3	1 weekly	
	Methyldopa	250–1000	2	
	Guanfacine	0.5–2	1	
Direct vasodilators	Hydralazine	100–200	2 or 3	These are associated with sodium and water retention and reflex tachycardia; use with a diuretic and beta blocker. Hydralazine is associated with drug-induced lupus-like syndrome at higher doses. Minoxidil is associated with hirsutism and requires a loop diuretic. Minoxidil can induce pericardial effusion.
	Minoxidil	5–100	1-3	

*Dosages may vary from those listed in the FDA-approved labeling (available at <https://dailymed.nlm.nih.gov/dailymed/>) From Chobanian et al JNC 7.^{58.1.4-15}

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BP, blood pressure; BPH, benign prostatic hyperplasia; CCB, calcium channel blocker; CKD, chronic kidney disease; CNS, central nervous system; CVD, cardiovascular disease; ER, extended release; GFR, glomerular filtration rate; HF, heart failure; HF/EF, heart failure with reduced ejection fraction; IHD, ischemic heart disease; IR, immediate release; LA, long-acting; and SR, sustained release.

assigned to different BP targets^{S8.1.5-3–S8.1.5-5} and in larger meta-analyses that additionally included trials that compared different intensities of treatment.^{S8.1.5-12} Data from the most recent meta-analysis (42 trials and 144 220 patients)^{S8.1.5-5} demonstrate a linear association between mean achieved SBP and risk of CVD mortality with the lowest risk at 120 to 124 mm Hg. The totality of the available information provides evidence that a lower BP target is generally better than a higher BP target and that some patients will benefit from an SBP treatment goal <120 mm Hg, especially those at high risk of CVD.^{S8.1.5-15} The specific inclusion and exclusion criteria of any RCT may limit extrapolation to a more general population with hypertension. In addition, all of the relevant trials have been efficacy studies in which BP measurements were more consistent with guideline recommendations than is common in clinical practice, resulting in lower absolute values for SBP. For both of these reasons, the SBP target recommended during BP lowering (<130 mm Hg) is higher than that which was used in SPRINT.

Recommendation-Specific Supportive Text

- 1. Meta-analysis and systematic review of trials that compare more intensive BP reduction to standard BP reduction report that more intense BP lowering significantly reduces the risk of stroke, coronary events, major cardiovascular events, and cardiovascular mortality.^{S8.1.5-1} In a stratified analysis of these data, achieving an additional 10-mm Hg reduction in SBP reduced CVD risk when compared with an average SBP of 158/82 to 143/76 mm Hg, 144/85 to 137/81 mm Hg, and 134/79 to 125/76 mm Hg. Patients with DM and CKD were included in the analysis.^{S8.1.5-1,S8.1.5-2,S8.1.5-11–S8.1.5-13,S8.1.5-18} (Specific management details are in Section 9.3 for CKD and Section 9.6 for DM.)
- 2. The treatment of patients with hypertension without elevated risk has been systematically understudied because lower-risk groups would require prolonged follow-up to have a sufficient number of clinical events to provide useful information. Although there is clinical trial evidence that both drug and nondrug therapy will interrupt the progressive course of hypertension,^{S8.1.5-6} there is no trial evidence that this treatment decreases CVD morbidity and mortality. The clinical trial evidence is strongest for a target BP of 140/90 mm Hg in this population. However, observational studies suggest that these individuals often have a high lifetime risk and would benefit from BP control earlier in life.^{S8.1.5-19,S8.1.5-20}

8.1.6. Choice of Initial Medication

Recommendation for Choice of Initial Medication		
References that support the recommendation are summarized in Online Data Supplement 27 and Systematic Review Report.		
COR	LOE	Recommendation
I	ASR	1. For initiation of antihypertensive drug therapy, first-line agents include thiazide diuretics, CCBs, and ACE inhibitors or ARBs. ^{S8.1.6-1,S8.1.6-2}

SR indicates systematic review.

Synopsis

The overwhelming majority of persons with BP sufficiently elevated to warrant pharmacological therapy may be best treated initially with 2 agents (see Section 8.1.6.1). When initiation of pharmacological therapy with a single medication is appropriate, primary consideration should be given to comorbid conditions (eg, HF, CKD) for which specific classes of BP-lowering medication are indicated (see Section 9).^{S8.1.6-1} In the largest head-to-head comparison of first-step drug therapy for hypertension,^{S8.1.6-3} the thiazide-type diuretic chlorthalidone was superior to the CCB amlodipine and the ACE inhibitor lisinopril in preventing HF, a BP-related outcome of increasing importance in the growing population of older persons with hypertension.^{S8.1.6-4–S8.1.6-7} Additionally, ACE inhibitors were less effective than thiazide diuretics and CCBs in lowering BP and in prevention of stroke. For black patients, ACE inhibitors were also notably less effective than CCBs in preventing HF^{S8.1.6-8} and in the prevention of stroke^{S8.1.6-9} (see Section 10.1). ARBs may be better tolerated than ACE inhibitors in black patients, with less cough and angioedema, but according to the limited available experience they offer no proven advantage over ACE inhibitors in preventing stroke or CVD in this population, making thiazide diuretics (especially chlorthalidone) or CCBs the best initial choice for single-drug therapy. For stroke, in the general population, beta blockers were less effective than CCBs (36% lower risk) and thiazide diuretics (30% lower risk). CCBs have been shown to be as effective as diuretics for reducing all CVD events other than HF, and CCBs are a good alternative choice for initial therapy when thiazide diuretics are not tolerated. Alpha blockers are not used as first-line therapy for hypertension because they are less effective for prevention of CVD than other first-step agents, such as thiazide diuretics.^{S8.1.6-3,S8.1.6-10}

Recommendation-Specific Supportive Text

- 1. The overall goal of treatment should be reduction in BP, in the context of underlying CVD risk. Five drug classes have been shown, in high-quality RCTs, to prevent CVD as compared with placebo (diuretics, ACE inhibitors, ARBs, CCBs, and beta blockers).^{S8.1.6-11,S8.1.6-12} In head-to-head comparisons of first-step therapy, different drug classes have been reported to provide somewhat divergent capacity to prevent specific CVD events. Interpretation of meta-analyses comparing agents from different drug classes is challenging because the relevant RCTs were conducted in different time periods, during which concurrent antihypertensive therapy was less or more common, and the efficacy of agents from certain drug classes may have changed. In recognition of this, some^{S8.1.6-2} but not all^{S8.1.6-11,S8.1.6-12} meta-analyses, as well as the largest individual RCT that compared first-step agents,^{S8.1.6-3} have suggested that diuretics, especially the long-acting thiazide-type agent chlorthalidone, may provide an optimal choice for first-step drug therapy of hypertension. In contrast, some meta-analyses have suggested that beta blockers may be less effective, especially for stroke prevention in older adults, but interpretation is hampered by inclusion of RCTs that used beta blockers that are now considered to be inferior for prevention of CVD.^{S8.1.6-13,S8.1.6-14} In a systematic review and network meta-analysis conducted for the present guideline, beta blockers were significantly less effective than diuretics

for prevention of stroke and cardiovascular events.^{S8.1.6-1} Diuretics were also significantly better than CCBs for prevention of HF. There were some other nonsignificant differences between diuretics, ACE inhibitors, ARBs, and CCBs, but the general pattern was for similarity in effect. As indicated in Section 8.1.6.1, most adults with hypertension require >1 drug to control their BP. As recommended in Section 10.1, for black adults with hypertension (without HF or CKD), initial antihypertensive treatment should include a thiazide diuretic or CCB.

8.1.6.1. Choice of Initial Monotherapy Versus Initial Combination Drug Therapy

Recommendations for Choice of Initial Monotherapy Versus Initial Combination Drug Therapy*		
COR	LOE	Recommendations
I	C-EO	1. Initiation of antihypertensive drug therapy with 2 first-line agents of different classes, either as separate agents or in a fixed-dose combination, is recommended in adults with stage 2 hypertension and an average BP more than 20/10 mm Hg above their BP target.
Ila	C-EO	2. Initiation of antihypertensive drug therapy with a single antihypertensive drug is reasonable in adults with stage 1 hypertension and BP goal <130/80 mm Hg with dosage titration and sequential addition of other agents to achieve the BP target.

*Fixed-dose combination antihypertensive medications are listed in [Online Data Supplement D](#).

Synopsis

Systematic review of the evidence comparing the initiation of antihypertensive treatment with monotherapy and sequential (stepped-care) titration of additional agents versus initiation of treatment with combination therapy (including fixed-dose combinations) did not identify any RCTs meeting the systematic review questions posed in the PICOTS format (P=population, I=intervention, C=comparator, O=outcome, T=timing, S=setting). However, in both ACCORD and SPRINT, 2-drug therapy was recommended for most participants in the intensive- but not standard-therapy groups.

Recommendation-Specific Supportive Text

1. Because most patients with hypertension require multiple agents for control of their BP and those with higher BPs are at greater risk, more rapid titration of antihypertensive medications began to be recommended in patients with BP >20/10 mm Hg above their target, beginning with the JNC 7 report.^{S8.1.6.1-1} In these patients, initiation of antihypertensive therapy with 2 agents is recommended. Evidence favoring this approach comes mostly from studies using fixed-dose combination products showing greater BP lowering with fixed-dose combination agents than with single agents, as well as better adherence to therapy.^{S8.1.6.1-2,S8.1.6.1-3} The safety and efficacy of this strategy have been demonstrated in adults to reduce BPs to <140/90 mm Hg though not compared with other strategies.^{S8.1.6.1-4–S8.1.6.1-6} In general, this approach is reasonable in older adults, those at high CVD risk, or those

who have a history of hypotension or drug-associated side effects. However, caution is advised in initiating antihypertensive pharmacotherapy with 2 drugs in older patients because hypotension or orthostatic hypotension may develop in some patients; BP should be carefully monitored.

2. The stepped-care approach defined by the initiation of antihypertensive drug therapy with a single agent followed by the sequential titration of the dose and addition of other agents has been the recommended treatment strategy since the first report of the National High Blood Pressure Education Program.^{S8.1.6.1-7} This approach is also reasonable in older adults or those at risk or who have a history of hypotension or drug-associated side effects. This strategy has been used successfully in nearly all hypertension treatment trials but has not been formally tested against other antihypertensive drug treatment strategies for effectiveness in achieving BP control or in preventing adverse outcomes.

8.2. Achieving BP Control in Individual Patients

Recommendations for lifestyle modifications and drug selection are specified in Sections 6.2, 8.1.4, and 8.1.6. Initial drug selections should be based on trial evidence of treatment efficacy, combined with recognition of compelling indications for use of an agent from a specific drug class, as well as the individual patient's lifestyle preferences and traits. For a subset of patients (25% to 50%),^{S8.2-1} the initial drug therapy will be well tolerated and effective in achieving the desired level of BP, with only the need for subsequent monitoring (see Section 8.3 for an appropriate follow-up schedule). For others, the initial drug will not be tolerated or will not be effective, requiring either a change in medication or addition of another medication, followed by BP monitoring.^{S8.2-2} Approximately 25% of patients will require additional treatment adjustments. In a minority of this group, achievement of goal BP can be challenging.

In patients who do not respond to or do not tolerate treatment with 2 to 3 medications or medication combinations, additional trials of treatment tend to be ineffective or poorly tolerated. Some patients may become disillusioned and lost to follow-up, whereas others will identify an alternative health-care provider, including nontraditional healers, or will try popular home remedies. Working with this more demanding subset requires provider expertise, patience, and a mechanism to respond efficiently and sensitively to concerns as they arise. In this setting, team-based care (see Section 12) may be effective, encouraging coupling of nonpharmacological and pharmacological treatments, while improving access to and communication with care providers.

In the setting of medication intolerance, consider allowing a defined period of time to evaluate the effects of lifestyle modification in patients with a relatively low CVD risk (10-year risk of ASCVD <10%, based on the ASCVD Risk Estimator [<http://tools.acc.org/ASCVD-Risk-Estimator>]), with scheduled follow-up visits for assessment of BP levels, including a review of HBPM data, and an appraisal of lifestyle change goal achievements. For patients with a higher level of CVD risk or with significant elevations in BP (SBP or DBP >20 or >10 mm Hg above target, respectively), medication is usually

started even while the patient is pursuing lifestyle change (see Section 8.1.2).

Consideration of patient comorbidities, lifestyle, and preferences may suggest better tolerance or greater effect from one class of medication versus other classes. For example, if hyponatremia is present, it would be important to avoid or stop thiazide diuretic therapy. In this case, a loop diuretic should be used if a diuretic is required. If hypokalemia is present, primary or secondary aldosteronism should be excluded, after which one should consider a potassium-sparing agent, such as spironolactone, eplerenone, triamterene, or amiloride. In addition, reducing dietary sodium intake will diminish urinary potassium losses. If the patient has chronic cough or a history of ACE inhibitor–induced cough or develops a cough or bronchial responsiveness while on an ACE inhibitor, one should use an ARB in place of an ACE inhibitor. For patients with bronchospastic lung disease, a beta-1-selective blocker (eg, bisoprolol, metoprolol) should be considered if beta-blocker therapy is required. A patient who is already adherent to lifestyle change recommendations, including diligent reduction in sodium intake, may show a greater response to a RAS blocker. Prior patient experience should be considered, as in the case of cough associated with prior use of an ACE inhibitor, which is likely to reoccur if an agent from the same class is prescribed.

8.3. Follow-Up of BP During Antihypertensive Drug Therapy

Appropriate follow-up and monitoring enable assessment of adherence (see Section 12.1) and response to therapy, help identify adverse responses to therapy and target organ damage, and allow assessment of progress toward treatment goals. High-quality RCTs have successfully and safely developed strategies for follow-up, monitoring, and reassessment from which recommendations can be made (Figure 4).^{S8.2-1,S8.2-2} A systematic approach to out-of-office BP assessment is an essential part of follow-up and monitoring of BP, to assess response to therapy; check for evidence of white coat hypertension, white coat effect, masked hypertension, or masked uncontrolled hypertension; and help achieve BP targets (see Sections 4 and 12).

8.3.1. Follow-Up After Initiating Antihypertensive Drug Therapy

Recommendation for Follow-Up After Initiating Antihypertensive Drug Therapy		
References that support the recommendation are summarized in Online Data Supplement 28.		
COR	LOE	Recommendation
I	B-R	1. Adults initiating a new or adjusted drug regimen for hypertension should have a follow-up evaluation of adherence and response to treatment at monthly intervals until control is achieved. ^{S8.3.1-1–S8.3.1-3}

Recommendation-Specific Supportive Text

1. Components of the follow-up evaluation should include assessment of BP control, as well as evaluation for

orthostatic hypotension, adverse effects from medication therapy, adherence to medication and lifestyle therapy, need for adjustment of medication dosage, laboratory testing (including electrolyte and renal function status), and other assessments of target organ damage.^{S8.3.1-1–S8.3.1-3}

8.3.2. Monitoring Strategies to Improve Control of BP in Patients on Drug Therapy for High BP

Recommendation for Monitoring Strategies to Improve Control of BP in Patients on Drug Therapy for High BP		
References that support the recommendation are summarized in Online Data Supplement 29.		
COR	LOE	Recommendation
I	A	1. Follow-up and monitoring after initiation of drug therapy for hypertension control should include systematic strategies to help improve BP, including use of HBPM, team-based care, and telehealth strategies. ^{S8.3.2-1–S8.3.2-6}

Recommendation-Specific Supportive Text

1. Systematic approaches to follow-up have been shown to improve hypertension control and can be adapted and incorporated into clinical practices according to local needs and resource availability (see Section 8.3.1 for time intervals for treatment follow-up and monitoring and Sections 12.2 and 12.3.2 on systematic strategies to improve BP control).

9. Hypertension in Patients With Comorbidities

Certain comorbidities may affect clinical decision-making in hypertension. These include ischemic heart disease, HF with reduced ejection fraction (HFrEF), HFpEF, CKD (including renal transplantation), cerebrovascular disease, AF, PAD, DM, and metabolic syndrome.^{S9-1} As noted in Section 8.1.2, this guideline generally recommends use of BP-lowering medications for secondary prevention of CVD in patients with clinical CVD (CHD, HF, and stroke) and an average BP ≥130/80 mm Hg and for primary prevention of CVD in adults with an estimated 10-year ASCVD risk of ≥10% and an average SBP ≥130 mm Hg or an average DBP ≥80 mm Hg. Although we recommend use of the ACC/AHA Pooled Cohort Equations (<http://tools.acc.org/ASCVD-Risk-Estimator/>) to estimate 10-year risk of ASCVD to establish the BP threshold for treatment, the vast majority of adults with a co-morbidity are likely to have a 10-year risk of ASCVD that exceeds 10%. In some instances, clinical trial confirmation of treatment in patients with comorbidities is limited to a target BP of 140/90 mm Hg. In addition, the selection of medications for use in treating high BP in patients with CVD is guided by their use for other compelling indications (eg, beta blockers after MI, ACE inhibitors for HFrEF), as discussed in specific guidelines for the clinical condition.^{S9-2–S9-4} The present guideline does not address the recommendations for treatment of hypertension occurring with acute coronary syndromes.

9.1. Stable Ischemic Heart Disease

Recommendations for Treatment of Hypertension in Patients With Stable Ischemic Heart Disease (SIHD)		
References that support recommendations are summarized in Online Data Supplements 30-32.		
COR	LOE	Recommendations
I	SBP: B-R	1. In adults with SIHD and hypertension, a BP target of less than 130/80 mm Hg is recommended. ^{S9.1-1-S9.1-5}
	DBP: C-E0	
I	SBP: B-R	2. Adults with SIHD and hypertension (BP ≥130/80 mm Hg) should be treated with medications (eg, GDMT ^{S9.1-6} beta blockers, ACE inhibitors, or ARBs) for compelling indications (eg, previous MI, stable angina) as first-line therapy, with the addition of other drugs (eg, dihydropyridine CCBs, thiazide diuretics, and/or mineralocorticoid receptor antagonists) as needed to further control hypertension. ^{S9.1-7-S9.1-10}
	DBP: C-E0	
I	B-NR	3. In adults with SIHD with angina and persistent uncontrolled hypertension, the addition of dihydropyridine CCBs to GDMT ^{S9.1-6} beta blockers is recommended. ^{S9.1-8,S9.1-11,S9.1-12}
IIa	B-NR	4. In adults who have had a MI or acute coronary syndrome, it is reasonable to continue GDMT (S9.1-6) beta blockers beyond 3 years as long-term therapy for hypertension. ^{S9.1-13,S9.1-14}
IIb	C-E0	5. Beta blockers and/or CCBs might be considered to control hypertension in patients with CAD (without HF/EF) who had an MI more than 3 years ago and have angina.

Synopsis

Hypertension is a major risk factor for ischemic heart disease. Numerous RCTs have demonstrated the benefits of antihypertensive drug therapy in reducing the risk of ischemic heart disease. The following recommendations apply only to management of hypertension in patients with SIHD without HF. See Section 9.2 for recommendations for the treatment of patients with SIHD and HF.

Figure 5 is an algorithm on management of hypertension in patients with SIHD.

Recommendation-Specific Supportive Text

1. In patients with increased cardiovascular risk, reduction of SBP to <130/80 mm Hg has been shown to reduce CVD complications by 25% and all-cause mortality by 27%.^{S9.1-1}
2. After 5 years of randomized therapy in high-CVD-risk patients, ramipril produced a 22% reduction in MI, stroke, or CVD compared with placebo.^{S9.1-10} No added benefit on CVD outcomes was seen when compared with CCBs and diuretics.^{S9.1-15,S9.1-16} After 4.2 years of randomized therapy in patients with SIHD, perindopril reduced CVD death, MI, or cardiac arrest by 20% compared with placebo.^{S9.1-7} Beta blockers are effective drugs for preventing angina pectoris, improving exercise time

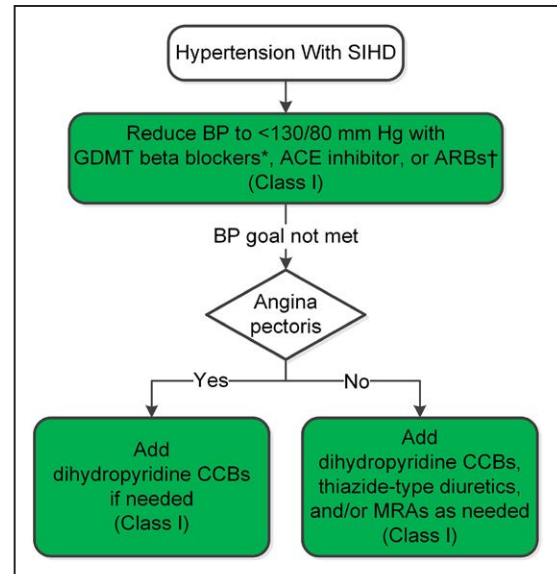


Figure 5. Management of hypertension in patients with SIHD. Colors correspond to Class of Recommendation in Table 1.

*GDMT beta blockers for BP control or relief of angina include carvedilol, metoprolol tartrate, metoprolol succinate, nadolol, bisoprolol, propranolol, and timolol. Avoid beta blockers with intrinsic sympathomimetic activity. The beta blocker atenolol should not be used because it is less effective than placebo in reducing cardiovascular events. †If needed for BP control. ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BP, blood pressure; CCB, calcium channel blocker; GDMT, guideline-directed management and therapy; and SIHD, stable ischemic heart disease.

until the onset of angina pectoris, reducing exercise-induced ischemic ST-segment depression, and preventing coronary events.^{S9.1-8,S9.1-17-S9.1-22} Because of their compelling indications for treatment of SIHD, these drugs are recommended as a first-line therapy in the treatment of hypertension when it occurs in patients with SIHD. GDMT beta blockers for SIHD that are also effective in lowering BP include carvedilol, metoprolol tartrate, metoprolol succinate, nadolol, bisoprolol, propranolol, and timolol. Atenolol is not as effective as other antihypertensive drugs in the treatment of hypertension.^{S9.1-23}

3. Dihydropyridine CCBs are effective antianginal drugs that can lower BP and relieve angina pectoris when added to beta blockers in patients in whom hypertension is present and angina pectoris persists despite beta-blocker therapy.^{S9.1-8,S9.1-17,S9.1-19-S9.1-22,S9.1-24,S9.1-25} GDMT beta blockers for SIHD that are also effective in lowering BP include carvedilol, metoprolol tartrate, metoprolol succinate, nadolol, bisoprolol, propranolol, and timolol.
4. In randomized long-term trials, use of beta blockers after MI reduced all-cause mortality by 23%.^{S9.1-13} Given the established efficacy of beta blockers for treating hypertension and SIHD, their use for treatment continuing beyond 3 years after MI is reasonable.^{S9.1-6,S9.1-25}
5. GDMT beta blockers and CCBs are effective antihypertensive and antianginal agents. CCBs include dihydropyridine and nondihydropyridine agents. CCBs can be

used separately or together with beta blockers beginning 3 years after MI in patients with CAD who have both hypertension and angina.

9.2. Heart Failure

Recommendation for Prevention of HF in Adults With Hypertension		
References that support the recommendation are summarized in Online Data Supplement 33.		
COR	LOE	Recommendation
I	SBP: B-R	1. In adults at increased risk of HF, the optimal BP in those with hypertension should be less than 130/80 mm Hg. ^{S9.2-1–S9.2-3}
	DBP: C-EO	

Synopsis

Antecedent hypertension is present in 75% of patients with chronic HF.^{S9.2-4} In the Cardiovascular Health Study^{S9.2-5} and the Health, Aging and Body Composition Study,^{S9.2-6} 11.2% of 4408 persons (53.1% women, with a mean age of 72.8 years, living in the community, and not receiving antihypertensive drugs at baseline) developed HF over 10 years.^{S9.2-7} Compared with those with an average SBP <120 mm Hg, the adjusted incidence of HF was increased 1.6, 2.2, and 2.6 times in those with average SBPs between 120 and 139 mm Hg, between 140 and 159 mm Hg, and ≥160 mm Hg, respectively.^{S9.2-7}

No RCTs are available that compare one BP-lowering agent to another for the management of patients with HF. The following recommendations for treatment of hypertension in HF are based on use of drugs that lower BP and also have compelling indications for management of HF (with HF_rEF or HF_pEF) as recommended in current ACC/AHA guidelines.^{S9.2-4,S9.2-8}

Recommendation-Specific Supportive Text

1. In adults with hypertension (SBP ≥130 mm Hg or DBP ≥80 mm Hg) and a high risk of CVD, a strong body of evidence supports treatment with antihypertensive medications (see Section 8.1.2) and more-intensive rather than less-intensive intervention (see Section 8.1.5). In SPRINT, a more intensive intervention that targeted an SBP <120 mm Hg significantly reduced the primary outcome (CVD composite) by about 25%.^{S9.2-9} The incidence of HF, a component of the primary outcome, was also substantially decreased (hazard ratio: 0.62; 95% confidence interval: 0.45–0.84). Meta-analyses of clinical trials have identified a similar beneficial effect of more-intensive BP reduction on the incidence of HF,^{S9.2-2,S9.2-10} but the body of information from studies confined to trials that randomly assigned participants to different BP targets is more limited and less compelling.^{S9.2-3} In addition, the available trials were efficacy studies in which BP measurements were more consistent with guideline recommendations than is common in clinical practice, resulting in lower absolute values for SBP. For both of these reasons, the SBP target recommended during BP lowering (<130 mm Hg) is higher than that used in SPRINT.

9.2.1. Heart Failure With Reduced Ejection Fraction

Recommendations for Treatment of Hypertension in Patients With HF _r EF		
References that support recommendations are summarized in Online Data Supplement 34.		
COR	LOE	Recommendations
I	C-EO	1. Adults with HF _r EF and hypertension should be prescribed GDMT ^{S9.2.1-2} titrated to attain a BP of less than 130/80 mm Hg.
III: No Benefit	B-R	2. Nondihydropyridine CCBs are not recommended in the treatment of hypertension in adults with HF _r EF. ^{S9.2.1-1}

Synopsis

Approximately 50% of patients with HF have HF_rEF.^{S9.2.1-2–S9.2.1-6} Numerous RCTs have shown that treatment of HF_rEF with GDMT reduces mortality and HF hospitalizations.^{S9.2.1-7} Large-scale RCTs have shown that antihypertensive drug therapy reduces the incidence of HF in patients with hypertension.^{S9.2.1-8–S9.2.1-11} In ALLHAT (Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial), chlorthalidone reduced the risk of HF_rEF more than amlodipine and doxazosin but similarly to lisinopril.^{S9.2.1-12,S9.2.1-13}

Recommendation-Specific Supportive Text

1. This recommendation is based on guidance in the 2017 ACC/AHA/HFSA guideline focused update on heart failure^{S9.2.1-14} (see figure from the HF focused update that is reproduced in Online Data Supplement A). Lifestyle modification, such as weight loss and sodium reduction, may serve as adjunctive measures to help these agents work better. No RCT evidence is available to support the superiority of one BP-lowering medication with compelling indications for treatment of HF_rEF over another. Medications with compelling indications for HF that may be used as first-line therapy to treat high BP include ACE inhibitors or ARBs, angiotensin receptor–neprilysin inhibitors, mineralocorticoid receptor antagonists, diuretics, and GDMT beta blockers (carvedilol, metoprolol succinate, or bisoprolol).
Clinical trials evaluating goal BP reduction and optimal BP-lowering agents in the setting of HF_rEF and concomitant hypertension have not been performed. However, in patients at higher CVD risk, BP lowering is associated with fewer adverse cardiovascular events.^{S9.2.1-7} GDMT for HF_rEF with agents known to lower BP should consider a goal BP reduction consistent with a threshold now associated with improved clinical outcomes but not yet proven by RCTs in an HF population.
2. Nondihydropyridine CCBs (verapamil, diltiazem) have myocardial depressant activity. Several clinical trials have demonstrated either no clinical benefit or even worse outcomes in patients with HF treated with these drugs.^{S9.2.1-1} Therefore, nondihydropyridine CCBs are not recommended in patients with hypertension and HF_rEF.

9.2.2. Heart Failure With Preserved Ejection Fraction

Recommendations for Treatment of Hypertension in Patients With HFpEF		
References that support recommendations are summarized in Online Data Supplements 35 and 36.		
COR	LOE	Recommendations
I	C-EO	1. In adults with HFpEF who present with symptoms of volume overload, diuretics should be prescribed to control hypertension.
I	C-LD	2. Adults with HFpEF and persistent hypertension after management of volume overload should be prescribed ACE inhibitors or ARBs and beta blockers titrated to attain SBP of less than 130 mm Hg. ^{S9.2.2-1–S9.2.2-6}

Synopsis

Approximately 50% of patients with HF have HFpEF.^{S9.2.2-2,S9.2.2-7–S9.2.2-11} The ejection fraction in these studies has varied from >40% to ≥55%.^{S9.2.2-2} Patients with HFpEF are usually older women with a history of hypertension. Obesity, CHD, DM, AF, and hyperlipidemia are also highly prevalent in patients with HFpEF.^{S9.2.2-2,S9.2.2-11,S9.2.2-12} Hypertension is the most important cause of HFpEF, with a prevalence of 60% to 89% in large RCTs, epidemiological studies, and HF registries.^{S9.2.2-2,S9.2.2-13} Patients with HFpEF also have an exaggerated hypertensive response to exercise.^{S9.2.2-14} Hypertensive acute pulmonary edema is an expression of HFpEF.^{S9.2.2-15}

BP control is important for prevention of HFpEF in patients with hypertension.^{S9.2.2-2,S9.2.2-16–S9.2.2-19} ALLHAT showed that treatment of hypertension with chlorthalidone reduced the risk of HF compared with amlodipine, doxazosin, and lisinopril.^{S9.2.2-19,S9.2.2-20} Improved BP control also reduces hospitalization, CVD events, and mortality.^{S9.2.2-2,S9.2.2-16–S9.2.2-19}

Recommendation-Specific Supportive Text

1. Diuretics are the only drugs used for the treatment of hypertension and HF that can adequately control the fluid retention of HF. Appropriate use of diuretics is also crucial to the success of other drugs used for the treatment of hypertension in the presence of HF. The use of inappropriately low doses of diuretics can result in fluid retention. Conversely, the use of inappropriately high doses of diuretics can lead to volume contraction, which can increase the risk of hypotension and renal insufficiency. Diuretics should be prescribed to all patients with hypertension and HFpEF who have evidence of, and to most patients with a prior history of, fluid retention.
2. In a trial of patients with HFpEF and MI, patients randomized to propranolol had at 32-month follow-up a 35% reduction in mortality rate.^{S9.2.2-3} After 21 months of treatment in patients with HFrEF and HFpEF, compared with placebo, those randomized to nebivolol had a 14% reduction in mortality or CVD hospitalization if they had HFrEF and a 19% reduction if they had HFpEF.^{S9.2.2-4} In patients with HFpEF, the primary outcome (a composite of CVD death or HF hospitalization) was observed in

22% for candesartan and 24% for placebo (11% reduction), but fewer patients receiving candesartan were hospitalized for HF.^{S9.2.2-5} The use of nitrates in the setting of HFpEF is associated with a signal of harm and in most situations should be avoided. For many other common antihypertensive agents, including alpha blockers, beta blockers, and calcium channel blockers, limited data exist to guide the choice of antihypertensive therapy in the setting of HFpEF.^{S9.2.2-21} Renin-angiotensin-aldosterone system inhibition, however, with ACE inhibitor or ARB and especially MRA would represent the preferred choice. A shared decision-making discussion, with the patient influenced by clinician judgment, should drive the ultimate choice of antihypertensive agents.

9.3. Chronic Kidney Disease

Recommendations for Treatment of Hypertension in Patients With CKD		
References that support recommendations are summarized in Online Data Supplements 37 and 38 and Systematic Review Report.		
COR	LOE	Recommendations
I	SBP: B-R ^{SR}	1. Adults with hypertension and CKD should be treated to a BP goal of less than 130/80 mm Hg. ^{S9.3-1–S9.3-6}
	DBP: C-EO	
Ila	B-R	2. In adults with hypertension and CKD (stage 3 or higher or stage 1 or 2 with albuminuria [≥300 mg/d, or ≥300 mg/g albumin-to-creatinine ratio or the equivalent in the first morning void]), treatment with an ACE inhibitor is reasonable to slow kidney disease progression. ^{S9.3-3,S9.3-7–S9.3-12}
Ilb	C-EO	3. In adults with hypertension and CKD (stage 3 or higher or stage 1 or 2 with albuminuria [≥300 mg/d, or ≥300 mg/g albumin-to-creatinine ratio in the first morning void]) (S9.3-7,S9.3-8), treatment with an ARB may be reasonable if an ACE inhibitor is not tolerated.

SR indicates systematic review.

Synopsis

Refer to the “Systematic Review for the 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults” for the complete systematic evidence review for additional data and analyses.^{S9.3-13} Hypertension is the most common comorbidity affecting patients with CKD. Hypertension has been reported in 67% to 92% of patients with CKD, with increasing prevalence as kidney function declines.^{S9.3-14} Hypertension may occur as a result of kidney disease, yet the presence of hypertension may also accelerate further kidney injury; therefore, treatment is an important means to prevent further kidney functional decline. This tight interaction has led to extensive debate about the optimal BP target for patients with CKD.^{S9.3-15–S9.3-18} Masked hypertension may occur in up to 30% of patients with CKD and portends higher risk of CKD

progression.^{S9.3-19–S9.3-23} CKD is an important risk factor for CVD,^{S9.3-24} and the coexistence of hypertension and CKD further increases the risk of adverse CVD and cerebrovascular events, particularly when proteinuria is present.^{S9.3-25} Even as the importance of hypertension treatment is widely accepted, data supporting BP targets in CKD are limited, as patients with CKD were historically excluded from clinical trials. Furthermore, CKD is not included in the CVD risk calculations used to determine suitability for most clinical trials.^{S9.3-26–S9.3-28}

Until publication of the SPRINT results, most guidelines for BP targets in patients with CKD favored treatment to a BP <140/90 mm Hg,^{S9.3-15} with consideration of the lower target of <130/80 mm Hg for those with more severe proteinuria (≥ 300 mg albuminuria in 24 hours or the equivalent), if tolerated.^{S9.3-16–S9.3-18} Patients with stage 3 to 4 CKD (eGFR of 20 to <60 mL/minute/1.73 m²) comprised 28% of the SPRINT study population, and in this group intensive BP management seemed to provide the same benefits for reduction in the CVD composite primary outcome and all-cause mortality as were seen in the full study cohort. Given that most patients with CKD die from CVD complications, this RCT evidence supports a lower target of <130/80 mm Hg for all patients with CKD (Figure 6). It is appropriate to acknowledge that many patients with CKD have additional comorbidities and evidence of frailty that caused them to be excluded from past clinical trials. Observational studies of CKD cohorts indicate a higher risk of mortality at lower systolic pressures and a flat relationship of SBP to event risk in elderly patients with CKD,^{S9.3-29,S9.3-30} which supports concerns that these complex patients may be at greater risk of complications from intensive BP treatment and may fail to achieve benefits from lower BP targets. In contrast, in the prespecified subgroup analysis of the elderly cohort in SPRINT, frail elderly patients did sustain benefit from the lower BP target, which supports a lower goal for all patients, including those with CKD.^{S9.3-31} In this setting, incremental BP reduction may be appropriate, with careful monitoring of physical and kidney function.

An ACE inhibitor (or an ARB, in case of ACE inhibitor intolerance) is a preferred drug for treatment of hypertension if albuminuria (≥ 300 mg/day or ≥ 300 mg/g creatinine by first morning void) is present, although the evidence is mixed^{S9.3-10,S9.3-11} (Figure 6). In the course of reducing intraglomerular pressure and thereby reducing albuminuria, serum creatinine may increase up to 30% because of concurrent reduction in GFR.^{S9.3-32} Further GFR decline should be investigated and may be related to other factors, including volume contraction, use of nephrotoxic agents, or renovascular disease.^{S9.3-33} The combination of an ACE inhibitor and an ARB should be avoided because of reported harms demonstrated in several large cardiology trials^{S9.3-34,S9.3-35} and in 1 diabetic nephropathy trial.^{S9.3-36} Because of the greater risk of hyperkalemia and hypotension and lack of demonstrated benefit, the combination of an ARB (or ACE inhibitor) and a direct renin inhibitor is also contraindicated during management of patients with CKD.^{S9.3-37}

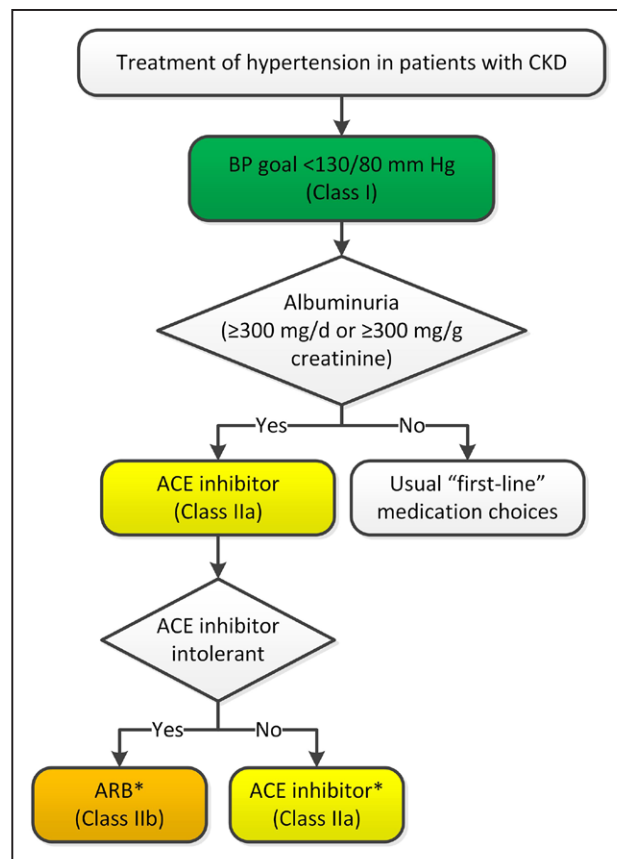


Figure 6. Management of hypertension in patients with CKD. Colors correspond to Class of Recommendation in Table 1. *CKD stage 3 or higher or stage 1 or 2 with albuminuria ≥ 300 mg/d or ≥ 300 mg/g creatinine. ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BP, blood pressure; and CKD, chronic kidney disease.

Figure 6 is an algorithm on management of hypertension in patients with CKD.

Recommendation-Specific Supportive Text

1. We recommend ASCVD risk assessment in all adults with hypertension, including those with CKD. As a matter of convenience, however, it can be assumed that the vast majority of patients with CKD have a 10-year ASCVD risk $\geq 10\%$, placing them in the high risk category that requires initiation of antihypertensive drug therapy at BP $\geq 130/80$ mm Hg (see Section 8.1.2, Figure 4 and Table 23 for BP thresholds for initiating antihypertensive drug treatment). In SPRINT, the participants with CKD who were randomized to intensive antihypertensive therapy (SBP target <120 mm Hg) appeared to derive the same beneficial reduction in CVD events and all-cause mortality that was seen in their counterparts without CKD at baseline. Likewise, intensive therapy was beneficial even in those ≥ 75 years of age with frailty or the slowest gait speed. There was no difference in the principal kidney outcome ($\geq 50\%$ decline in eGFR or ESRD) between the intensive-and standard-therapy (SBP target <140 mm Hg) groups.^{S9.3-26} Three other RCTs^{S9.3-1–S9.3-3} have evaluated the effect

of differing BP goals of <140/90 mm Hg versus 125–130/75–80 mm Hg on CKD progression in patients with CKD. None of these trials demonstrated a benefit for more intensive BP reduction, although post hoc follow-up analyses favored the lower targets in patients with more severe proteinuria,^{S9.3-38,S9.3-39} and these trials were underpowered to detect differences in CVD event rates. Recent meta-analyses and systematic reviews that included patients with CKD from SPRINT support more intensive BP treatment^{S9.3-40–S9.3-42} to reduce cardiovascular events but do not demonstrate a reduction in the rate of progression of kidney disease (doubling of serum creatinine or reaching ESRD). More intensive BP treatment may result in a modest reduction in GFR, which is thought to be primarily due to a hemodynamic effect and may be reversible. Electrolyte abnormalities are also more likely during intensive BP treatment. More intensive BP lowering in patients with CKD is also supported by a BP Lowering Treatment Trialists' Collaboration meta-analysis of RCTs in patients with CKD.^{S9.3-43}

- Evidence comes from AASK (The African American Study of Kidney Disease and Hypertension), 2 small trials (1 positive, 1 negative), and a meta-analysis.^{S9.3-3,S9.3-6,S9.3-10,S9.3-11} Albuminuria is quantified by 24-hour urine collection. A 10% to 25% increase in serum creatinine may occur in some patients with CKD as a result of ACE inhibitor therapy.
- ARBs were shown to be noninferior to ACE inhibitors in clinical trials in the non-CKD population.^{S9.3-35} A 10% to 25% increase in serum creatinine may occur in some patients with CKD as a result of ARB therapy.

9.3.1. Hypertension After Renal Transplantation

Recommendations for Treatment of Hypertension After Renal Transplantation		
References that support recommendations are summarized in Online Data Supplements 39 and 40.		
COR	LOE	Recommendations
IIa	SBP: B-NR	1. After kidney transplantation, it is reasonable to treat patients with hypertension to a BP goal of less than 130/80 mm Hg. ^{S9.3.1-1}
	DBP: C-EO	
IIa	B-R	2. After kidney transplantation, it is reasonable to treat patients with hypertension with a calcium antagonist on the basis of improved GFR and kidney survival. ^{S9.3.1-2}

Synopsis

After kidney transplantation, hypertension is common because of preexisting kidney disease, the effects of immunosuppressive medications, and the presence of allograft pathology.^{S9.3.1-3} Transplant recipients frequently harbor multiple CVD risk factors and are at high risk of CVD events. Hypertension may accelerate target organ damage and kidney function decline, particularly when proteinuria is present.^{S9.3.1-4–S9.3.1-6}

Use of calcineurin inhibitor–based immunosuppression regimens after transplantation is associated with a high (70% to 90%) prevalence of hypertension.^{S9.3.1-7} Hypertension is

less common when calcineurin inhibitors have been used without corticosteroids in liver transplantation patients,^{S9.3.1-8} although prevalence rates have not differed in steroid minimization trials after kidney transplantation.^{S9.3.1-9,S9.3.1-10} Reports from long-term belatacept-based immunosuppression studies indicate higher GFR and preservation of kidney function. However, hypertension was still present in the majority of patients, although fewer agents were needed to achieve BP goals.^{S9.3.1-11} Severity of hypertension and intensity of treatment may differ somewhat depending on the type of organ transplanted; however, most concepts relevant to kidney transplant recipients will apply to the other solid organ recipients as well.

BP targets change over time after transplantation. Initially, it is important to maintain ample organ perfusion with less stringent BP targets (<160/90 mm Hg) to avoid hypotension and risk of graft thrombosis. Beyond the first month, BP should be controlled to prevent target organ damage as in the nontransplantation setting.^{S9.3.1-12,S9.3.1-13} Hypertension after transplantation is often associated with altered circadian BP rhythm with loss of the normal nocturnal BP fall^{S9.3.1-14,S9.3.1-15} and, in some, a nocturnal BP rise. These changes may return to normal after a longer period of follow-up.^{S9.3.1-16}

Recommendation-Specific Supportive Text

- Although treatment targets for hypertension after transplantation should probably be similar to those for other patients with CKD, there are no trials in post-transplantation patients comparing different BP targets. As kidney transplant recipients generally have a single functioning kidney and CKD, BP targets should be similar to those for the general CKD population.
- Limited studies have compared drug choice for initial antihypertensive therapy in patients after kidney transplantation. On the basis of a Cochrane analysis,^{S9.3.1-2} most studies favor CCBs to reduce graft loss and maintain higher GFR, with some evidence suggesting potential harm from ACE inhibitors because of anemia, hyperkalemia, and lower GFR. In recognition of this concern, RAS inhibitors may be reserved for the subset of patients with hypertension and additional comorbidities that support the need for ACE inhibitor therapy (ie, proteinuria or HF after transplantation). With appropriate potassium and creatinine monitoring, this has been demonstrated to be safe.^{S9.3.1-17}

9.4. Cerebrovascular Disease

Stroke is a leading cause of death, disability, and dementia.^{S9.4-1} Because of its heterogeneous causes and hemodynamic consequences, the management of BP in adults with stroke is complex and challenging.^{S9.4-2} To accommodate the variety of important issues pertaining to BP management in the stroke patient, treatment recommendations require recognition of stroke acuity, stroke type, and therapeutic objectives. Future studies should target more narrowly defined questions, such as optimal BP-reduction timing and target, as well as ideal antihypertensive agent therapeutic class by patient type and event type.

9.4.1. Acute Intracerebral Hemorrhage

Recommendations for Management of Hypertension in Patients With Acute Intracerebral Hemorrhage (ICH)		
References that support recommendations are summarized in Online Data Supplement 41.		
COR	LOE	Recommendations
Ila	C-EO	1. In adults with ICH who present with SBP greater than 220 mm Hg, it is reasonable to use continuous intravenous drug infusion (Table 19) and close BP monitoring to lower SBP.
III: Harm	A	2. Immediate lowering of SBP (Table 19) to less than 140 mm Hg in adults with spontaneous ICH who present within 6 hours of the acute event and have an SBP between 150 mm Hg and 220 mm Hg is not of benefit to reduce death or severe disability and can be potentially harmful. ^{S9.4.1-1,S9.4.1-2}

Synopsis

Spontaneous, nontraumatic ICH is a significant global cause of morbidity and mortality.^{S9.4.1-3} Elevated BP is highly prevalent in the setting of acute ICH and is linked to greater hematoma expansion, neurological worsening, and death and dependency after ICH.

Figure 7 is an algorithm on management of hypertension in patients with acute ICH.

Recommendation-Specific Supportive Text

1. Information about the safety and effectiveness of early intensive BP-lowering treatment is least well established for patients with markedly elevated BP (sustained SBP >220 mm Hg) on presentation, patients with large and severe ICH, or patients requiring surgical decompression. However, given the consistent nature of the data linking high BP with poor clinical outcomes^{S9.4.1-4-S9.4.1-6} and some suggestive data for treatment in patients with modestly high initial SBP levels,^{S9.4.1-1,S9.4.1-7} early

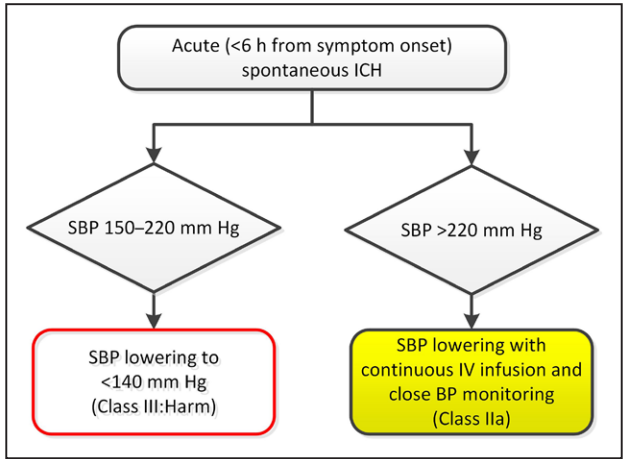


Figure 7. Management of hypertension in patients with acute ICH. Colors correspond to Class of Recommendation in Table 1. BP indicates blood pressure; ICH, intracerebral hemorrhage; IV, intravenous; and SBP, systolic blood pressure.

lowering of SBP in ICH patients with markedly high SBP levels (>220 mm Hg) might be sensible. A secondary endpoint in 1 RCT and an overview of data from 4 RCTs indicate that intensive BP reduction, versus BP-lowering guideline treatment, is associated with greater functional recovery at 3 months.^{S9.4.1-1,S9.4.1-7}

2. RCT data have suggested that immediate BP lowering (to <140/90 mm Hg) within 6 hours of an acute ICH was feasible and safe,^{S9.4.1-1,S9.4.1-8,S9.4.1-9} may be linked to greater attenuation of absolute hematoma growth at 24 hours,^{S9.4.1-7} and might be associated with modestly better functional recovery in survivors.^{S9.4.1-1,S9.4.1-7} However, a recent RCT^{S9.4.1-2} that examined immediate BP lowering within 4.5 hours of an acute ICH found that treatment to achieve a target SBP of 110 to 139 mm Hg did not lead to a lower rate of death or disability than standard reduction to a target of 140 to 179 mm Hg. Moreover, there were significantly more renal adverse events within 7 days after randomization in the intensive-treatment group than in the standard-treatment group.^{S9.4.1-2} Put together, neither of the 2 key trials^{S9.4.1-1,S9.4.1-2} evaluating the effect of lowering SBP in the acute period after spontaneous ICH met their primary outcomes of reducing death and severe disability at 3 months.

9.4.2. Acute Ischemic Stroke

Recommendations for Management of Hypertension in Patients With Acute Ischemic Stroke		
References that support recommendations are summarized in Online Data Supplement 42.		
COR	LOE	Recommendations
I	B-NR	1. Adults with acute ischemic stroke and elevated BP who are eligible for treatment with intravenous tissue plasminogen activator should have their BP slowly lowered to less than 185/110 mm Hg before thrombolytic therapy is initiated. ^{S9.4.2-1,S9.4.2-2}
I	B-NR	2. In adults with an acute ischemic stroke, BP should be less than 185/110 mm Hg before administration of intravenous tissue plasminogen activator and should be maintained below 180/105 mm Hg for at least the first 24 hours after initiating drug therapy. ^{S9.4.2-3}
Ila	B-NR	3. Starting or restarting antihypertensive therapy during hospitalization in patients with BP greater than 140/90 mm Hg who are neurologically stable is safe and reasonable to improve long-term BP control, unless contraindicated. ^{S9.4.2-4,S9.4.2-5}
IIb	C-EO	4. In patients with BP of 220/120 mm Hg or higher who did not receive intravenous alteplase or endovascular treatment and have no comorbid conditions requiring acute antihypertensive treatment, the benefit of initiating or reinitiating treatment of hypertension within the first 48 to 72 hours is uncertain. It might be reasonable to lower BP by 15% during the first 24 hours after onset of stroke.

Recommendations for Management of Hypertension in Patients With Acute Ischemic Stroke (Continued)		
COR	LOE	Recommendations
III: No Benefit	A	5. In patients with BP less than 220/120 mm Hg who did not receive intravenous thrombolysis or endovascular treatment and do not have a comorbid condition requiring acute antihypertensive treatment, initiating or reinitiating treatment of hypertension within the first 48 to 72 hours after an acute ischemic stroke is not effective to prevent death or dependency. ^{S9.4.2-4–S9.4.2-9}

Synopsis

Elevated BP is common during acute ischemic stroke (occurring in up to 80% of patients), especially among patients with a history of hypertension.^{S9.4.2-10} However, BP often decreases spontaneously during the acute phase of ischemic stroke, as soon as 90 minutes after the onset of symptoms. Countervailing theoretical concerns about arterial hypertension during acute ischemic stroke include aiming to enhance cerebral perfusion of the ischemic tissue while minimizing the exacerbation of brain edema and hemorrhagic transformation of the ischemic tissue.^{S9.4.2-11,S9.4.2-12} Some studies have shown a U-shaped relationship between the admission BP and favorable clinical outcomes, with an optimal SBP and DBP ranging from 121 to 200 mm Hg and 81 to 110 mm Hg, respectively.^{S9.4.2-13} It is conceivable that an optimal arterial BP range exists during acute ischemic stroke on an individual basis, contingent on the ischemic

stroke subtype and other patient-specific comorbidities. Early initiation or resumption of antihypertensive treatment after acute ischemic stroke is indicated only in specific situations: 1) patients treated with tissue-type plasminogen activator,^{S9.4.2-1,S9.4.2-2} and 2) patients with SBP >220 mm Hg or DBP >120 mm Hg. For the latter group, it should be kept in mind that cerebral autoregulation in the ischemic penumbra of the stroke is grossly abnormal and that systemic perfusion pressure is needed for blood flow and oxygen delivery. Rapid reduction of BP, even to lower levels within the hypertensive range, can be detrimental. For all other acute ischemic stroke patients, the advantage of lowering BP early to reduce death and dependency is uncertain,^{S9.4.2-4–S9.4.2-9} but restarting antihypertensive therapy to improve long-term BP control is reasonable after the first 24 hours for patients who have preexisting hypertension and are neurologically stable.^{S9.4.2-4,S9.4.2-5,S9.4.2-14}

Figure 8 is an algorithm on management of hypertension in patients with acute ischemic stroke.

Recommendation-Specific Supportive Text

1. These BP cutoffs correspond to study inclusion criteria in pivotal clinical trials of intravenous thrombolysis for acute ischemic stroke.^{S9.4.2-1}
2. In a large observational study of patients with acute ischemic stroke who received intravenous tissue-type plasminogen activator, high BP during the initial 24 hours was linked to greater risk of symptomatic ICH.^{S9.4.2-3}
3. For the goal of antihypertensive therapy, see Section 8.1.5.
4. Extreme arterial hypertension is detrimental because it can lead to encephalopathy, cardiac compromise, and

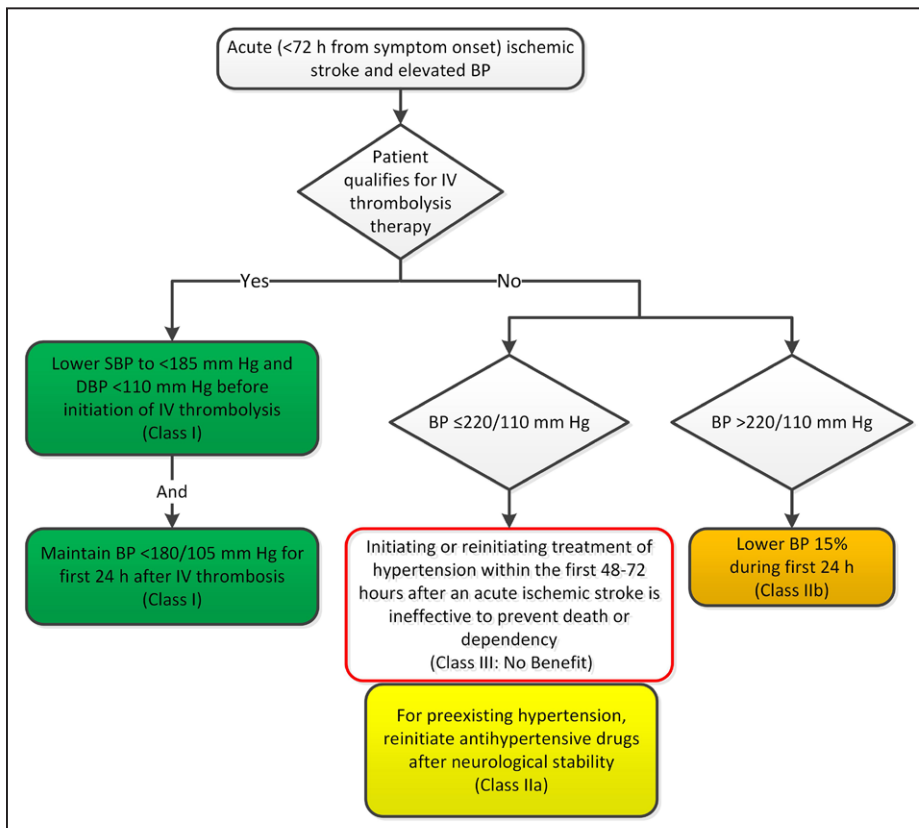


Figure 8. Management of hypertension in patients with acute ischemic stroke. Colors correspond to Class of Recommendation in Table 1. BP indicates blood pressure; DBP, diastolic blood pressure; IV, intravenous; and SBP, systolic blood pressure.

renal damage. However, hypotension, especially when too rapidly achieved, is potentially harmful because it abruptly reduces perfusion to multiple organs, including the brain.

5. Data from 2 RCTs,^{S9.4.2-5,S9.4.2-9} as well as systematic reviews and meta-analyses,^{S9.4.2-6-S9.4.2-8} indicate that antihypertensive agents reduce BP during the acute phase of an ischemic stroke but do not confer benefit with regard to short- and long-term dependency and mortality rate. One RCT did not demonstrate a benefit of continuing pre-stroke antihypertensive drugs during the first few days after an acute stroke, but it was substantially underpowered to answer the question.^{S9.4.2-4}

9.4.3. Secondary Stroke Prevention

Recommendations for Treatment of Hypertension for Secondary Stroke Prevention		
References that support recommendations are summarized in Online Data Supplements 43 and 44.		
COR	LOE	Recommendations
I	A	1. Adults with previously treated hypertension who experience a stroke or transient ischemic attack (TIA) should be restarted on antihypertensive treatment after the first few days of the index event to reduce the risk of recurrent stroke and other vascular events. ^{S9.4.3-1-S9.4.3-3}
I	A	2. For adults who experience a stroke or TIA, treatment with a thiazide diuretic, ACE inhibitor, or ARB, or combination treatment consisting of a thiazide diuretic plus ACE inhibitor, is useful. ^{S9.4.3-1,S9.4.3-3-S9.4.3-5}
I	B-R	3. Adults not previously treated for hypertension who experience a stroke or TIA and have an established BP of 140/90 mm Hg or higher should be prescribed antihypertensive treatment a few days after the index event to reduce the risk of recurrent stroke and other vascular events. ^{S9.4.3-1-S9.4.3-3}
I	B-NR	4. For adults who experience a stroke or TIA, selection of specific drugs should be individualized on the basis of patient comorbidities and agent pharmacological class. ^{S9.4.3-6}
IIb	B-R	5. For adults who experience a stroke or TIA, a BP goal of less than 130/80 mm Hg may be reasonable. ^{S9.4.3-6,S9.4.3-7}
IIb	B-R	6. For adults with a lacunar stroke, a target SBP goal of less than 130 mm Hg may be reasonable. ^{S9.4.3-8}
IIb	C-LD	7. In adults previously untreated for hypertension who experience an ischemic stroke or TIA and have a SBP less than 140 mm Hg and a DBP less than 90 mm Hg, the usefulness of initiating antihypertensive treatment is not well established. ^{S9.4.3-9}

Synopsis

Each year in the United States, >750000 adult patients experience a stroke, of which up to 25% are recurrent strokes.^{S9.4.3-10} For an individual who experiences an initial stroke or TIA, the annual risk of a subsequent or “secondary” stroke is approximately 4%,^{S9.4.3-11} and the case mortality rate is 41% after a recurrent stroke versus 22% after an initial stroke.^{S9.4.3-12} Among patients with a recent stroke or TIA, the prevalence of pre-morbid hypertension is approximately 70%.^{S9.4.3-13} Risk of recurrent stroke is heightened by presence of elevated BP, and guideline-recommended antihypertensive drug treatment to lower BP has been linked to a reduction in 1-year recurrent stroke risk.^{S9.4.3-14} RCT meta-analyses show an approximately 30% decrease in recurrent stroke risk with BP-lowering therapies.^{S9.4.3-1-S9.4.3-3} An issue frequently raised by clinicians is whether the presence of clinically asymptomatic cerebral infarction incidentally noted on brain imaging (computed tomography or MRI scan) in patients without a history of or symptoms of a stroke or TIA warrants implementation of secondary stroke prevention measures. Clinically asymptomatic vascular brain injury is increasingly being considered as an entry point for secondary stroke prevention therapies, because these apparently “silent” brain infarctions are associated with typical stroke risk factors, accumulatively lead to subtle neurological impairments, and bolster risk of future symptomatic stroke events.^{S9.4.3-15} Although the evidence for using antihypertensive treatment to prevent recurrent stroke in stroke patients with elevated BP is compelling,^{S9.4.3-1-S9.4.3-3} questions remain about when precisely after an index stroke to initiate it, what specific agent(s) to use (if any), which therapeutic targets to aim for, and whether the treatment approach should vary by index stroke mechanism and baseline level of BP.^{S9.4.3-16}

Figure 9 is an algorithm on management of hypertension in patients with a previous history of stroke (secondary stroke prevention).

Recommendation-Specific Supportive Text

1. Two overviews of RCTs published through 2009 showed that antihypertensive medications lowered the risk of recurrent vascular events in patients with stroke or TIA.^{S9.4.3-1-S9.4.3-3}
2. Specific agents that have shown benefit in either dedicated RCTs or systematic reviews of RCT data include diuretics, ACE inhibitors, and ARBs.
3. Support for this recommendation is based on data from 2 dedicated RCTs, as well as a systematic review and meta-analysis, among patients with a history of stroke or TIA.^{S9.4.3-1-S9.4.3-3}
4. Reduction in BP appears to be more important than the choice of specific agents used to achieve this goal. Thus, if diuretic and ACE inhibitor or ARB treatment do not achieve BP target, other agents, such as CCB and/or mineralocorticoid receptor antagonist, may be added.
5. An overview of RCTs showed that larger reductions in SBP tended to be associated with greater reduction in risk of recurrent stroke. However, a separate overview of RCTs in patients who experienced a stroke noted that achieving an SBP level <130 mm Hg was not associated with a lower stroke risk, and several observational

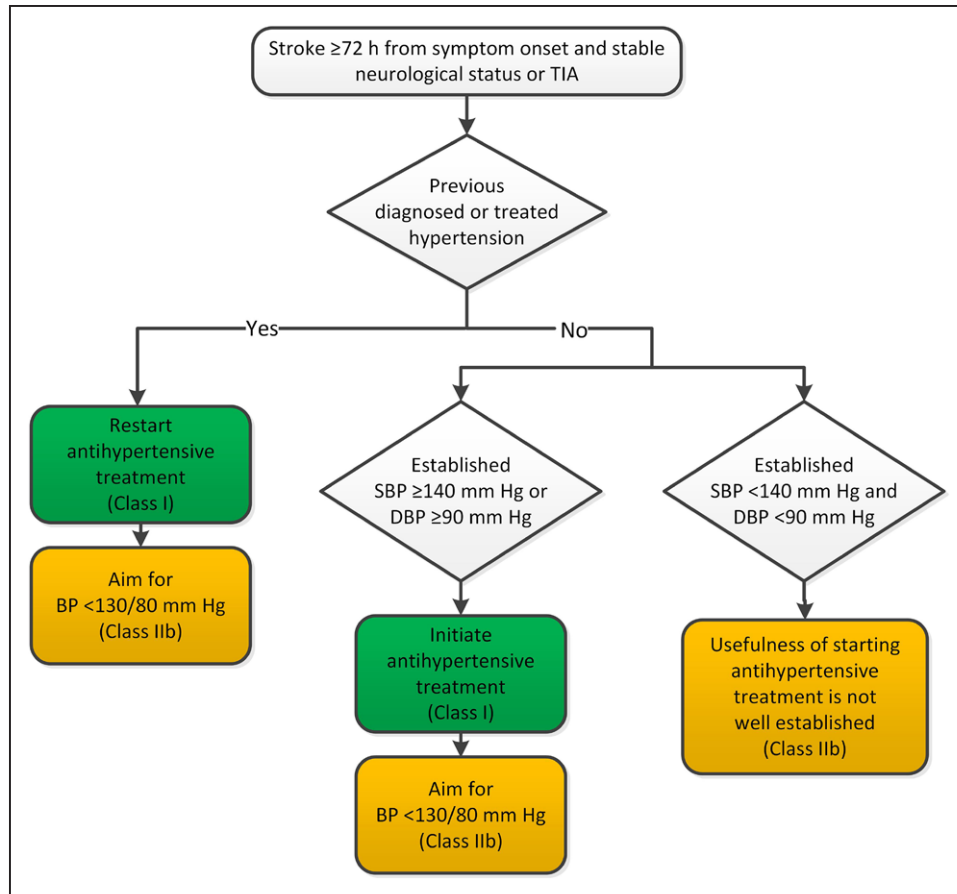


Figure 9. Management of hypertension in patients with a previous history of stroke (secondary stroke prevention). Colors correspond to Class of Recommendation in Table 1. DBP indicates diastolic blood pressure; SBP, systolic blood pressure; and TIA, transient ischemic attack.

studies did not show benefit with achieved SBP levels <120 mm Hg.^{S9.4.3-5}

6. Patients with a lacunar stroke treated to an SBP target of <130 mm Hg versus 130 to 140 mm Hg may be less likely to experience a future ICH.
7. No published RCTs have specifically addressed this question, but a post hoc analysis of an RCT suggests that the effectiveness of antihypertensive treatment for secondary stroke prevention diminishes as initial baseline BP declines.^{S9.4.3-9}

9.5. Peripheral Artery Disease

Recommendation for Treatment of Hypertension in Patients With PAD

References that support the recommendation are summarized in Online Data Supplement 45.

COR	LOE	Recommendation
I	B-NR	1. Adults with hypertension and PAD should be treated similarly to patients with hypertension without PAD. ^{S9.5-1–S9.5-4}

Synopsis

Patients with PAD are at increased risk of CVD and stroke. Hypertension is a major risk factor for PAD, so these patients

are commonly enrolled in trials of antihypertensive drug therapy. However, patients with PAD typically comprise a small fraction of participants, so in the few trials that report results in patients with PAD, subgroup analyses are generally underpowered.

Recommendation-Specific Supportive Text

1. There is no major difference in the relative risk reduction in CVD from BP-lowering therapy between patients with hypertension and PAD and patients without PAD.^{S9.5-1} There is also no evidence that any one class of antihypertensive medication or strategy is superior.^{S9.5-2–S9.5-4} In the INVEST (International Verapamil-Trandolapril) study, the beta blocker atenolol (with or without hydrochlorothiazide) was compared with the CCB verapamil (with or without perindopril). The study showed no significant difference in CVD outcomes between the 2 drug regimens in patients with and without PAD.^{S9.5-3} No trials have reported the effects of a higher versus a lower BP goal in patients with PAD. In the 1 trial (ALLHAT) that reported the effects of different classes of BP medications on PAD as an outcome, there was no significant difference by medication class.^{S9.5-5}

9.6. Diabetes Mellitus

Recommendations for Treatment of Hypertension in Patients With DM		
References that support recommendations are summarized in Online Data Supplements 46 and 47 and Systematic Review Report.		
COR	LOE	Recommendations
I	SBP: B-R ^{SR}	1. In adults with DM and hypertension, antihypertensive drug treatment should be initiated at a BP of 130/80 mm Hg or higher with a treatment goal of less than 130/80 mm Hg. ^{S9.6-1–S9.6-8}
	DBP: C-EO	
I	A ^{SR}	2. In adults with DM and hypertension, all first-line classes of antihypertensive agents (ie, diuretics, ACE inhibitors, ARBs, and CCBs) are useful and effective. ^{S9.6-1,S9.6-9,S9.6-10}
I b	B-NR	3. In adults with DM and hypertension, ACE inhibitors or ARBs may be considered in the presence of albuminuria. ^{S9.6-11,S9.6-12}

SR indicates systematic review.

Synopsis

Refer to the “Systematic Review for the 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults” for the complete systematic evidence review for additional data and analyses.^{S9.6-13} The prevalence of hypertension among adults with DM is approximately 80%, and hypertension is at least twice as common in persons with type 2 DM than in age-matched individuals without DM.^{S9.6-14–S9.6-16} The coexistence of hypertension and DM markedly increases the risk of developing CVD damage, resulting in a higher incidence of CHD, HF, PAD, stroke, and CVD mortality,^{S9.6-17} and may increase risk of microvascular disease, such as nephropathy or retinopathy.^{S9.6-16,S9.6-18}

There is limited quality evidence to determine a precise BP target in adults with DM. No RCTs have explicitly 1) documented whether treatment to an SBP goal <140 mm Hg versus a higher goal improves clinical outcomes in adults with hypertension and DM or 2) directly evaluated clinical outcomes associated with SBP <130 mm Hg.^{S9.6-2} However, 2 high-quality systematic reviews of RCTs support an SBP target of <140 mm Hg.^{S9.6-4,S9.6-7}

There is little or no available RCT evidence supporting a specific DBP threshold for initiation of pharmacological therapy. Several RCTs, including the HOT (Hypertension Optimal Treatment) trial, UKPDS (United Kingdom Prospective Diabetes Study), and ABCD (Appropriate Blood Pressure Control in Diabetes) trial,^{S9.6-19–S9.6-22} are often cited to support a lower DBP target (eg, ≤85 or 80 mm Hg) for adults with hypertension and DM. However, these trials were conducted when the diagnostic criteria for DM were more conservative than they are currently (2 fasting glucose levels >140 mg/dL as opposed to 126 mm/dL today).

Recommendation-Specific Supportive Text

1. We recommend ASCVD risk assessment in all adults with hypertension, including adults with DM. As a

matter of convenience, however, it can be assumed that the vast majority of adults with DM have a 10-year ASCVD risk ≥ 10%, placing them in the high risk category that requires initiation of antihypertensive drug therapy at BP ≥ 130/80 mm Hg (see Section 8.1.2, Figure 4 and Table 23 for BP thresholds for initiating antihypertensive drug treatment). The ACCORD trial,^{S9.6-5} which compared CVD outcomes in adults with DM and hypertension who were randomized to an SBP target of <140 mm Hg (standard therapy) or <120 mm Hg (intensive therapy), did not document a significant reduction in the primary outcome (CVD composite) with the lower BP goal, but the trial was underpowered to detect a statistically significant difference between the 2 treatment arms. The ACCORD trial demonstrated a small reduction in absolute risk (1.1%) for stroke, but there were few such events. More adverse events (2% increase in absolute risk) were identified in the lower BP group, especially self-reported hypotension and a reduction in estimated GFR, but these did not result in an excess of stroke or ESRD. The ACCORD trial was a factorial study; secondary analysis demonstrated a significant outcome benefit in the intensive BP/standard glycemic group,^{S9.6-3} but benefit in the intensive BP/intensive glycemic control group was no better than in the intensive BP/standard glycemic control group, which suggests a floor benefit beyond which the combined intensive interventions were ineffective.^{S9.6-5} An ACCORD secondary analysis suggested that an SBP <120 mm Hg is superior to standard BP control in reducing LVH.^{S9.6-6}

A meta-analysis of 73 913 patients with DM reported that an SBP <130 mm Hg reduced stroke by 39%. However, there was no significant risk reduction for MI.^{S9.6-23} Two meta-analyses addressing target BP in adults with DM restricted the analysis to RCTs that randomized patients to different BP levels.^{S9.6-4,S9.6-7} Target BP of 133/76 mm Hg provided significant benefit compared with that of 140/81 mm Hg for major cardiovascular events, MI, stroke, albuminuria, and retinopathy progression.^{S9.6-4} Several meta-analyses of RCTs included all trials with a difference in BP,^{S9.6-24,S9.6-25} but 2 restricted their analyses to trials in which participants were randomized to different BP target levels.^{S9.6-4,S9.6-7}

SPRINT demonstrated cardiovascular benefit from intensive treatment of BP to a goal of <120 mm Hg as compared with <140 mm Hg but did not include patients with DM. However, the results of ACCORD and SPRINT were generally consistent.^{S9.6-26} In addition, a SPRINT substudy demonstrated that patients with prediabetes derived a benefit similar to that of patients with normoglycemia.^{S9.6-8} Previous trials have shown similar quantitative benefits from lowering BP in persons with and without DM.^{S9.6-9}

2. BP control is more difficult to achieve in patients with DM than in those without DM, necessitating use of combination therapy in the majority of patients.^{S9.6-27} All major antihypertensive drug classes (ie, ACE inhibitors, ARBs, CCBs, and diuretics) are useful in the treatment of hypertension in DM.^{S9.6-1,S9.6-9} However, in ALLHAT, doxazosin was clearly inferior to chlorthalidone, which

also reduced some events more than amlodipine or lisinopril.^{S9.6-28}

3. ACE inhibitors and ARBs have the best efficacy among the drug classes on urinary albumin excretion^{S9.6-12} (see Section 9.3). Therefore, an ACE inhibitor or ARB may be considered as part of the combination. A meta-analysis of RCTs of primary prevention of albuminuria in patients with DM demonstrated a significant reduction in progression of moderately to severely increased albuminuria with the use of ACE inhibitors or ARBs.^{S9.6-11}

9.7. Metabolic Syndrome

Metabolic syndrome is a state of metabolic dysregulation characterized by visceral fat accumulation, insulin resistance, hyperinsulinemia, and hyperlipidemia, as well as predisposition to type 2 DM, hypertension, and atherosclerotic CVD.^{S9.7-1-S9.7-3} According to data from the NHANES III and NHANES 1999–2006,^{S9.7-1,S9.7-4} the prevalence of metabolic syndrome in the United States was 34.2% in 2006 and has likely increased substantially since that time. The metabolic syndrome is linked to several other disorders, including non-alcoholic steatohepatitis, polycystic ovary syndrome, certain cancers, CKD, Alzheimer's disease, Cushing's syndrome, lipodystrophy, and hyperalimentation.^{S9.7-5,S9.7-6}

Lifestyle modification, with an emphasis on improving insulin sensitivity by means of dietary modification, weight reduction, and exercise, is the foundation of treatment of the metabolic syndrome. The optimal antihypertensive drug therapy for patients with hypertension in the setting of the metabolic syndrome has not been clearly defined.^{S9.7-1} Although caution exists with regard to the use of thiazide diuretics in this population because of their ability to increase insulin resistance, dyslipidemia, and hyperuricemia and to accelerate conversion to overt DM, no data are currently available demonstrating deterioration in cardiovascular or renal outcomes in patients treated with these agents.^{S9.7-1} Indeed, as shown in follow-up of ALLHAT, chlorthalidone use was associated with only a small increase in fasting glucose levels (1.5–4.0 mg/dL), and this increase did not translate into increased CVD risk at a later date.^{S9.7-7-S9.7-10}

In addition, in post hoc analysis of the nearly two thirds of participants in ALLHAT that met criteria for the metabolic syndrome, chlorthalidone was unsurpassed in reducing CVD and renal outcomes compared with lisinopril, amlodipine, or doxazosin.^{S9.7-9,S9.7-11} Similarly, high-dose ARB therapy reduces arterial stiffness in patients with hypertension with the metabolic syndrome, but no outcomes data are available from patients in which this form of treatment was used.^{S9.7-12} Use of traditional beta blockers may lead to dyslipidemia or deterioration of glucose tolerance, and ability to lose weight.^{S9.7-2} In several large clinical trials, the risk of developing DM as a result of traditional beta-blocker therapy was 15% to 29%.^{S9.7-2} However, the newer vasodilating beta blockers (eg, labetalol, carvedilol, nebivolol) have shown neutral or favorable effects on metabolic profiles compared with the traditional beta blockers.^{S9.7-13} Trials using vasodilator beta blockers have not been performed to demonstrate effects on CVD outcomes.

9.8. Atrial Fibrillation

Recommendation for Treatment of Hypertension in Patients With AF

References that support the recommendation are summarized in Online Data Supplement 48.

COR	LOE	Recommendation
Ila	B-R	1. Treatment of hypertension with an ARB can be useful for prevention of recurrence of AF. ^{S9.8-1,S9.8-2}

Synopsis

AF and hypertension are common and often coexistent conditions, both of which increase in frequency with age. AF occurs in 3% to 4% of the population >65 years of age.^{S9.8-3} Hypertension is present in >80% of patients with AF and is by far the most common comorbid condition, regardless of age.^{S9.8-4} AF is associated with systemic thromboembolism, as recognized in the CHADS2 and CHA2DS2-VASc scoring systems for stroke risk.^{S9.8-5} It is also associated with gradual worsening of ventricular function, the subsequent development of HF, and increased mortality.

Hypertension has long been recognized as a risk factor for AF because it is associated with LVH, decreased diastolic function with impaired LV filling, rising left atrial pressures with left atrial hypertrophy and enlargement, increased atrial fibrosis, and slowing of intra-atrial and interatrial electrical conduction velocities. Such a distortion of atrial anatomy and physiology increases the incidence of AF.^{S9.8-6} Left atrial pressure also increases with ischemic or valvular heart disease and myopathies that are often associated with systemic hypertension, potentially leading to AF.

Although management of AF will continue to revolve around restoration of sinus rhythm when appropriate, rate control when it is not, and anticoagulation, control of hypertension is a key component of therapy.^{S9.8-1,S9.8-2}

Treatment of hypertension may prevent new-onset AF, especially in patients with LVH or LV dysfunction.^{S9.8-1} Five RCTs have compared the value of antihypertensive agents for reduction of new-onset AF.^{S9.8-7-S9.8-11} One study suggested superiority of RAS blockade over a CCB,^{S9.8-8} and another reported superiority of RAS blockade over a beta blocker that is no longer recommended for treatment of hypertension.^{S9.8-9} In the largest trial, there was no difference in incident AF among adults with hypertension assigned to first-step therapy with a diuretic, ACE inhibitor, or CCB.^{S9.8-10} In ALLHAT, the incidence of AF was 23% higher during first-step antihypertensive therapy with the alpha-receptor blocker doxazosin than with chlorthalidone. Furthermore, the occurrence of AF or atrial flutter during the study, either new onset or recurrent, was associated with an increase in mortality of nearly 2.5-fold.^{S9.8-10}

Recommendation-Specific Supportive Text

1. Although RAS blockade in theory is the treatment of choice for hypertension in patients with prior AF, relative to other classes of agents, all of the trials that have shown clinical superiority of ARBs over other agents were comparisons with CCBs or beta blockers that are no longer recommended

as first-line agents for treatment of hypertension.^{S9,8-2} There are no available trials comparing ACE inhibitors with other drugs or any RAS-blocking agents with diuretics.

9.9. Valvular Heart Disease

Recommendations for Treatment of Hypertension in Patients With Valvular Heart Disease		
References that support recommendations are summarized in Online Data Supplements 49 and 50.		
COR	LOE	Recommendations
I	B-NR	1. In adults with asymptomatic aortic stenosis, hypertension should be treated with pharmacotherapy, starting at a low dose and gradually titrating upward as needed. ^{S9,9-1-S9,9-4}
Ila	C-LD	2. In patients with chronic aortic insufficiency, treatment of systolic hypertension with agents that do not slow the heart rate (ie, avoid beta blockers) is reasonable. ^{S9,9-5,S9,9-6}

Recommendation-Specific Supportive Text

1. Hypertension is a risk factor for the development of aortic stenosis (stage A [eg, aortic sclerosis or bicuspid aortic valve]) and asymptomatic aortic stenosis (stage B [progressive asymptomatic aortic stenosis]). The combination of hypertension and aortic stenosis, “2 resistors in series,” increases the rate of complications. In patients with asymptomatic mild-to-moderate aortic stenosis, hypertension has been associated with more abnormal LV structure and increased cardiovascular morbidity and mortality.^{S9,9-1} There is no evidence that antihypertensive medications will produce an inordinate degree of hypotension in patients with aortic stenosis. Nitroprusside infusion in hypertensive patients with severe aortic stenosis lowers pulmonary and systemic resistance, with improvements in stroke volume and LV end-diastolic pressure.^{S9,9-2} Thus, careful use of antihypertensive agents to achieve BP control in patients with hypertension and aortic stenosis is beneficial. Although there are no specific trials comparing various classes of antihypertensive agents, RAS blockade may be advantageous because of the potentially beneficial effects on LV fibrosis,^{S9,9-3} control of hypertension, reduction of dyspnea, and improved effort tolerance.^{S9,9-4} Diuretics should be used sparingly in patients with small LV chamber dimensions. Beta blockers may be appropriate for patients with aortic stenosis who have reduced ejection fraction, prior MI, arrhythmias, or angina pectoris. In patients with moderate or severe aortic stenosis, consultation or co-management with a cardiologist is preferred for hypertension management.
2. Vasodilator therapy can reduce the LV volume and mass and improve LV performance in patients with aortic regurgitation,^{S9,9-5} but improvement of long-term clinical outcomes, such as time to valve replacement, have been variable.^{S9,9-5,S9,9-6} Beta blockers may result in increased diastolic filling period because of bradycardia, potentially causing increased aortic insufficiency. Marked reduction in DBP may lower coronary perfusion pressure in patients with chronic severe aortic regurgitation

(stage B [progressive asymptomatic aortic regurgitation] and stage C [asymptomatic severe AR]). However, there are no outcomes data to support these theoretical concerns.

9.10. Aortic Disease

Recommendation for Management of Hypertension in Patients With Aortic Disease		
COR	LOE	Recommendation
I	C-EO	1. Beta blockers are recommended as the preferred antihypertensive agents in patients with hypertension and thoracic aortic disease. ^{S9,10-1,S9,10-2}

Synopsis

Thoracic aortic aneurysms are generally asymptomatic until a person presents with a sudden catastrophic event, such as an aortic dissection or rupture, which is rapidly fatal in the majority of patients.^{S9,10-3,S9,10-4} The rationale for antihypertensive therapy is based largely on animal and observational studies associating hypertension with aortic dissection.^{S9,10-5,S9,10-6} RCTs specifically addressing hypertension and aortic disease are not available, and trials in patients with primary hypertension do not provide insight on either the optimal BP target or choice of antihypertensive drug therapy in patients with thoracic aortic aneurysm, aortic dissection, or aortic disease.^{S9,10-7,S9,10-8} A study in 20 humans with hypertension suggested that hypertension is associated with significant changes in the mechanical properties of the aortic wall, with more strain-induced stiffening in hypertension than in normotension, which may reflect destruction of elastin and predisposition to aortic dissection in the presence of hypertension.^{S9,10-9} In a retrospective observational study, high BP variability was an independent risk factor for the prognosis of aortic dissection.^{S9,10-10} Recommendations for treatment of acute aortic dissection are provided in Section 11.2.

Recommendation-Specific Supportive Text

1. In patients with chronic aortic dissection, observational studies suggest lower risk for operative repair with beta-blocker therapy.^{S9,10-1} In a series of patients with type A and type B aortic dissections, beta blockers were associated with improved survival in both groups, whereas ACE inhibitors did not improve survival.^{S9,10-2}

10. Special Patient Groups

Special attention is needed for specific patient subgroups.

10.1. Race and Ethnicity

In the United States, at any decade of life, blacks have a higher prevalence of hypertension than that of Hispanic Americans, whites, Native Americans, and other subgroups defined by race and ethnicity (see Section 3.3). Hypertension control rates are lower for blacks, Hispanic Americans, and Asian Americans than for whites.^{S10,1-1} Among men with hypertension, non-Hispanic white (53.8%) adults had a higher prevalence of controlled high blood pressure than did non-Hispanic black (43.8%), non-Hispanic Asian (39.9%), and Hispanic (43.5%) adults. For women with hypertension, the percentage

of non-Hispanic white (59.1%) adults with controlled high blood pressure was higher than among non-Hispanic black (52.3%) and non-Hispanic Asian (46.8%) adults.^{S10.1-1} In Hispanic Americans, the lower control rates result primarily from lack of awareness and treatment,^{S10.1-2,S10.1-3} whereas in blacks, awareness and treatment are at least as high as in whites, but hypertension is more severe and some agents are less effective at BP control.^{S10.1-4} Morbidity and mortality attributed to hypertension are also more common in blacks and Hispanic Americans than in whites. Blacks have a 1.3-times greater risk of nonfatal stroke, 1.8-times greater risk of fatal strokes, 1.5-times greater risk of HF, and 4.2-times greater risk of ESRD.^{S10.1-4} Hispanic Americans have lower rates of hypertension awareness and treatment than those of whites and blacks, as well as a high prevalence of comorbid CVD risk factors (eg, obesity, DM). In 2014, age-adjusted hypertension-attributable mortality rates per 1000 persons for non-Hispanic white, non-Hispanic black, and Hispanic-American men and women were 19.3 and 15.8, 50.1 and 35.6, and 19.1 and 14.6, respectively.^{S10.1-5} However, Hispanics in the United States are a heterogeneous subgroup, and rates of both hypertension and its consequences vary according to whether their ancestry is from the Caribbean, Mexico, Central or South America, or Europe.^{S10.1-6–S10.1-8} Hispanics from Mexico and Central America have lower CVD rates than US whites, whereas those of Caribbean origin have higher rates. Thus, pooling of data for Hispanics may not accurately reflect risk in a given patient. Finally, the excess risk of CKD outcomes in at least some blacks with hypertension may be due to the presence of high-risk APOL1 (apolipoprotein L1) genetic variants.^{S10.1-9–S10.1-11} The rate of renal decline associated with this genotype appears to be largely unresponsive to either BP lowering or RAS inhibition.^{S10.1-9–S10.1-12}

10.1.1. Racial and Ethnic Differences in Treatment

Recommendations for Race and Ethnicity		
References that support recommendations are summarized in Online Data Supplement 51.		
COR	LOE	Recommendations
I	B-R	1. In black adults with hypertension but without HF or CKD, including those with DM, initial antihypertensive treatment should include a thiazide-type diuretic or CCB. ^{S10.1.1-1–S10.1.1-4}
I	C-LD	2. Two or more antihypertensive medications are recommended to achieve a BP target of less than 130/80 mm Hg in most adults with hypertension, especially in black adults with hypertension. ^{S10.1.1-5–S10.1.1-7}

Synopsis

Lifestyle modification (ie, weight reduction, dietary modification, and increased physical activity) is particularly important in blacks and Hispanic Americans for prevention and first-line or adjunctive therapy of hypertension (see Sections 12.1.2 and 12.1.3). However, the adoption of lifestyle recommendations is often challenging in ethnic minority patients because of poor social support, limited access to exercise opportunities and

healthy foods, and financial considerations. The greater prevalence of lower socioeconomic status may impede access to basic living necessities,^{S10.1.1-8} including medical care and medications. Consideration must also be given to learning styles and preference, personal beliefs, values, and culture.^{S10.1.1-9,S10.1.1-10}

The principles of antihypertensive drug selection discussed in Sections 8.1.4 through 8.1.6 apply to ethnic minorities with a few caveats. In blacks, thiazide-type diuretics and CCBs are more effective in lowering BP when given as monotherapy or as initial agents in multidrug regimens.^{S10.1.1-11–S10.1.1-13} In addition, thiazide-type agents are superior to drugs that inhibit the RAS (ie, ACE inhibitors, ARBs, renin inhibitors, and beta blockers) for prevention of selected clinical outcomes in blacks.^{S10.1.1-2,S10.1.1-14–S10.1.1-16} For optimum endpoint protection, the thiazide chlorthalidone should be administered at a dose of 12.5 to 25 mg/day (or 25–50 mg/d for hydrochlorothiazide) because lower doses are either unproven or less effective in clinical outcome trials.^{S10.1.1-2,S10.1.1-16} The CCB amlodipine is as effective as chlorthalidone and more effective than the ACE inhibitor lisinopril in reducing BP, CVD, and stroke events but less effective in preventing HF. Blacks have a greater risk of angioedema with ACE inhibitors,^{S10.1.1-2,S10.1.1-3} and Asian Americans have a higher incidence of ACE inhibitor-induced cough.^{S10.1.1-17} ACE inhibitors and ARBs are recommended more generally as components of multidrug antihypertensive regimens in blacks with CKD (see Section 9.3), with the addition of beta blockers in those with HF (see Section 9.2). Beta blockers are recommended for treatment of patients with CHD who have had a MI. Most patients with hypertension, especially blacks, require ≥2 antihypertensive medications to achieve adequate BP control. A single-tablet combination that includes either a diuretic or a CCB may be particularly effective in achieving BP control in blacks. Racial and ethnic differences should not be the basis for excluding any class of antihypertensive agent in combination therapy.

Recommendation-Specific Supportive Text

1. In blacks, thiazide diuretics or CCBs are more effective in lowering BP than are RAS inhibitors or beta blockers and more effective in reducing CVD events than are RAS inhibitors or alpha blockers. RAS inhibitors are recommended in black patients with hypertension, DM, and nephropathy, but they offer no advantage over diuretics or CCBs in hypertensive patients with DM without nephropathy or HF.
2. Four drug classes (thiazide diuretic, CCB, ACE inhibitor, or ARB) lower BP and reduce cardiovascular or renal outcomes.^{S10.1.1-18–S10.1.1-21} Thus, except for the combination of ACE inhibitors and ARBs, regimens containing a combination of these classes are reasonable to achieve the BP target.^{S10.1.1-16,S10.1.1-21} Furthermore, the combination of an ACE inhibitor or ARB with a CCB or thiazide diuretic produces similar BP lowering in blacks as in other racial or ethnic groups. For blacks who do not achieve control with 3 drugs, see resistant hypertension (see Section 11.1).

10.2. Sex-Related Issues

The prevalence of hypertension is lower in women than in men until about the fifth decade but is higher later in life.^{S10.2-1} Other

than special recommendations for management of hypertension during pregnancy, there is no evidence that the BP threshold for initiating drug treatment, the treatment target, the choice of initial antihypertensive medication, or the combination of medications for lowering BP differs for women versus men.^{S10.2.2,S10.2.3}

10.2.1. Women

A potential limitation of RCTs, including SPRINT, is that they are not specifically powered to determine the value of intensive SBP reduction in subgroups, including women in the case of SPRINT. However, in prespecified analyses, there was no evidence of an interaction between sex and treatment effect. Furthermore, no significant differences in CVD outcomes were observed between men and women in a large meta-analysis that included 31 RCTs with about 100 000 men and 90 000 women with hypertension.^{S10.2.1-1} Some have called for a SPRINT-like trial with sufficient power to assess the effects of intensive SBP reduction in women.^{S10.2.1-2} In meta-analyses, there was no convincing evidence that different antihypertensive drug classes exerted sex-related differences in BP lowering or provided distinct CVD protection.^{S10.2.1-1} Calcium antagonists offered slightly greater benefits for stroke prevention than did ACE inhibitors for women than for men, whereas calcium antagonists reduced all-cause deaths compared with placebo in men but not in women. However, these sex-related differences might have been due to chance because of the large number of statistical comparisons that were performed. The Heart Attack Trial and Hypertension Care Computing Project reported that beta blockers were associated with reduced mortality in men but not in women, but this finding was likely due to the low event rates in women.^{S10.2.1-3} Similarly, in the open-label Second Australian National BP study, a significant reduction in CVD events was demonstrated in men but not in women with ACE inhibitors versus diuretics.^{S10.2.1-4}

Adverse effects of antihypertensive therapy were noted twice as often in women as in men in the TOMHS study.^{S10.2.1-5} A higher incidence of ACE inhibitor–induced cough and of edema with calcium antagonists was observed in women than in men.^{S10.2.1-6} Women were more likely to experience hypokalemia and hyponatremia and less likely to experience gout with diuretics.^{S10.2.1-7} Hypertension in pregnancy has special requirements (see Section 10.2.2).

10.2.2. Pregnancy

Recommendations for Treatment of Hypertension in Pregnancy		
References that support recommendations are summarized in Online Data Supplement 53.		
COR	LOE	Recommendations
I	C-LD	1. Women with hypertension who become pregnant, or are planning to become pregnant, should be transitioned to methyldopa, nifedipine, and/or labetalol during pregnancy. ^{S10.2.2-1}
III: Harm	C-LD	2. Women with hypertension who become pregnant should not be treated with ACE inhibitors, ARBs, or direct renin inhibitors. ^{S10.2.2-4–S10.2.2-6}

Synopsis

BP usually declines during the first trimester of pregnancy and then slowly rises. Hypertension management during pregnancy includes 4 general areas: 1) the newly pregnant mother with existing hypertension; 2) incident hypertension; 3) preeclampsia (a dangerous form of hypertension with proteinuria that has the potential to result in serious adverse consequences for the mother [stroke, HF] and fetus [small for gestational age, premature birth]); and 4) severe hypertension, often in the setting of preeclampsia, requiring urgent treatment to prevent HF, stroke, and adverse fetal outcomes. Hypertension during pregnancy and preeclampsia are recognized as risk factors for future hypertension and CVD.^{S10.2.2-7–S10.2.2-9} BP management during pregnancy is complicated by the fact that many commonly used antihypertensive agents, including ACE inhibitors and ARBs, are contraindicated during pregnancy because of potential harm to the fetus.^{S10.2.2-2,S10.2.2-3} The goal of antihypertensive treatment during pregnancy includes prevention of severe hypertension and the possibility of prolonging gestation to allow the fetus more time to mature before delivery.

There are 3 Cochrane database reviews of treatment for mild-to-moderate hypertension during pregnancy.^{S10.2.2-10–S10.2.2-12} With regard to the treatment of mild-to-moderate hypertension (SBP of 140–169 or DBP of 90–109 mm Hg), antihypertensive treatment reduces the risk of progression to severe hypertension by 50% compared with placebo but has not been shown to prevent preeclampsia, preterm birth, small for gestational age, or infant mortality. Beta blockers and CCBs appear superior to alpha-methyldopa in preventing preeclampsia.^{S10.2.2-10} An earlier review of 2 small trials did not show improved outcomes with more comprehensive treatment of BP to a target of <130/80 mm Hg.^{S10.2.2-11} Consistent with the results of the Cochrane reviews, a large multinational RCT of treatment in pregnant women with mild-to-moderate hypertension also reported that treatment prevented progression to severe hypertension, but other maternal and infant outcomes were unaffected by the intensity of treatment.^{S10.2.2-13} An earlier review confined to assessing the effect of beta blockers found them generally safe and effective but of no benefit for newborn outcomes, either in placebo-controlled studies or when compared with other antihypertensive agents. There was a suggestion that beta-blocker therapy might be associated with small for gestational age and neonatal bradycardia.^{S10.2.2-12} The largest experience for beta blockers is with labetalol; the largest experience for CCBs is with nifedipine. Methyldopa and hydralazine may also be used. A review of treatment for pregnancy-associated severe hypertension found insufficient evidence to recommend specific agents; rather, clinician experience was recommended in this setting.^{S10.2.2-14}

Preeclampsia is a potentially dangerous condition for the pregnant woman and fetus, occurring in 3.8% of pregnancies, and preeclampsia and eclampsia account for 9% of maternal deaths in the United States.^{S10.2.2-15} Preeclampsia is associated with an increased risk of preterm delivery, intrauterine growth restriction, placental abruption, and perinatal mortality and is twice as likely to occur in the first pregnancy. The US Preventive Services Task Force has recommended screening

all pregnant women for preeclampsia by measuring BP at every prenatal visit.^{S10.2.2-16}

It is beyond the scope of the present guideline to address the management of hypertension during pregnancy in detail. Several international guidelines provide guidance on management of hypertension during pregnancy.^{S10.2.2-2,S10.2.2-3,S10.2.2-17} The American College of Obstetricians and Gynecologists has issued a task force report that includes recommendations for prevention (aspirin in selected cases) and treatment (magnesium for severe hypertension) of hypertension in pregnancy.^{S10.2.2-2} A report detailing treatment of hypertensive emergencies during pregnancy and postpartum has also been released.^{S10.2.2-2,S10.2.2-17,S10.2.2-18}

Recommendation-Specific Supportive Text

1. ACE inhibitors and ARBs are not approved for use during pregnancy; they are fetotoxic. Among the agents recommended, no specific agent is first choice because there are no data supporting one over another. Therapeutic classes are not recommended because potential toxicity differs among agents within classes.
2. ACE inhibitors and ARBs are fetotoxic in the second and third trimesters of pregnancy. Adverse effects in the first trimester of pregnancy may be secondary to hypertension or the medication.^{S10.2.2-4,S10.2.2-5} Adverse events in the later trimesters have been suggested by observational data and meta-analyses.^{S10.2.2-6} For ARBs, case reports with effects similar to ACE inhibitors have been published.^{S10.2.2-19}

10.3. Age-Related Issues

10.3.1. Older Persons

Recommendations for Treatment of Hypertension in Older Persons		
References that support recommendations are summarized in Online Data Supplement 54.		
COR	LOE	Recommendations
I	A	1. Treatment of hypertension with a SBP treatment goal of less than 130 mm Hg is recommended for noninstitutionalized ambulatory community-dwelling adults (≥65 years of age) with an average SBP of 130 mm Hg or higher. ^{S10.3.1-1}
IIa	C-E0	2. For older adults (≥65 years of age) with hypertension and a high burden of comorbidity and limited life expectancy, clinical judgment, patient preference, and a team-based approach to assess risk/benefit is reasonable for decisions regarding intensity of BP lowering and choice of antihypertensive drugs.

Synopsis

Because of its extremely high prevalence in older adults, hypertension is not only a leading cause of preventable morbidity and mortality but, perhaps more importantly, is under-recognized as a major contributor to premature disability and institutionalization.^{S10.3.1-2-S10.3.1-5} Both SBP and DBP increase linearly up to the fifth or sixth decade of life, after which DBP gradually

decreases while SBP continues to rise.^{S10.3.1-6} Thus, isolated systolic hypertension is the predominant form of hypertension in older persons.^{S10.3.1-7,S10.3.1-8} RCTs have clearly demonstrated that BP lowering in isolated systolic hypertension (defined as SBP ≥160 mm Hg with variable DBP ≤90, ≤95, or ≤110 mm Hg) is effective in reducing the risk of fatal and nonfatal stroke (primary outcome), cardiovascular events, and death.^{S10.3.1-9-S10.3.1-12}

Cross-sectional and longitudinal epidemiologic studies in older adults have raised questions about the benefits of more intensive antihypertensive treatment and the relationship between BP lowering and risk of falls.^{S10.3.1-13} Treatment of elevated BP in older persons is challenging because of a high degree of heterogeneity in comorbidity, as well as poly-pharmacy, frailty, cognitive impairment, and variable life expectancy. However, over the past 3 decades, RCTs of antihypertensive therapy have included large numbers of older persons, and in every instance, including when the SBP treatment goal was <120 mm Hg, more intensive treatment has safely reduced the risk of CVD for persons over the ages of 65, 75, and 80 years.^{S10.3.1-1,S10.3.1-14} Both HYVET (Hypertension in the Very Elderly Trial) and SPRINT included those who were frail but still living independently in the community,^{S10.3.1-1,S10.3.1-14} and both were stopped early for benefit (HYVET after 1.8 years and SPRINT after 3.26 years). In fact, BP-lowering therapy is one of the few interventions shown to reduce mortality risk in frail older individuals. RCTs in noninstitutionalized community-dwelling older persons have also demonstrated that improved BP control does not exacerbate orthostatic hypotension and has no adverse impact on risk of injurious falls.^{S10.3.1-1,S10.3.1-15,S10.3.1-16} It should be noted, however, that SPRINT excluded those with low (<110 mm Hg) standing BP on study entry. Older persons need to be carefully monitored for orthostatic hypotension during treatment. Intensive BP control increases the risk of acute kidney injury, but this is no different from the risk seen in younger adults.^{S10.3.1-1} In summary, despite the complexity of management in caring for older persons with hypertension, RCTs have demonstrated that in many community-dwelling older adults, even adults >80 years of age, BP-lowering goals during antihypertensive treatment need not differ from those selected for persons <65 years of age.^{S10.3.1-17} Importantly, no randomized trial of BP lowering in persons >65 years of age has ever shown harm or less benefit for older versus younger adults. However, clinicians should implement careful titration of BP lowering and monitoring in persons with high comorbidity burden; large RCTs have excluded older persons at any age who live in nursing homes, as well as those with prevalent dementia and advanced HF.

Recommendation-Specific Supportive Text

1. We recommend ASCVD risk assessment in all adults with hypertension, including older persons. As a matter of convenience, however, it can be assumed that the vast majority of older adults have a 10-year ASCVD risk ≥ 10%, placing them in the high risk category that requires initiation of antihypertensive drug therapy at BP ≥ 130/80 mm Hg (see Section 8.1.2, Figure 4 and Table 23 for BP thresholds for initiating antihypertensive

drug treatment). Large RCTs using medications to reduce hypertension-related CVD risk with a mean follow-up of ≥ 2 years have now included a large number of adults ≥ 65 years of age. These trials have enrolled a broad range of ages ≥ 65 years, including persons in their 90s and even 100s, as well as those with mild-to-moderate frailty but who were ambulatory and able to travel to a treatment clinic. In these patients, RCTs have shown that BP lowering decreased CVD morbidity and mortality but did not increase the risk of orthostatic hypotension or falls.^{S10.3.1-1, S10.3.1-15, S10.3.1-16} Analysis of the NHANES (2011–2014) data set indicates that 88% of US adults (98% men and 80% women) ≥ 65 years old have a 10-year predicted ASCVD risk $\geq 10\%$ or have a history of CVD (CHD, stroke, or HF). For persons ≥ 75 years of age, 100% have an ASCVD risk score $\geq 10\%$ or a history of CVD. Therefore, the BP target of $\leq 130/80$ mm Hg would be appropriate (see Section 8.1.2). Initiation of antihypertensive therapy with 2 agents should be undertaken cautiously in older persons, and they need to be monitored carefully for orthostatic hypotension and history of falls. In SPRINT, the benefit was for an SBP goal of <120 mm Hg. Older persons may present with neurogenic orthostatic hypotension associated with supine hypertension. This is particularly common in Parkinson's disease and other neurodegenerative disorders. For management of this problem, the reader is referred to the recommendations of a 2017 consensus panel.^{S10.3.1-18}

- Patients with prevalent and frequent falls, advanced cognitive impairment, and multiple comorbidities may be at risk of adverse outcomes with intensive BP lowering, especially when they require multiple BP-lowering medications. Older persons in this category typically reside in nursing homes and assisting living facilities, are unable to live independently in the community, and have not been represented in RCTs.

10.3.2. Children and Adolescents

Pediatric guidelines are available from other organizations.^{S10.3.2-1, S10.3.2-2} The 2011 report updates the 2004 report for publications through 2008 (antihypertensive medication trials, normative data on pediatric BP) but is otherwise unchanged. In the 2011 guideline,^{S10.3.2-3} BP was stratified into normal, prehypertension (90th percentile to 95th percentile), stage 1 hypertension (95th percentile to >99 th percentile), and stage 2 hypertension (above stage 1) by using age-, sex-, and height-based tables beginning at 1 year of age, which were based on the distribution of BP in $>60\,000$ healthy children in various population-based studies.^{S10.3.2-1} These definitions were designed to be analogous to definitions in the extant JNC 7 report; for older adolescents (≥ 14 years), the JNC 7 thresholds generally apply.^{S10.3.2-4} Treatment recommendations are based on hypertension severity, published short-term clinical trials of antihypertensive treatment, age, coexisting CVD risk factors, and risk stratification by presence of LVH on echocardiogram. The treatment goal is to achieve BP <90 th percentile. New tables for ambulatory BP distribution in children have been developed. A classification of BP that is based on these ambulatory BP results has been proposed.^{S10.3.2-5, S10.3.2-6} A new pediatric BP guideline was published in late 2017.^{S10.3.2-7}

11. Other Considerations

11.1. Resistant Hypertension

The diagnosis of resistant hypertension is made when a patient takes 3 antihypertensive medications with complementary mechanisms of action (a diuretic should be 1 component) but does not achieve control or when BP control is achieved but requires ≥ 4 medications.^{S11.1-1} On the basis of the previous cutoff of 140/90 mm Hg, the prevalence of resistant hypertension is approximately 13% in the adult population.^{S11.1-2, S11.1-3} Multiple single-cohort studies have indicated that common risk factors for resistant hypertension include older age, obesity, CKD, black race, and DM. Estimates suggest the prevalence would be about 4% higher with the newly recommended control target of $<130/80$ mm Hg (subject to validation in future study). The prognosis of resistant hypertension (by the previous definition),^{S11.1-1} compared with the prognosis of those who more readily achieve control, has not been fully ascertained; however, risk of MI, stroke, ESRD, and death in adults with resistant hypertension and CHD may be 2- to 6-fold higher than in hypertensive adults without resistant hypertension.^{S11.1-4–S11.1-6} The evaluation of resistant hypertension involves consideration of many patient characteristics, pseudoresistance (BP technique, white coat hypertension, and medication compliance), and screening for secondary causes of hypertension (Figure 10; Section 5.4; Table 13). The term “refractory hypertension” has been used to refer to an extreme phenotype of antihypertensive treatment failure, defined as failure to control BP despite use of at least 5 antihypertensive agents of different classes, including a long-acting thiazide-type diuretic, such as chlorthalidone, and a mineralocorticoid receptor antagonist, such as spironolactone.^{S11.1-7} The prevalence of refractory hypertension is low; patients with refractory hypertension experience high rates of CVD complications, including LVH, HF, and stroke.

Treatment of resistant hypertension involves improving medication adherence, improving detection and correction of secondary hypertension, and addressing other patient characteristics.^{S11.1-8–S11.1-10} Pharmacological therapy with combinations of medications with complementary mechanisms of action provides an empirical approach that enhances BP control while mitigating untoward effects of potent vasodilators (eg, fluid retention and reflex tachycardia). CCBs, inhibitors of RAS, and chlorthalidone comprise a common 3-drug regimen.^{S11.1-11} Considerable evidence indicates that the addition of spironolactone to multidrug regimens provides substantial BP reduction^{S11.1-12} when compared with placebo. Substantial data also demonstrate the advantage of spironolactone as compared with other active drugs.^{S11.1-8, S11.1-13–S11.1-15} In particular, the recent PATHWAY-2 (Optimum Treatment for Drug-Resistant Hypertension) RCT demonstrated the superiority of spironolactone over alpha and beta blockers.^{S11.1-13} There is also clinical trial evidence that the addition of hydralazine or minoxidil is effective in achieving BP control in patients resistant to usual combination therapy.^{S11.1-8, S11.1-12–S11.1-16} The dosing of multidrug regimens, occasionally including nighttime dosing, may be best optimized by hypertension specialists.

Several studies have investigated devices that interrupt sympathetic nerve activity (carotid baroreceptor pacing and catheter ablation of renal sympathetic nerves); however, these studies have not provided sufficient evidence to

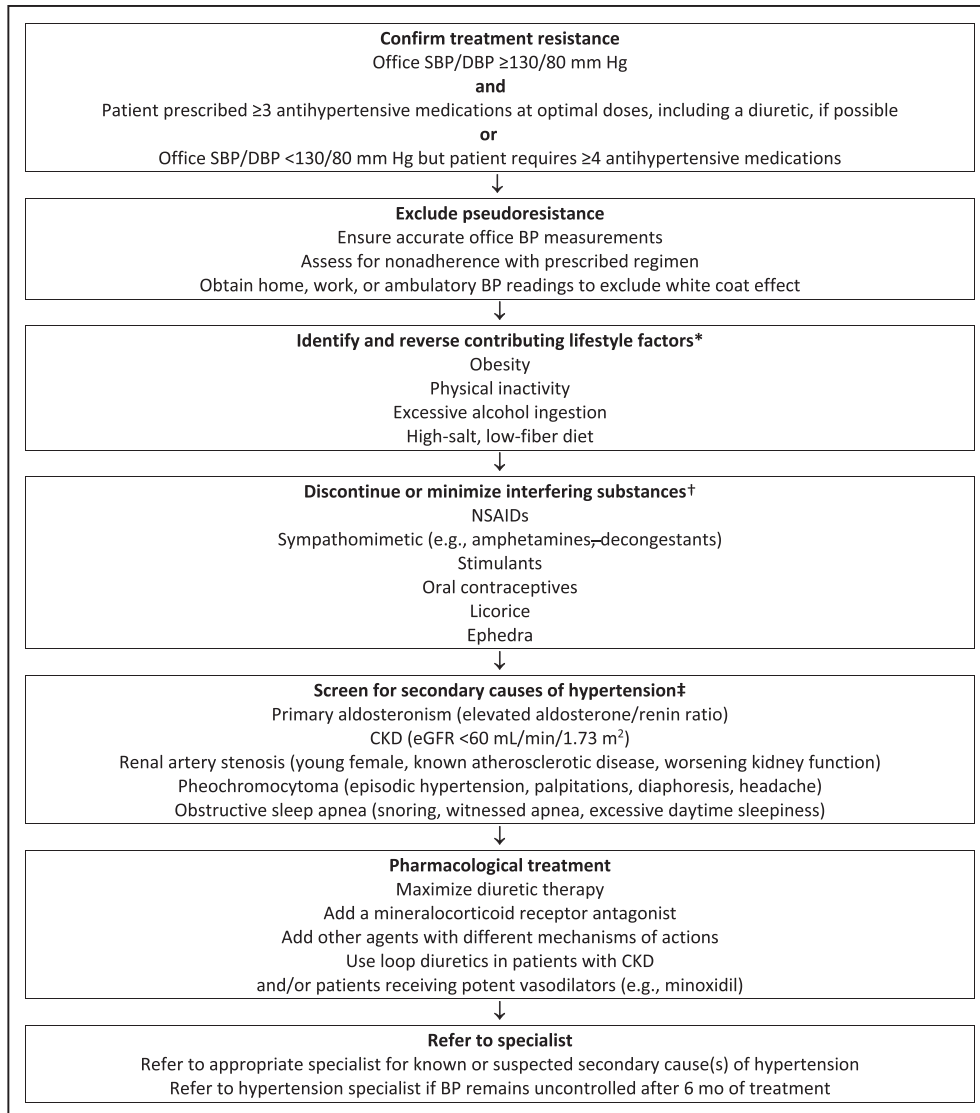


Figure 10. Resistant hypertension: diagnosis, evaluation, and treatment. *See additional details in Section 6, Nonpharmacological Intervention. †See Section 5.4.1 and Table 14 for complete list of drugs that elevate BP. ‡See Section 5.4 and Table 13 for secondary hypertension. BP indicates blood pressure; CKD, chronic kidney disease; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; NSAIDs, nonsteroidal anti-inflammatory drugs; and SBP, systolic blood pressure. Adapted with permission from Calhoun et al^{S11.1-1} (American Heart Association, Inc.).

recommend the use of these device in managing resistant hypertension.^{S11.1-8–S11.1-10} In particular, 2 RCTs of renal sympathetic nerve ablation have been negative.^{S11.1-8,S11.1-9}

11.2. Hypertensive Crises—Emergencies and Urgencies

Recommendations for Hypertensive Crises and Emergencies

References that support recommendations are summarized in Online Data Supplement 55.

COR	LOE	Recommendations
I	B-NR	1. In adults with a hypertensive emergency, admission to an intensive care unit is recommended for continuous monitoring of BP and target organ damage and for parenteral administration of an appropriate agent (Tables 19 and 20). ^{S11.2-1,S11.2-2}

Recommendations for Hypertensive Crises and Emergencies (Continued)

COR	LOE	Recommendations
I	C-EO	2. For adults with a compelling condition (ie, aortic dissection, severe preeclampsia or eclampsia, or pheochromocytoma crisis), SBP should be reduced to less than 140 mm Hg during the first hour and to less than 120 mm Hg in aortic dissection.
I	C-EO	3. For adults without a compelling condition, SBP should be reduced by no more than 25% within the first hour; then, if stable, to 160/100 mm Hg within the next 2 to 6 hours; and then cautiously to normal during the following 24 to 48 hours.

Synopsis

Hypertensive emergencies are defined as severe elevations in BP ($>180/120$ mm Hg) associated with evidence of new or worsening target organ damage.^{S11.2-3–S11.2-6} The 1-year death rate associated with hypertensive emergencies is $>79\%$, and the median survival is 10.4 months if the emergency is left untreated.^{S11.2-7} The actual BP level may not be as important as the rate of BP rise; patients with chronic hypertension can often tolerate higher BP levels than previously normotensive individuals. Hypertensive emergencies demand immediate reduction of BP (not necessarily to normal) to prevent or limit further target organ damage. Examples of target organ damage include hypertensive encephalopathy, ICH, acute ischemic stroke, acute MI, acute LV failure with pulmonary edema, unstable angina pectoris, dissecting aortic aneurysm, acute renal failure, and eclampsia. In general, use of oral therapy is discouraged for hypertensive emergencies. Hypertensive emergencies in patients with acute ICH and acute ischemic stroke are discussed in Section 9.4.

In contrast, hypertensive urgencies are situations associated with severe BP elevation in otherwise stable patients without acute or impending change in target organ damage or dysfunction. Many of these patients have withdrawn from or are noncompliant with antihypertensive therapy and do not have clinical or laboratory evidence of acute target organ damage. These patients should not be considered as having a

hypertensive emergency and instead are treated by reinstitution or intensification of antihypertensive drug therapy and treatment of anxiety as applicable. There is no indication for referral to the emergency department, immediate reduction in BP in the emergency department, or hospitalization for such patients.

Figure 11 is an algorithm on diagnosis and management of a hypertensive crisis. Tables 19 and 20 summarize intravenous antihypertensive drugs for treatment of hypertensive emergencies.

Recommendation-Specific Supportive Text

1. There is no RCT evidence that antihypertensive drugs reduce morbidity or mortality in patients with hypertensive emergencies.^{S11.2-8} However, from clinical experience, it is highly likely that antihypertensive therapy is an overall benefit in a hypertensive emergency.^{S11.2-9} There is also no high-quality RCT evidence to inform clinicians as to which first-line antihypertensive drug class provides more benefit than harm in hypertensive emergencies.^{S11.2-8} This lack of evidence is related to the small size of trials, the lack of long-term follow-up, and failure to report outcomes. However, 2 trials have demonstrated that nicardipine may be better than labetalol in achieving the short-term BP target.^{S11.2-1, S11.2-10–S11.2-12} Several antihypertensive agents in various pharmacological classes are available for the treatment of hypertensive

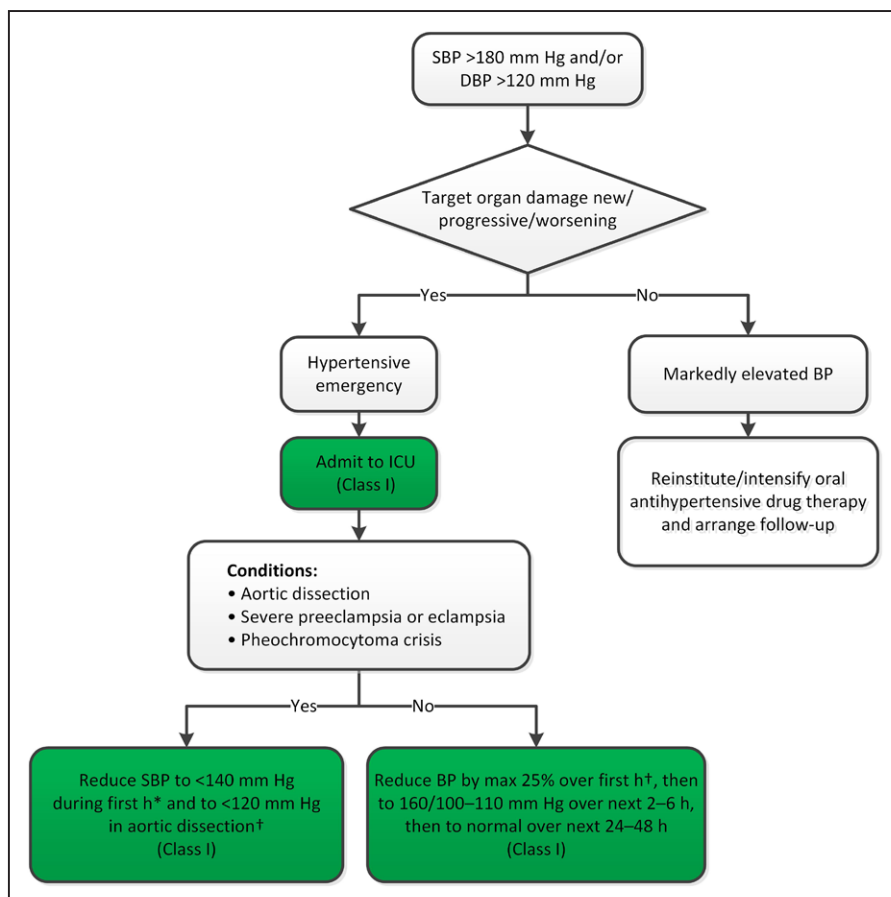


Figure 11. Diagnosis and management of a hypertensive crisis. Colors correspond to Class of Recommendation in Table 1. *Use drug(s) specified in Table 19. †If other comorbidities are present, select a drug specified in Table 20. BP indicates blood pressure; DBP, diastolic blood pressure; ICU, intensive care unit; and SBP, systolic blood pressure.

Table 19. Intravenous Antihypertensive Drugs for Treatment of Hypertensive Emergencies

Class	Drug(s)	Usual Dose Range	Comments
CCB—dihydropyridines	Nicardipine	Initial 5 mg/h, increasing every 5 min by 2.5 mg/h to maximum 15 mg/h.	Contraindicated in advanced aortic stenosis; no dose adjustment needed for elderly.
	Clevidipine	Initial 1–2 mg/h, doubling every 90 s until BP approaches target, then increasing by less than double every 5–10 min; maximum dose 32 mg/h; maximum duration 72 h.	Contraindicated in patients with soybean, soy product, egg, and egg product allergy and in patients with defective lipid metabolism (eg, pathological hyperlipidemia, lipid nephrosis or acute pancreatitis). Use low-end dose range for elderly patients.
Vasodilators—Nitric-oxide dependent	Sodium nitroprusside	Initial 0.3–0.5 mcg/kg/min; increase in increments of 0.5 mcg/kg/min to achieve BP target; maximum dose 10 mcg/kg/min; duration of treatment as short as possible. For infusion rates ≥ 4 –10 mcg/kg/min or duration > 30 min, thiosulfate can be coadministered to prevent cyanide toxicity.	Intra-arterial BP monitoring recommended to prevent “overshoot.” Lower dosing adjustment required for elderly. Tachyphylaxis common with extended use. Cyanide toxicity with prolonged use can result in irreversible neurological changes and cardiac arrest.
	Nitroglycerin	Initial 5 mcg/min; increase in increments of 5 mcg/min every 3–5 min to a maximum of 20 mcg/min.	Use only in patients with acute coronary syndrome and/or acute pulmonary edema. Do not use in volume-depleted patients.
Vasodilators—direct	Hydralazine	Initial 10 mg via slow IV infusion (maximum initial dose 20 mg); repeat every 4–6 h as needed.	BP begins to decrease within 10–30 min, and the fall lasts 2–4 h. Unpredictability of response and prolonged duration of action do not make hydralazine a desirable first-line agent for acute treatment in most patients.
Adrenergic blockers— β_1 receptor selective antagonist	Esmolol	Loading dose 500–1000 mcg/kg/min over 1 min followed by a 50-mcg/kg/min infusion. For additional dosing, the bolus dose is repeated and the infusion increased in 50-mcg/kg/min increments as needed to a maximum of 200 mcg/kg/min.	Contraindicated in patients with concurrent beta-blocker therapy, bradycardia, or decompensated HF. Monitor for bradycardia. May worsen HF. Higher doses may block β_2 receptors and impact lung function in reactive airway disease.
Adrenergic blockers—combined α_1 and nonselective beta receptor antagonist	Labetalol	Initial 0.3–1.0-mg/kg dose (maximum 20 mg) slow IV injection every 10 min or 0.4–1.0-mg/kg/h IV infusion up to 3 mg/kg/h. Adjust rate up to total cumulative dose of 300 mg. This dose can be repeated every 4–6 h.	Contraindicated in reactive airways disease or chronic obstructive pulmonary disease. Especially useful in hyperadrenergic syndromes. May worsen HF and should not be given in patients with second- or third-degree heart block or bradycardia.
Adrenergic blockers—nonselective α receptor antagonist	Phentolamine	IV bolus dose 5 mg. Additional bolus doses every 10 min as needed to lower BP to target.	Used in hypertensive emergencies induced by catecholamine excess (pheochromocytoma, interactions between monamine oxidase inhibitors and other drugs or food, cocaine toxicity, amphetamine overdose, or clonidine withdrawal).
Dopamine $_1$ -receptor selective agonist	Fenoldopam	Initial 0.1–0.3 mcg/kg/min; may be increased in increments of 0.05–0.1 mcg/kg/min every 15 min until target BP is reached. Maximum infusion rate 1.6 mcg/kg/min.	Contraindicated in patients at risk of increased intraocular pressure (glaucoma) or intracranial pressure and those with sulfite allergy.
ACE inhibitor	Enalaprilat	Initial 1.25 mg over a 5-min period. Doses can be increased up to 5 mg every 6 h as needed to achieve BP target.	Contraindicated in pregnancy and should not be used in acute MI or bilateral renal artery stenosis. Mainly useful in hypertensive emergencies associated with high plasma renin activity. Dose not easily adjusted. Relatively slow onset of action (15 min) and unpredictability of BP response.

BP indicates blood pressure; CCB, calcium channel blocker; HF, heart failure; IV, intravenous; and MI, myocardial infarction.

emergencies (Table 19). Because autoregulation of tissue perfusion is disturbed in hypertensive emergencies, continuous infusion of short-acting titratable antihypertensive agents is often preferable to prevent further target organ damage.^{S11.2-5,S11.2-6} The selection of an antihypertensive agent should be based on the drug’s pharmacology, pathophysiological factors underlying the patient’s hypertension (as well as they can be rapidly determined),

degree of progression of target organ damage, the desirable rate of BP decline, and the presence of comorbidities (Table 20). The therapeutic goal is to minimize target organ damage safely by rapid recognition of the problem and early initiation of appropriate antihypertensive treatment.

2. Compelling conditions requiring rapid lowering of SBP, usually to < 140 mm Hg, in the first hour of treatment

Table 20. Intravenous Antihypertensive Drugs for Treatment of Hypertensive Emergencies in Patients With Selected Comorbidities

Comorbidity	Preferred Drug(s) [*]	Comments
Acute aortic dissection	Esmolol, labetalol	Requires rapid lowering of SBP to ≤ 120 mm Hg. Beta blockade should precede vasodilator (eg, nicardipine or nitroprusside) administration, if needed for BP control or to prevent reflex tachycardia or inotropic effect; SBP ≤ 120 mm Hg should be achieved within 20 min.
Acute pulmonary edema	Clevidipine, nitroglycerin, nitroprusside	Beta blockers contraindicated.
Acute coronary syndromes	Esmolol, [†] labetalol, nicardipine, nitroglycerin [†]	Nitrates given in the presence of PDE-5 inhibitors may induce profound hypotension. Contraindications to beta blockers include moderate-to-severe LV failure with pulmonary edema, bradycardia (<60 bpm), hypotension (SBP <100 mm Hg), poor peripheral perfusion, second- or third-degree heart block, and reactive airways disease.
Acute renal failure	Clevidipine, fenoldopam, nicardipine	N/A
Eclampsia or preeclampsia	Hydralazine, labetalol, nicardipine	Requires rapid BP lowering. ACE inhibitors, ARBs, renin inhibitors, and nitroprusside contraindicated.
Perioperative hypertension (BP $\geq 160/90$ mm Hg or SBP elevation $\geq 20\%$ of the preoperative value that persists for >15 min)	Clevidipine, esmolol, nicardipine, nitroglycerin	Intraoperative hypertension is most frequently seen during anesthesia induction and airway manipulation.
Acute sympathetic discharge or catecholamine excess states (eg, pheochromocytoma, post-carotid endarterectomy status)	Clevidipine, nicardipine, phentolamine	Requires rapid lowering of BP.
Acute ICH	Section 9.4.1	Section 9.4.1
Acute ischemic stroke	Section 9.4.2	Section 9.4.2

^{*}Agents are listed in alphabetical order, not in order of preference.

[†]Agent of choice for acute coronary syndromes.

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BP, blood pressure; bpm, beats per minute; ICH, intracerebral hemorrhage; LV, left ventricular; PDE-5, phosphodiesterase type-5; and SBP, systolic blood pressure.

include aortic dissection, severe preeclampsia or eclampsia, and pheochromocytoma with hypertensive crisis.

- There is no RCT evidence comparing different strategies to reduce BP, except in patients with ICH.^{S11.2-9,S11.2-13} Neither is there RCT evidence to suggest how rapidly or how much BP should be lowered in a hypertensive emergency.^{S11.2-9} However, clinical experience indicates that excessive reduction of BP may cause or contribute to renal, cerebral, or coronary ischemia and should be avoided. Thus, comprehensive dosing of intravenous or even oral antihypertensive agents to rapidly lower BP is not without risk. Oral loading doses of antihypertensive agents can engender cumulative effects, causing hypotension after discharge from the emergency department or clinic.

11.3. Cognitive Decline and Dementia

Recommendation for Prevention of Cognitive Decline and Dementia

References that support the recommendation are summarized in [Online Data Supplement 56](#).

COR	LOE	Recommendation
Ila	B-R	1. In adults with hypertension, BP lowering is reasonable to prevent cognitive decline and dementia. ^{S11.3-1-S11.3-6}

Synopsis

Dementia is a leading cause of mortality and placement into nursing homes and assisted living facilities, affecting >46 million individuals globally and 5 million persons in the United States, a number that is expected to double by 2050.^{S11.3-7} A 5-year delay in onset of dementia would likely decrease the number of cases of incident dementia by about 50% after several decades.^{S11.3-8} Vascular disease and its risk factors are implicated in a large proportion of patients with dementia, including those with Alzheimer's dementia.^{S11.3-9-S11.3-11} Hypertension is also the primary risk factor for small-vessel ischemic disease and cortical white matter abnormalities.^{S11.3-12-S11.3-15} Most observational studies have suggested that better control of SBP may reduce Alzheimer's disease and other dementias, and the evidence is stronger for BP lowering in middle age than in older adults.^{S11.3-9,S11.3-16} Clinical trials with dementia assessment have evaluated all-cause dementia but not Alzheimer's disease specifically. However, all of these trials have methodological issues, such as low power, insufficient follow-up length, and inadequately designed dementia assessment batteries.

Recommendation-Specific Supportive Text

- Five clinical trials of BP lowering have included assessment for incident dementia. Of these 5 trials, 4

demonstrated a reduction in dementia incidence, with 2 of these 4 demonstrating statistical significance (746-751). SYST-EUR (Systolic Hypertension in Europe)^{S11.3-17} and PROGRESS (Perindopril Protection Against Recurrent Stroke)^{S11.3-18} both showed statistically significant reductions in incident dementia. SYST-EUR achieved an SBP of 152 mm Hg in the treatment arm (8.3 mm Hg lower than placebo arm) during its blinded phase and an SBP of 149 mm Hg (7.0 mm Hg lower than comparison group) during its open-label follow-up phase.^{S11.3-2,S11.3-3} PROGRESS achieved an SBP of 138 mm Hg in the treatment group (9 mm Hg lower than the placebo group) and demonstrated dementia prevention in patients with a recent stroke.^{S11.3-5} The trial showing no benefit in the direction of dementia reduction achieved an SBP reduction of only 3.2 mm Hg, whereas the other 4 trials achieved SBP reductions of 7 to 15 mm Hg (746-751). When the rate of cognitive decline (not dementia) has been a trial outcome, 7 clinical trials of BP-lowering therapy have been completed, and 2 of these have shown benefit.^{S11.3-4-S11.3-6,S11.3-19-S11.3-22} No randomized trial of BP lowering has demonstrated an adverse impact on dementia incidence or cognitive function. However, the anticipated results from SPRINT, the first adequately powered RCT to test whether intensive BP control reduces dementia, may help clarify this issue in the near future.

11.4. Sexual Dysfunction and Hypertension

An association among sexual dysfunction, atherosclerosis, and hypertension can be constructed from several epidemiology surveys, clinical trials, and cohort studies. Although these data converge to suggest that endothelial dysfunction is a common denominator, the story is incomplete. Sexual dysfunction represents several domains in desire or interest, as well as physical limitations such as erectile dysfunction. In addition, beta blockers, mineralocorticoid receptor antagonists, and other antihypertensive drugs can have negative effects on libido and erectile function. There are emerging data on the association between erectile dysfunction and CVD compared with other domains of sexual dysfunction. Experimental and clinical studies describe a role for angiotensin II, endothelin, and hydrogen sulfide on cavernous tissue function.^{S11.4-1} Many of the signaling pathways for the increased production of oxidative stress and the subsequent deleterious effects of oxidative stress on vascular tissue have been described. Accordingly, it is reasonable to suggest that hypertension might lead to vascular changes that cause erectile dysfunction but, conversely, erectile dysfunction may be part of the causal pathway to CVD.^{S11.4-1} Although there is insufficient evidence to recommend screening for CVD risk factors in all men with erectile dysfunction, it has been reported as a sole precursor for CVD in men.^{S11.4-2-S11.4-6}

With the introduction of the phosphodiesterase-5 inhibitors, which can be coadministered with antihypertensive medications, there is now effective therapy for erectile dysfunction that has implications for systemic vascular

disease.^{S11.4-7} These drugs have additive effects on lowering BP and are recommended as a primary therapy for pulmonary hypertension.^{S11.4-8} Although data are available to suggest that some antihypertensive medications affect erectile dysfunction more than others, the use of phosphodiesterase-5 inhibitors make drug class distinctions for erectile dysfunction less relevant.^{S11.4-9} The long-term safety and efficacy of chronic administration of phosphodiesterase-5 inhibitors for the mitigation of CVD has yet to be determined and represents an important knowledge gap.

11.5. Patients Undergoing Surgical Procedures

Recommendations for Treatment of Hypertension in Patients Undergoing Surgical Procedures

References that support recommendations are summarized in Online Data Supplements 57 and 58.

COR	LOE	Recommendations
Preoperative		
I	B-NR	1. In patients with hypertension undergoing major surgery who have been on beta blockers chronically, beta blockers should be continued. ^{S11.5-1-S11.5-7}
IIa	C-EO	2. In patients with hypertension undergoing planned elective major surgery, it is reasonable to continue medical therapy for hypertension until surgery.
IIb	B-NR	3. In patients with hypertension undergoing major surgery, discontinuation of ACE inhibitors or ARBs perioperatively may be considered. ^{S11.5-8-S11.5-10}
IIb	C-LD	4. In patients with planned elective major surgery and SBP of 180 mm Hg or higher or DBP of 110 mm Hg or higher, deferring surgery may be considered. ^{S11.5-11,S11.5-12}
III: Harm	B-NR	5. For patients undergoing surgery, abrupt preoperative discontinuation of beta blockers or clonidine is potentially harmful. ^{S11.5-2,S11.5-13}
III: Harm	B-NR	6. Beta blockers should not be started on the day of surgery in beta blocker-naïve patients. ^{S11.5-14}
Intraoperative		
I	C-EO	7. Patients with intraoperative hypertension should be managed with intravenous medications (Table 19) until such time as oral medications can be resumed.

Synopsis

Hypertension in the perioperative period increases the risk of CVD, cerebrovascular events, and bleeding.^{S11.5-15,S11.5-16}

As many as 25% of patients who undergo major noncardiac surgery^{S11.5-17} and 80% of patients who have cardiac surgery experience perioperative hypertension.^{S11.5-16,S11.5-18}

In general, the level of risk is related to the severity of the hypertension.

No high-quality RCTs were identified relating to the treatment of hypertension in patients undergoing major surgical procedures. One analysis evaluated data from 3 prospective, randomized, open-label, parallel-comparison studies in patients undergoing cardiac surgery and concluded that clevidipine is a safe and effective treatment for acute hypertension in patients undergoing cardiac surgery.^{S11.5-19} Another systematic review and meta-analysis, including 4 studies, concluded that clevidipine is more effective than other antihypertensive drugs in the management of perioperative hypertension without adverse events.^{S11.5-20} Several general strategies and principles based on experience and observation are recommended for this section. In the management of patients with perioperative hypertension, it is important to assess other potential contributing factors, such as volume status, pain control, oxygenation, and bladder distention, when the use of pharmacological therapy to control BP is under consideration. Uncontrolled hypertension is associated with increased perioperative and postoperative complications. Certain medications (eg, beta blockers, clonidine) may be associated with rebound hypertension if discontinued abruptly.^{S11.5-13} Therefore, several general strategies and principles based on experience and observation are recommended for this section.

These recommendations for beta blockers, ACE inhibitors, and ARBs are generally consistent with the “2014 ACC/AHA Guideline on Perioperative Cardiovascular Evaluation and Management of Patients Undergoing Noncardiac Surgery” and are provided to assist in the management of patients undergoing major noncardiac surgical procedures.^{S11.5-21}

Recommendation-Specific Supportive Text

1. If well tolerated, beta blockers should be continued in patients who are currently receiving them for longitudinal reasons, particularly when longitudinal treatment is provided according to GDMT, such as for MI.^{S11.5-22} Multiple observational studies support the benefits of continuing beta blockers in patients who are undergoing surgery and who are on these agents for longitudinal indications.^{S11.5-1–S11.5-7}
2. In the absence of conclusive RCTs, the expert opinion of this writing committee is that control of BP to levels recommended by the present guideline (BP <130/80 mm Hg) or other target levels specified for a particular individual is reasonable before undertaking major elective procedures in either the inpatient or outpatient setting. If the patient is unable to take oral medications, it is reasonable to use intravenous medications (Table 19) as necessary to control BP. Special consideration of parenteral therapy usually occurs for patients taking clonidine or beta blockers because of the risk of stopping these medications acutely. Withdrawal syndromes, accompanied by sympathetic discharge and acute hypertension, can occur on cessation of these agents.^{S11.5-13}
3. Data on the potential risk and benefit of ACE inhibitors in the perioperative setting are limited to observational analyses, and this area is controversial. Recent

evidence from a large cohort study demonstrates that patients who stopped their ACE inhibitors or ARBs 24 hours before noncardiac surgery were less likely to suffer the primary composite outcome (all-cause death, stroke, or myocardial injury) and intraoperative hypotension than were those continuing these medications until surgery.^{S11.5-10}

4. JNC 6^{S11.5-23} noted conflicting evidence for patients with DBP >110 mm Hg and recommended delay of surgery for gradual reduction in DBP before proceeding with surgery. In a systematic review and meta-analysis of 30 observational studies, preoperative hypertension was associated with a 35% increase in cardiovascular complications.^{S11.5-12} An increase in complications, including dysrhythmias, myocardial ischemia or infarction, neurological complications, and renal failure, has been reported in patients with DBP ≥110 mm Hg immediately before surgery.^{S11.5-24} In contrast, patients with DBP <110 mm Hg do not appear to be at significantly increased risk.^{S11.5-25} The relationship of systolic hypertension to surgical risk is less certain. Among patients undergoing carotid endarterectomy, increased risk of postoperative hypertension and neurological defects were observed,^{S11.5-26} and an increased risk of CVD morbidity after coronary artery bypass graft surgery has been observed in patients with isolated systolic hypertension.^{S11.5-27} During induction of anesthesia for surgery, sympathetic action can result in a 20- to 30-mm Hg increase in BP and a 15- to 20-bpm increase in heart rate among patients with normal BP.^{S11.5-24} Exaggerated responses may occur in patients with poorly treated or untreated hypertension by as much as 90 mm Hg and 40 bpm.^{S11.5-24} With further anesthesia, the accompanying inhibition of the sympathetic nervous system and loss of baroreceptor control may result in intraoperative hypotension. Lability in BP appears more likely in patients with poorly controlled hypertension,^{S11.5-25} whereas studies have observed that patients with controlled hypertension respond similarly to those who are normotensive.^{S11.5-28} Early work indicated that patients with severe hypertension (SBP >210 mm Hg and DBP >105 mm Hg) had exaggerated responses in BP during the induction of anesthesia.^{S11.5-28}
5. Although few studies describe risks of withdrawing beta blockers in the perioperative time period,^{S11.5-2,S11.5-5} longstanding evidence from other settings suggests that abrupt withdrawal of long-term beta blockers is harmful.^{S11.5-29–S11.5-31} There are fewer data to describe whether short-term (1 to 2 days) perioperative use of beta blockers, followed by rapid discontinuation, is harmful.^{S11.5-5,S11.5-14,S11.5-21,S11.5-30}
6. The 2014 ACC/AHA perioperative guideline specifically recommends against starting beta blockers on the day of surgery in beta-blocker-naïve patients,^{S11.5-5,S11.5-21,S11.5-30} particularly at high initial doses, in long-acting form, and if there are no plans for dose titration or monitoring for adverse events. Data from the POISE (Perioperative Ischemic Evaluation) study demonstrate the risk of initiating long-acting beta blockers on the day of surgery.^{S11.5-14}

7. Several antihypertensive agents in a variety of pharmacological classes are available for the treatment of hypertensive emergencies (Table 19).

12. Strategies to Improve Hypertension Treatment and Control

In addition to promoting pharmacological and nonpharmacological treatment adherence in individual patients with hypertension, several population-based systems approaches can play an important role in treatment goals.

12.1. Adherence Strategies for Treatment of Hypertension

Therapeutic nonadherence (not following recommended medical or health advice, including failure to “persist” with medications and recommended lifestyle modifications) is a major contributor to poor control of hypertension and a key barrier to reducing CVD deaths. Adherence rates vary substantially in different populations and, in general, are lower for lifestyle change and more behaviorally demanding regimens.

12.1.1. Antihypertensive Medication Adherence Strategies

Recommendations for Antihypertensive Medication Adherence Strategies		
References that support recommendations are summarized in Online Data Supplements 59 and 60.		
COR	LOE	Recommendations
I	B-R	1. In adults with hypertension, dosing of antihypertensive medication once daily rather than multiple times daily is beneficial to improve adherence. ^{S12.1.1-1-S12.1.1-3}
Ila	B-NR	2. Use of combination pills rather than free individual components can be useful to improve adherence to antihypertensive therapy. ^{S12.1.1-4-S12.1.1-7}

Synopsis

Up to 25% of patients do not fill their initial prescription for antihypertensive therapy.^{S12.1.1-8-S12.1.1-10} During the first year of treatment, the average patient has possession of antihypertensive medications only 50% of the time, and only 1 in 5 patients has sufficiently high adherence to achieve the benefits observed in clinical trials.^{S12.1.1-11,S12.1.1-12}

Factors contributing to poor adherence are myriad, complex, and multilevel.^{S12.1.1-11,S12.1.1-13,S12.1.1-14} Therefore, solutions to improve adherence may be introduced at patient, provider, and healthcare system levels.^{S12.1.1-13,S12.1.1-15,S12.1.1-16} Several systematic reviews and meta-analyses have assessed the impact of interventions on adherence to antihypertensive medications, including modification of antihypertensive therapy.^{S12.1.1-1-S12.1.1-7,S12.1.1-11,S12.1.1-15,S12.1.1-16} No single intervention is uniquely effective, and a sustained, coordinated effort that targets all barriers to adherence in an individual is likely to be the most effective approach. See [Online Data](#)

[Supplement F](#) for barriers to medication adherence and the most successful interventions.

The creation of an encouraging, blame-free environment in which patients are recognized for achieving treatment goals and given “permission” to answer questions related to their treatment honestly is essential to identify and address nonadherence. Patient medication adherence assessment tools^{S12.1.1-17} are presented in [Online Data Supplement A](#). Members of the hypertension care team may use these self-report tools in a nonthreatening fashion to identify barriers and facilitate behaviors associated with improved adherence to antihypertensive medications. Use of more objective methods (eg, pill counts, data on medication refills) to assess adherence along with self-report methods is optimal.

Recommendation-Specific Supportive Text

1. Remembering to take medication is often challenging, particularly for regimens that must be dosed several times daily. Taking medications several times throughout the day requires greater attention to scheduling, as well as additional issues such as transportation or storage, which can be challenging for some patients. The impact of once-daily dosing of antihypertensive drugs versus dosing multiple times daily has been evaluated in several meta-analyses.^{S12.1.1-1-S12.1.1-3} Medication adherence was greatest with once-daily dosing (range 71% to 94%) and declined as dosing frequency increased.^{S12.1.1-1,S12.1.1-2}
2. Assessment and possible modification of drug therapy regimens can improve suboptimal adherence. Simplifying medication regimens, either by less frequent dosing (ie, once daily versus multiple times daily) or use of combination drug therapy, improves adherence. Available fixed-dose combination drug therapy is listed in [Online Data Supplement D](#).

12.1.2. Strategies to Promote Lifestyle Modification

Recommendation for Strategies to Promote Lifestyle Modification		
References that support the recommendation are summarized in Online Data Supplement 61.		
COR	LOE	Recommendation
I	C-E0	1. Effective behavioral and motivational strategies to achieve a healthy lifestyle (ie, tobacco cessation, weight loss, moderation in alcohol intake, increased physical activity, reduced sodium intake, and consumption of a healthy diet) are recommended for adults with hypertension. ^{S12.1.2-1,S12.1.2-2}

Synopsis

The primary lifestyle modification interventions that can help reduce high BP are outlined in Section 6 (healthy diet, weight loss, exercise and moderate alcohol intake). In addition, tobacco cessation is crucial for CVD risk reduction. These modifications are central to good health and require

specific motivational and cognitive intervention strategies designed to promote adherence to these healthy behaviors. High-quality evidence supporting some of these strategies is provided in [Online Data Supplement G](#). Additionally, interventions such as goal setting, provision of feedback, self-monitoring, follow-up, motivational interviewing, and promotion of self-sufficiency are most effective when combined. Most individuals have clear expectations about what a new lifestyle will provide; if their experiences do not match these expectations, they will be dissatisfied and less motivated to maintain a lifestyle change, particularly in environments that do not support healthy choices. Other factors that may influence adoption and maintenance of new physical activity or dietary behaviors include age, sex, baseline health status, and body mass index, as well as the presence of comorbid conditions and depression, which negatively affect adherence to most lifestyle change regimens.^{S12.1.2-1} Primary strategies include cognitive-behavioral strategies for promoting behavior change, intervention processes and delivery strategies, and addressing cultural and social context variables that influence behavioral change.

Recommendation-Specific Supportive Text

1. It is crucial to translate and implement into practice the most effective evidence-based strategies for adherence to nonpharmacological treatment for hypertension. Both adoption and maintenance of new CVD risk-reducing behaviors pose challenges for many individuals. Success requires consideration of race, ethnicity, and socioeconomic status, as well as individual, provider, and environmental factors that may influence the design of such interventions.^{S12.1.2-1} High-quality evidence has shown that even modest sustained lifestyle changes can substantially reduce CVD morbidity and mortality.^{S12.1.2-1} Because many beneficial effects of lifestyle changes accrue over time, long-term adherence maximizes individual and population benefits. Interventions targeting sodium restriction, other dietary patterns, weight reduction, and new physical activity habits often result in impressive rates of initial behavior changes but frequently are not translated into long-term behavioral maintenance.

12.1.3. Improving Quality of Care for Resource-Constrained Populations

The availability of financial, informational, and instrumental support resources can be important though not sole determinants of hypertension control.^{S12.1.3-1, S12.1.3-2} The management of hypertension in resource-constrained populations poses a challenge that will require the implementation of all recommendations discussed in Section 13 (Table 21), with specific sensitivity to challenges posed by limited financial resources, including those related to health literacy, alignment of and potential need to realign healthcare priorities by patients, the convenience and complexity of the management strategy, accessibility to health care, and health-related costs (including medications). Resource-constrained populations are also populations with high representation of groups most likely to manifest health disparities, including racial and ethnic

Table 21. Clinician's Sequential Flow Chart for the Management of Hypertension

Clinician's Sequential Flow Chart for the Management of Hypertension	
Measure office BP accurately	Section 4
Detect white coat hypertension or masked hypertension by using ABPM and HBPM	Section 4
Evaluate for secondary hypertension	Section 5
Identify target organ damage	Sections 5 and 7
Introduce lifestyle interventions	Section 6
Identify and discuss treatment goals	Sections 7 and 8
Use ASCVD risk estimation to guide BP threshold for drug therapy	Section 8.1.2
Align treatment options with comorbidities	Section 9
Account for age, race, ethnicity, sex, and special circumstances in antihypertensive treatment	Sections 10 and 11
Initiate antihypertensive pharmacological therapy	Section 8
Insure appropriate follow-up	Section 8
Use team-based care	Section 12
Connect patient to clinician via telehealth	Section 12
Detect and reverse nonadherence	Section 12
Detect white coat effect or masked uncontrolled hypertension	Section 4
Use health information technology for remote monitoring and self-monitoring of BP	Section 12

ABPM indicates ambulatory blood pressure monitoring; ASCVD, atherosclerotic cardiovascular disease; BP, blood pressure; and HBPM, home blood pressure monitoring.

minorities (see Section 10.1), residents located in rural areas, and older adults. The more comprehensive BP targets proposed in the present guideline will present added challenges in these populations.

It is crucial to invest in measures to enhance health literacy and reinforce the importance of adhering to treatment strategies, while paying attention to cultural sensitivities. These measures may include identification of and partnering with community resources and organizations devoted to hypertension control and cardiovascular health. Although comparative-effectiveness data documenting efficacy of various interventions are limited, multidisciplinary team-based approaches and the use of community health workers (see Sections 12.1.1 and 12.2) have shown some utility, as has the use of out-of-office BP monitoring (or no-cost BP control visits), particularly among resource-constrained populations.^{S12.1.3-3–S12.1.3-5} Long-acting once-daily medications (eg, chlorthalidone, amlodipine) that are now available generically and often on discount formularies can often be used to reduce complexity of the regimen and promote adherence by decreasing the effect of missed medication dosages. When possible, prescriptions requiring longer than 30-day refills should be considered, especially once a

stable regimen is achieved. Where appropriate, using scored tablets and pill cutters can decrease the cost of medication for patients.

12.2. Structured, Team-Based Care Interventions for Hypertension Control

Recommendation for Structured, Team-Based Care Interventions for Hypertension Control

References that support the recommendation are summarized in [Online Data Supplement 62](#).

COR	LOE	Recommendation
I	A	1. A team-based care approach is recommended for adults with hypertension. ^{S12.2-1–S12.2-7}

Synopsis

Team-based care to improve BP control is a health systems-level, organizational intervention that incorporates a multidisciplinary team to improve the quality of hypertension care for patients.^{S12.2-8–S12.2-10} Various team-based hypertension care models have been demonstrated to increase the proportion of individuals with controlled BP and to reduce both SBP and DBP.^{S12.2-1–S12.2-7, S12.2-11, S12.2-12} A team-based care approach is patient centered and is frequently implemented as part of a multifaceted approach, with systems support for clinical decision making (ie, treatment algorithms), collaboration, adherence to prescribed regimen, BP monitoring, and patient self-management. Team-based care for hypertension includes the patient, the patient's primary care provider, and other professionals, such as cardiologists, nurses, pharmacists, physician assistants, dietitians, social workers, and community health workers. These professionals complement the activities of the primary care provider by providing process support and sharing the responsibilities of hypertension care. Section 13 contains a comprehensive, patient-centered plan of care that should be the basis of all team-based care for hypertension.

Team-based care aims to achieve effective control of hypertension by application of the strategies outlined in [Online Data Supplement H](#).^{S12.2-3} Delineation of individual team member roles on the basis of knowledge, skill set, and availability, as well as the patient's needs, allows the primary care provider to delegate routine matters to the team, thereby permitting more time to manage complex and critical patient-care issues. Important implementation aspects, such as type of team member added, role of team members related to medication management, and number of team members, influence BP outcomes.^{S12.2-3, S12.2-13} Team member roles should be clear to all team members and to patients and families.

Team-based care often requires organizational change and reallocation of resources.^{S12.2-14, S12.2-15} Systems-level support, such as use of electronic health records (EHR) (see Section 12.3.1), clinical decision support (ie, treatment algorithms), technology-based remote monitoring (see Section 12.3.2), self-management support tools, and monitoring of

performance, are likely to augment and intensify team-based care efforts to reduce high BP.

Recommendation-Specific Supportive Text

1. RCTs and meta-analyses of RCTs of team-based hypertension care involving nurse or pharmacist intervention demonstrated reductions in SBP and DBP and/or greater achievement of BP goals when compared with usual care.^{S12.2-1, S12.2-2, S12.2-4, S12.2-5} Similarly, systematic reviews of team-based care, including a review of studies that included community health workers, for patients with primary hypertension showed reductions in SBP and DBP and improvements in BP control, appointment keeping, and hypertension medication adherence as compared with usual care.^{S12.2-3, S12.2-12}

12.3. Health Information Technology–Based Strategies to Promote Hypertension Control

12.3.1. EHR and Patient Registries

Recommendations for EHR and Patient Registries

References that support recommendations are summarized in [Online Data Supplement 63](#).

COR	LOE	Recommendations
I	B-NR	1. Use of the EHR and patient registries is beneficial for identification of patients with undiagnosed or undertreated hypertension. ^{S12.3.1-1–S12.3.1-3}
I	B-NR	2. Use of the EHR and patient registries is beneficial for guiding quality improvement efforts designed to improve hypertension control. ^{S12.3.1-1–S12.3.1-3}

Synopsis

A growing number of health systems are developing or using registries and EHR that permit large-scale queries to support population health management strategies to identify undiagnosed or undertreated hypertension. Such innovations are implemented as ongoing quality improvement initiatives in clinical practice. To reduce undiagnosed hypertension and improve hypertension management, a multipronged approach may include 1) application of hypertension screening algorithms to EHR databases to identify at-risk patients, 2) contacting at-risk patients to schedule BP measurements, 3) monthly written feedback to clinicians about at-risk patients who have yet to complete a BP measurement, and 4) electronic prompts for BP measurements whenever at-risk patients visit the clinic.^{S12.3.1-1, S12.3.1-2}

Recommendation-Specific Supportive Text

1. A growing number of health systems have implemented secure EHR and are developing databases that permit large-scale queries to support population health management strategies for more effective and accurate identification of patients with hypertension.^{S12.3.1-1–S12.3.1-3}
2. A growing number of health systems have implemented secure EHR and are developing databases that permit large-scale quality improvement initiative–designed

queries to support population health management strategies for more effective management and control of hypertension.^{S12.3.1-1–S12.3.1-3}

12.3.2. Telehealth Interventions to Improve Hypertension Control

Recommendation for Telehealth Interventions to Improve Hypertension Control		
References that support the recommendation are summarized in Online Data Supplement 64 .		
COR	LOE	Recommendation
Ila	A	1. Telehealth strategies can be useful adjuncts to interventions shown to reduce BP for adults with hypertension. ^{S12.3.2-1–S12.3.2-5}

Synopsis

Telehealth strategies, such as telemedicine, digital health (“eHealth”), and use of mobile computing and communication technologies (“mHealth”), are new and innovative tools to facilitate improvements in managing patients with hypertension. mHealth interventions show promise in reducing SBP in patients with hypertension but with large variability in behavioral targets, intervention components, delivery modalities, and patient engagement.^{S12.3.2-5} In addition, there are important implications for the role of social networks, social media, and electronic technology as viable components of weight management and other lifestyle modification and disease management programs.^{S12.3.2-6}

Commonly used telehealth interventions for hypertension management are listed in [Online Data Supplement I](#). Wireless technologies ([Online Data Supplement I](#)) allow linking BP devices and other measurement devices to telephone- or Internet-based transmission systems or to Wi-Fi access points available in users’ homes and in communities. Some systems require patients to manually enter data, which is then forwarded to a remote computer or the mobile device of the telehealth provider through a telephone line or the Internet.^{S12.3.2-7} When data are received, they are stored and analyzed, and reports are generated, including variations and averages in BP and other parameters over the recording period.

Recommendation-Specific Supportive Text

1. Meta-analyses of RCTs of different telehealth interventions have demonstrated greater SBP and DBP reductions^{S12.3.2-1,S12.3.2-2,S12.3.2-4} and a larger proportion of patients achieving BP control^{S12.3.2-2} than those achieved with usual care without telehealth. The effect of various telehealth interventions on BP lowering was significantly greater than that of BP self-monitoring without transmission of BP data, which suggests a possible added value of the teletransmission approach.^{S12.3.2-1,S12.3.2-3} Although mHealth interventions in general showed promise in reducing SBP in patients with hypertension, results were inconsistent.^{S12.3.2-5} It is unclear which combination of telehealth intervention features is most effective, and telehealth has not been demonstrated to be effective as a standalone strategy for improving hypertension control.

12.4. Improving Quality of Care for Patients With Hypertension

12.4.1. Performance Measures

Recommendation for Performance Measures		
References that support the recommendation are summarized in Online Data Supplement 65 .		
COR	LOE	Recommendation
Ila	B-NR	1. Use of performance measures in combination with other quality improvement strategies at patient-, provider-, and system-based levels is reasonable to facilitate optimal hypertension control. ^{S12.4.1-1–S12.4.1-3}

Synopsis

Efforts to improve suboptimal medical care include the use of performance measures, which are defined as standardized, validated approaches to assess whether correct healthcare processes are being performed and that desired patient outcomes are being achieved.^{S12.4.1-4} Performance measures are often combined with other quality improvement strategies, such as certification or financial incentives tied to higher-quality care.^{S12.4.1-5} Guidelines help define clinical care standards that can be used to develop performance measures. As guidelines evolve over time to incorporate new evidence, related performance measures may also evolve.

Because identification, treatment, and control of hypertension are suboptimal, performance measures for hypertension control have been developed and recommended for use in quality improvement projects aimed at improving hypertension control and related outcomes in clinical practice.^{S12.4.1-6–S12.4.1-8} Because the specific methods used in performance measures can have an impact on their accuracy and ultimate impact (eg, the method of BP measurement used in the assessment), they should be developed, tested, and implemented according to published standards.^{S12.4.1-9} See [Online Data Supplement J](#) for publicly available performance measures to assess the quality of hypertension care (generally using JNC 7 criteria).

Recommendation-Specific Supportive Text

1. RCTs on the impact of performance measures on hypertension control are lacking; RCTs of quality improvement protocols have shown improvements in hypertension control.^{S12.4.1-1,S12.4.1-2} Furthermore, a large observational study showed that a systematic approach to hypertension control, including the use of performance measures, was associated with significant improvement in hypertension control compared with historical control groups.^{S12.4.1-3}

12.4.2. Quality Improvement Strategies

Recommendation for Quality Improvement Strategies		
References that support the recommendation are summarized in Online Data Supplements 66 and 67 .		
COR	LOE	Recommendation
Ila	B-R	1. Use of quality improvement strategies at the health system, provider, and patient levels to improve identification and control of hypertension can be effective. ^{S12.4.2-1–S12.4.2-8}

Synopsis

High-quality BP management is multifactorial and requires the engagement of patients, families, providers, and healthcare delivery systems.^{S12.4.2-9} The difference between patient outcomes achieved with current hypertension treatment methods and patient outcomes thought to be possible with best-practice treatment methods is known as a quality gap, and such gaps are at least partly responsible for the loss of thousands of lives each year.^{S12.4.2-10} This includes expanding patient and healthcare provider awareness, appropriate lifestyle modifications, access to care, evidence-based treatment, a high level of medication adherence, and adequate follow-up.^{S12.4.2-9} Quality improvement strategies or interventions aimed at reducing the quality gap for a group of patients who are representative of those encountered in routine practice have been effective in improving the hypertension care and outcomes across a wide variety of clinic and community settings.^{S12.4.2-1–S12.4.2-4,S12.4.2-6,S12.4.2-8,S12.4.2-10}

Hypertension quality improvement strategies, with examples of substrategies that have been demonstrated to reduce BP and improve BP, are provided in [Online Data Supplement E](#). Because the effects of the different quality improvement strategies varied across trials, and most trials included >1 quality improvement strategy, it is not possible to discern which specific quality improvement strategies have the greatest effects. Team-based care (see Section 12.4) and an organized system of regular review, with antihypertensive drug therapy implemented via a stepped-care protocol, had a clinically significant effect on reducing SBP and DBP and improving BP control. The assessed strategies in [Online Data Supplement E](#) may be beneficial under some circumstances and in varying combinations.^{S12.4.2-1–S12.4.2-5} National initiatives such as Million Hearts Make Control Your Goal Blood Pressure Toolkit and Team Up Pressure Down provide quality improvement tools to support hypertension care in communities and clinical settings.^{S12.4.2-11} For other national and regional initiatives to improve hypertension, see [Online Data Supplement G](#).

Recommendation-Specific Supportive Text

1. Systematic review and meta-analyses of trials of quality improvement interventions at health system, provider, and patient levels have demonstrated greater SBP and DBP reductions and a larger proportion of patients achieving BP control than those observed with no intervention or usual care. Multicomponent and multilevel strategies at the local community and healthcare delivery system levels have been shown to improve BP control.^{S12.4.2-6,S12.4.2-7}

12.5. Financial Incentives

Recommendations for Financial Incentives		
References that support recommendations are summarized in Online Data Supplement 68 .		
COR	LOE	Recommendations
Ia	B-R	1. Financial incentives paid to providers can be useful in achieving improvements in treatment and management of patient populations with hypertension. ^{S12.5-1–S12.5-3}

Recommendations for Financial Incentives (Continued)		
COR	LOE	Recommendations
Ia	B-NR	2. Health system financing strategies (eg, insurance coverage and copayment benefit design) can be useful in facilitating improved medication adherence and BP control in patients with hypertension. ^{S12.5-4}

Synopsis

With the evolution of the US health system to reward “value over volume,” payment systems have focused on financial incentives to improve quality of care. Use of performance measures promulgated by national organizations, governmental payers, and commercial payers have fostered greater attention to control of high BP among healthcare providers and their patients. These performance measures have formed the basis for determining financial incentives for pay for performance initiatives, commercial insurer “pay-for-value” contracts, and the Medicare Shared Savings Programs developed by the Centers for Medicare & Medicaid Services Innovation for Accountable Care Organizations. In addition, the Centers for Medicare and Medicaid Services has developed The Million Hearts: Cardiovascular Disease Risk Reduction Model, which is an RCT designed to identify and test scalable models of care delivery that reduce CVD risk.^{S12.5-5}

Greater attention is being paid to the influence of health insurance coverage and benefit designs focused on reducing patient copayments for antihypertensive medications.

Recommendation-Specific Supportive Text

1. Moderate-quality evidence with mixed results suggests that population-based payment incentive programs can play an important role in achieving better BP control.^{S12.5-1–S12.5-3}
2. Reduced copayments for health care, including for medications, and improved outcomes of hypertension care have been identified in several US studies and in single studies in Finland, Israel, and Brazil.^{S12.5-4} This is consistent with other evidence on how copayments reduce uptake of care and has implications for policy makers, particularly because the balance of evidence does not suggest that reducing medication copayments leads to an increase in overall healthcare expenditure.

13. The Plan of Care for Hypertension

Recommendation for the Plan of Care for Hypertension		
COR	LOE	Recommendation
I	C-EO	1. Every adult with hypertension should have a clear, detailed, and current evidence-based plan of care that ensures the achievement of treatment and self-management goals, encourages effective management of comorbid conditions, prompts timely follow-up with the healthcare team, and adheres to CVD GDMT (Table 22).

Synopsis

A specific plan of care for hypertension is essential and should reflect understanding of the modifiable and nonmodifiable determinants of health behaviors, including the social determinants of risk and outcomes. A clinician's sequential flow chart for management of hypertension is presented in Table 21. Detailed evidence-based elements of the plan of care are listed in Table 22. The determinants will vary among demographic subgroups (see Section 10 for additional information).

Recommendation-Specific Supportive Text

1. Studies demonstrate that implementation of a plan of care for hypertension can lead to sustained reduction of BP and attainment of BP targets over several years.^{S13-1–S13-6} Meta-analysis of RCTs shows reductions in BP of patients with hypertension and achievement of BP goals at 6 months and 1 year when compared with usual care.

13.1. Health Literacy

Communicating alternative behaviors that support self-management of healthy BP in addition to medication adherence is important. This should be done both verbally and in writing. Today, mobile phones have a recording option. For patients with mobile phones, the phone can be used to inform patients and family members of medical instructions after the doctor's visit as an additional level of communication. Inclusion of a family member or friend that can help interpret and encourage self-management treatment goals is suggested when appropriate. Examples of needed communication for alternative behaviors include a specific regimen relating to physical activity; a specific sodium-reduced meal plan indicating selections for breakfast, lunch, and dinner; lifestyle recommendations relating to sleep, rest, and relaxation; and finally, suggestions and alternatives to environmental barriers, such as barriers that prevent healthy food shopping or limit reliable transportation to and from appointments with health providers and pharmacy visits.

13.2. Access to Health Insurance and Medication Assistance Plans

Health insurance and medication plan assistance for patients is especially important to improving access to and affordability of medical care and BP medications. Learning how the patient financially supports and budgets for his or her medical care and medications offers the opportunity to share additional insight relating to cost reductions, including restructured payment plans. Ideally, this would improve the patient's compliance with medication adherence and treatment goals.

13.3. Social and Community Services

Health care can be strengthened through local partnerships. Hypertensive patients, particularly patients with lower incomes, have more opportunity to achieve treatment goals with the assistance of strong local partnerships. In patients with low socioeconomic status or patients who are challenged by social situations, integration of social and community services offers complementary reinforcement of clinically identified treatment goals. Social and community services are helpful when explicitly related to medical care. However, additional

Table 22. Evidence-Based Elements of the Plan of Care for Patients With Hypertension

Plan of Care	Associated Section(s) of Guideline and Other Reference(s)
Pharmacological and nonpharmacological treatments	
Medication selection (initial and ongoing)	Section 8.1
Monitoring for adverse effects and adherence	Sections 8.3.1, 8.3.2, 12.1.1
Nonpharmacological interventions Diet Exercise Weight loss if overweight Moderate alcohol consumption	Sections 6, 12.1.2 ^{S13.1-2}
Management of common comorbidities and conditions	
Ischemic heart disease	Section 9.1 ^{S13.1-3, S13.1-4}
Heart failure Reduced ejection fraction Preserved ejection fraction	Section 9.2 ^{S13.1-5}
Diabetes mellitus	Section 9.6 ^{S13.1-6}
Chronic kidney disease	Section 9.3
Cerebrovascular disease	Section 9.4
Peripheral artery disease	Section 9.5
Atrial fibrillation	Section 9.8
Valvular heart disease	Section 9.9
Left ventricular hypertrophy	Section 7.3
Thoracic aortic disease	Section 9.10
Patient and family education	
Achieving BP control and self-monitoring	Sections 4.2, 8.2
Risk assessment and prognosis	Section 8.1.2
Sexual activity and dysfunction	Section 11.4
Special patient groups	
Pregnancy	Section 10.2.2
Older persons	Section 10.3.1
Children and adolescents	Section 10.3.2
Metabolic syndrome	Section 9.7
Possible secondary causes of hypertension	Section 5.4
Resistant hypertension	Section 11.1
Patients with hypertension undergoing surgery	Section 11.5
Renal transplantation	Section 9.3.1
Psychosocial factors	
Sex-specific issues	Section 10.2
Culturally sensitive issues (race and ethnicity)	Section 10.1
Resource constraints	Section 12.5

(Continued)

Table 22. Continued

Plan of Care	Associated Section(s) of Guideline and Other Reference(s)
Clinician follow-up, monitoring, and care coordination	
Follow-up visits	Sections 8.1.3, 8.3.1, 8.3.2
Team-based care	Section 12.2
Electronic health record	Section 12.3.1
Health information technology tools for remote and self-monitoring	Section 12.3.2
Socioeconomic and cultural factors	
Health literacy	Section 13.1.3
Access to health insurance and medication assistance plans	Section 13.1.3
Social services	Section 13.1.3
Community services	Section 13.1.3

BP indicates blood pressure.

financial support and financial services are incredibly beneficial to patients, some of whom may choose to skip a doctor's appointment to pay a residential utility bill.

14. Summary of BP Thresholds and Goals for Pharmacological Therapy

Several different BP thresholds and goals for the long-term treatment of hypertension with pharmacological therapy are recommended in this guideline. To provide a quick reference for practicing clinicians, these are summarized for hypertensive patients in general and for those with specific comorbidities in Table 23.

15. Evidence Gaps and Future Directions

In the present guideline, the writing committee was able to call on the large body of literature on BP and hypertension to make strong recommendations across a broad range of medical conditions. Nonetheless, significant gaps in knowledge exist.

Importantly, there are areas where epidemiological and natural history studies suggest that hypertension prevention or earlier treatment of hypertension might substantially improve outcomes, but clinical trials are lacking to provide guidance. The combination of epidemiological data showing a graded relationship between BP and outcomes, particularly above a BP of 120/80 mm Hg, and the results of the SPRINT trial showing benefit of more comprehensive treatment to a target BP of <120/80 mm Hg, suggests that a lifelong BP below that level will substantially lower CVD and CKD incidence. This is especially the case for younger individuals, those with DM, and those with high lifetime CVD risk based on the presence of multiple risk factors, including high BP. If hard, cardiovascular outcome clinical trials remain the sole driver of evidence-based guidelines, then determining the full benefit of earlier intervention may not be possible because of the cost and length of time needed for intervention. Outcomes may be different if antihypertensive treatment is initiated earlier in the natural history

Table 23. BP Thresholds for and Goals of Pharmacological Therapy in Patients With Hypertension According to Clinical Conditions

Clinical Condition(s)	BP Threshold, mm Hg	BP Goal, mm Hg
General		
Clinical CVD or 10-year ASCVD risk $\geq 10\%$	$\geq 130/80$	<130/80
No clinical CVD and 10-year ASCVD risk <10%	$\geq 140/90$	<130/80
Older persons (≥ 65 years of age; noninstitutionalized, ambulatory, community-living adults)	≥ 130 (SBP)	<130 (SBP)
Specific comorbidities		
Diabetes mellitus	$\geq 130/80$	<130/80
Chronic kidney disease	$\geq 130/80$	<130/80
Chronic kidney disease after renal transplantation	$\geq 130/80$	<130/80
Heart failure	$\geq 130/80$	<130/80
Stable ischemic heart disease	$\geq 130/80$	<130/80
Secondary stroke prevention	$\geq 140/90$	<130/80
Peripheral artery disease	$\geq 130/80$	<130/80

ASCVD indicates atherosclerotic cardiovascular disease; BP, blood pressure; CVD, cardiovascular disease; and SBP, systolic blood pressure.

of CVD. DM may provide a population in whom to test this hypothesis. Composite outcomes that include both prevention of events and surrogates, such as prevention of decline in renal function or amelioration of measures of subclinical atherosclerosis, vascular stiffness, or LV structure and function, should be considered. Otherwise, these younger individuals may be undertreated and experience mortality or CVD events before being old enough to enter hard outcome-driven trials such as SPRINT. Replication of SPRINT, especially in younger patients with DM and in countries where nonischemic stroke is the predominant cause of CVD, is highly desirable. Likewise, implementation studies that demonstrate the practicality of SPRINT-like interventions in resource-constrained practice settings are needed.

More information is urgently needed relating hypertensive target organ damage to CVD risk and outcomes. Should the identification of target organ damage and hypertensive heart disease prompt more aggressive BP management (ie, increase the rationale for instituting pharmacological therapy earlier or more intensively)? Should all patients with hypertension be screened with echocardiogram for LVH? Should echocardiography be repeated once LVH is noted? Is it important to document LVH regression? At present, there are no RCT data to inform guideline recommendations.

ABPM and HBPM provide enhanced ability to both diagnose hypertension and monitor treatment. Although evidence is sufficient to recommend incorporating these tools into clinical practice, more knowledge about them is required. Areas of inquiry include closer mapping of the relationship of outcomes to ambulatory and home BP measurements, so that definitions of hypertension and hypertension severity based on

these measures can be developed, including the importance of masked hypertension, white coat hypertension, and nocturnal hypertension. Reproducibility of ambulatory and home BPs must be studied, and cohorts should include a broader range of ethnicities. Trials with entry criteria and treatment goals based on ambulatory or home BP measures should be conducted, including studies of masked and white coat hypertension. The practicality and cost of incorporating ABPM into EHR and routine care should be assessed. The existence of these techniques should not hamper efforts to investigate ways to improve accuracy in the measurement of clinic BP. Further research on improving accuracy of office BP measurements, including number of measurements, training of personnel measuring BP, and device comparisons, will help standardize care and thus improve outcomes. Technology for measurement of BP continues to evolve with the emergence of cuffless devices and other strategies that provide the opportunity for continuous noninvasive assessment of BP. The accuracy, cost, and usefulness of these new technologies will need to be assessed.

The contemporary healthcare environment is dramatically different from the era in which awareness of hypertension as a risk factor and benefits of treatment were discovered. With the advent of the EHR, complex calculations of CVD risk and renal function can be incorporated into routine reports, and many new avenues to support intervention strategies are available to clinicians. Optimizing these approaches will require continued focused research. Recognition that simply applying what we know about BP control would have a large impact on population health, observations on inefficiencies and excessive cost in the US healthcare system, and the growth of information technology have led to promising studies of ways to improve and monitor hypertension care. Results of this research are reflected in this guideline, but further work is required. Examples for study include the effectiveness of multidisciplinary healthcare teams to achieve BP treatment goals at lower cost, social media to maintain contact with patients, information technology to monitor outcomes and decrease practice variability, and incentives to providers to achieve better outcomes for patients. A key goal of these efforts should be to demonstrate reduction in healthcare disparities across ethnicity, sex, social and economic class, and age barriers.

More research on the prevention of the development of hypertension and the benefit of lifetime low BP should be conducted. In this regard, elucidation of genetic expression, epigenetic effects, transcriptomics, and proteomics that link genotypes with longitudinal databases may add considerable knowledge about beneficial outcomes of lifelong lower BP, determinants of rise in BP over time, and identification of new treatment targets through understanding the underlying pathophysiological mechanisms. Research should be directed toward the development of therapies that directly counteract the mechanisms accounting for the development of hypertension and disease progression. Additional research aimed at development of practical approaches to implementation of clinical and population-based strategies to prevent obesity, increase physical fitness, and control excess salt and sugar intake could have significant public health impact. In addition, there are minimal, if any, data on whether treatment of hypertension during pregnancy mitigates risk; thus, there

is a need for further research in this area, considering both proximate (during the pregnancy and postpartum period) and distant (CVD prevention) outcomes.^{S15-1}

In the very old, frailty and higher risk of medication side effects complicate treatment. Additional knowledge of the effects of antihypertensive treatment for patients with dementia and patients who reside in long-term-care facility settings is needed. The best approach to older persons who have supine hypertension but postural hypotension needs to be clarified.

Further research related to shared decision-making with patients and their families is needed. Examples include areas where evidence does not clearly identify one treatment or goal as substantially better than another, where improved patient knowledge (or improved provider knowledge of the patient's circumstances) might improve compliance, where reliance on patient collaboration improves achievement of outcomes (eg, HBPM, use of social media), and where there are competing health concerns (eg, older individuals with frailty).

Finally, clinical guidelines are increasingly required to manage the large body of accumulated knowledge related to diagnosis and management of high BP. However, guidelines often cause controversy and confusion when competing recommendations are made by different "expert" groups or when changes in definitions, treatments, or treatment goals are introduced. Now may be the time to begin the investigation of the impact of guidelines on clinical practice, costs, and patient outcomes, as well as ways to facilitate communication and collaboration between different guideline-developing organizations. This document is, as its name implies, a guide. In managing patients, the responsible clinician's judgment remains paramount.

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1. Introduction

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8. Treatment of High BP

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KEY WORDS: AHA Scientific Statements ■ ambulatory care ■ antihypertensive agents ■ behavior modification ■ blood pressure ■ chronic kidney disease ■ diabetes ■ hypertension ■ hypertension emergency ■ lifestyle measures ■ measurement ■ nonpharmacologic treatment ■ resistant hypertension ■ risk reduction ■ secondary hypertension ■ systems of care ■ treatment adherence ■ treatment outcomes

Appendix 1. Author Relationships With Industry and Other Entities (Relevant)—2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults (October 2017)

Committee Member	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Salary
Paul K. Whelton, Chair	Tulane University School of Hygiene and Tropical Medicine—Show Chwan Professor of Global Public Health	None	None	None	None	None	None	None
Robert M. Carey, Vice Chair	University of Virginia School of Medicine—Dean, Emeritus, and Professor of Medicine	None	None	None	• Daiichi Sankyo Inc†	None	None	None
Wilbert S. Aronow	Westchester Medical Center and New York Medical College—Professor of Medicine	None	None	None	None	None	None	None
Donald E. Casey, Jr	Thomas Jefferson College of Population Health—Adjunct Faculty; Alvarez & Marsal Ipo4health—Principal and Founder	None	None	None	None	None	None	None
Karen J. Collins	Collins Collaboration—President	None	None	None	None	None	None	None
Cheryl Dennison Himmelfarb	John Hopkins University—Professor of Nursing and Medicine, Institute for Clinical and Translational Research	None	None	None	None	None	None	None
Sondra M. DePalma	PinnacleHealth CardioVascular Institute—Physician Assistant; American Academy of PAs—Director, Regulatory and Professional Practice	None	None	None	None	None	None	None
Samuel Gidding	Alfred I. Dupont Hospital for Children—Chief, Division of Pediatric Cardiology, Nemours Cardiac Center	None	None	None	None	None	None	None
David C. Goff, Jr*	Colorado School of Public Health—Professor and Dean, Department of Epidemiology	None	None	None	None	None	None	None
Kenneth A. Jamerson	University of Michigan Health System—Professor of Internal Medicine and Frederick G.L. Huetwell Collegiate Professor of Cardiovascular Medicine	None	None	None	None	None	None	None
Daniel W. Jones	University of Mississippi Medical Center—Professor of Medicine and Physiology; Metabolic Diseases and Nutrition—University Sanderson Chair in Obesity Mississippi Center for Obesity Research—Director, Clinical and Population Science	None	None	None	None	None	None	None
Eric J. MacLaughlin	Texas Tech University Health Sciences Center—Professor and Chair, Department of Pharmacy Practice, School of Pharmacy	None	None	None	None	None	None	None
Paul Muntner	University of Alabama at Birmingham—Professor, Department of Epidemiology	None	None	None	None	None	None	None
Bruce Ovbiagele	Medical University of South Carolina—Pihl Professor and Chairman of Neurology	None	• Boehringer Ingelheim Korea Ltd	None	None	None	None	None
Sidney C. Smith, Jr	University of North Carolina at Chapel Hill—Professor of Medicine; Center for Cardiovascular Science and Medicine—Director	None	None	None	None	None	None	None
Crystal C. Spencer	Spencer Law, PA—Attorney at Law	None	None	None	None	None	None	None

(Continued)

Appendix 1. Continued

Committee Member	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Salary
Randall S. Stafford	Stanford Prevention Research Center—Professor of Medicine; Program on Prevention Outcomes—Director	None	None	None	None	None	None	None
Sandra J. Taler	Mayo Clinic—Professor of Medicine, College of Medicine	None	None	None	None	None	None	None
Randal J. Thomas	Mayo Clinic—Medical Director, Cardiac Rehabilitation Program	None	None	None	None	None	None	None
Kim A. Williams, Sr	Rush University Medical Center—James B. Herrick Professor; Division of Cardiology—Chief	None	None	None	None	None	None	None
Jeff D. Williamson	Wake Forest Baptist Medical Center—Professor of Internal Medicine; Section on Gerontology and Geriatric Medicine—Chief	None	None	None	None	None	None	None
Jackson T. Wright, Jr	Case Western Reserve University—Professor of Medicine; William T. Dahms MD Clinical Research Unit—Program Director; University Hospitals Case Medical Center—Director, Clinical Hypertension Program	None	• Amgen†	None	None	None	None	None

This table represents the relationships of committee members with industry and other entities that were determined to be relevant to this document. These relationships were reviewed and updated in conjunction with all meetings and/or conference calls of the writing committee during the document development process. The table does not necessarily reflect relationships with industry at the time of publication. A person is deemed to have a significant interest in a business if the interest represents ownership of $\geq 5\%$ of the voting stock or share of the business entity, or ownership of $\geq \$5000$ of the fair market value of the business entity; or if funds received by the person from the business entity exceed 5% of the person's gross income for the previous year. Relationships that exist with no financial benefit are also included for the purpose of transparency. Relationships in this table are modest unless otherwise noted.

According to the ACC/AHA, a person has a *relevant* relationship IF: a) the *relationship or interest* relates to the same or similar subject matter, intellectual property or asset, topic, or issue addressed in the *document*; or b) the *company/entity* (with whom the relationship exists) makes a drug, drug class, or device addressed in the document, or makes a competing drug or device addressed in the *document*; or c) the *person or a member of the person's household*, has a reasonable potential for financial, professional or other personal gain or loss as a result of the issues/content addressed in the *document*.

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*Dr. David C. Goff resigned from the writing committee in December 2016 because of a change in employment before the recommendations were balloted. The writing committee thanks him for his contributions, which were extremely beneficial to the development of the draft.

†Significant relationship.

AAPA indicates American Academy of Physician Assistants; ACC, American College of Cardiology; ACPM, American College of Preventive Medicine; AGS, American Geriatrics Society; AHA, American Heart Association; APHA, American Pharmacists Association; ASH, American Society of Hypertension; ASPC, American Society for Preventive Cardiology; ABC, Association of Black Cardiologists; NMA, National Medical Association; and PCNA, Preventive Cardiovascular Nurses Association.

Appendix 2. Reviewer Relationships With Industry and Other Entities (Comprehensive)—2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults (October 2017)

Reviewer	Representation	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Salary
Kim K. Birtcher	Official Reviewer—TFPG Lead Reviewer	University of Houston College of Pharmacy—Clinical Professor, Department of Pharmacy Practice and Translational Research	• Jones & Bartlett Learning	None	None	None	• Accreditation Council for Clinical Lipidology†	None	• Walgreens*
Roger Blumenthal	Official Reviewer—Prevention Subcommittee	Johns Hopkins Hospital—Kenneth Jay Pollin Professor of Cardiology; Ciccarone Center for the Prevention of Heart Disease—Director	None	None	None	None	None	None	None
Anna Dominiczak	Official Reviewer—AHA	University of Glasgow—Regius Professor of Medicine; Vice-Principal and Head of College of Medical, Veterinary and Life Sciences	None	None	None	None	None	None	None
Carlos M. Ferrario	Official Reviewer—AHA	Wake Forest School of Medicine—Professor, of Physiology and Pharmacology; Hypertension and Vascular Disease Center—Director	None	None	None	None	None	None	None
Eugene Yang	Official Reviewer—ACC-BOG	University of Washington School of Medicine—Associate Clinical Professor of Medicine; UW Medicine Eastside Specialty Center—Medical Director	• RubiconMD* • Regeneron*	None	None	• Amgen Inc.* • Gilead Sciences, Inc. (DSMB)*	None	• Third party, CAD, 2016*	None
Robert Jay Amrien	Organizational Reviewer—AAPA	Massachusetts General Hospital—Clinical Physician Assistant, Chelsea Health Center; Bryant University—Physician Assistant Program	None	None	None	None	None	• Defendant, aortic dissection, 2016*	None
Greg Holzman	Organizational Reviewer—ACPM	Montana Department of Public Health and Human Services—State Medical Officer	None	None	None	None	• American Academy of Family Medicine† • American College of Preventive Medicine†	None	None
Martha Gulati	Organizational Reviewer—ASPC	University of Arizona College of Medicine—Professor of Medicine; Chief, Division of Cardiology; University Medicine Cardiovascular Institute in Phoenix—Physician Executive Director, Banner	None	None	None	None	• REATA (spouse)*	None	None
Wallace Johnson	Organizational Reviewer—NMA	University of Maryland Medical Center—Assistant Professor of Medicine	None	None	None	Amgen†	None	None	None
Nancy Houston Miller	Organizational Reviewer—PCNA	The Lifecare Company—Associate Director	• Moving Analytics*	None	None	None	None	None	None

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Reviewer	Representation	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Salary
Aldo J. Peixoto	Organizational Reviewer—ASH	Yale University School of Medicine—Professor of Medicine (Nephrology); Associate Chair for Ambulatory Services Operations and Quality, Department of Internal Medicine; Clinical Chief, Section of Nephrology	• Lundbeck Inc.	None	None	• Bayer Healthcare Pharmaceuticals†	• Bayer Healthcare Pharmaceuticals	None	None
Carlos Rodriguez	Organizational Reviewer—ABC	Wake Forest University—Professor, Epidemiology and Prevention	• Amgen Inc.	None	None	None	None	None	None
Joseph Saseen	Organizational Reviewer—APhA	University of Colorado Anschutz Medical Campus—Vice-Chair, Department of Clinical Pharmacy, Skaggs School of Pharmacy and Pharmaceutical Sciences	None	None	None	None	• National Lipid Association†	• Defendant, statin use, 2016	None
Mark Supiano	Organizational Reviewer—AGS	University of Utah School of Medicine—D. Keith Barnes, MD, and Dottie Barnes Presidential Endowed Chair in Medicine; Chief, Division of Geriatrics; VA Salt Lake City Geriatric Research—Director, Education, and Clinical Center; University of Utah Center on Aging Executive—Director	None	None	None	None	• American Geriatrics Society† • Division Chief† • McGraw-Hill Medical	None	None
Sana M. Al-Khatib	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	Duke Clinical Research Institute—Professor of Medicine	None	None	None	• AHRQ* • FDA* • PCORI* • VA Health System (DSMB)	• Elsevier* • NIH, NHLBI	• Third party, implantable cardioverter defibrillators, 2017	None
George Bakris	Content Reviewer	University of Chicago Medicine—Professor of Medicine; Director, Hypertensive Diseases Unit	None	None	None	• AbbVie, Inc. • Janssen, Bayer, Relypsa	None	None	None
Jan Basile	Content Reviewer	Medical University of South Carolina—Professor of Medicine, Seinsheimer Cardiovascular Health Program; Ralph H Johnson VA Medical Center—Internist	None	• Amgen Inc. • Arbor • Janssen Pharmaceuticals, Inc	None	• Eli Lilly and Company • NHLBI	None	None	None
Joshua A. Beckman	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	Vanderbilt University Medical Center: Director, Cardiovascular Fellowship Program,	• AstraZeneca* • Merck* • SANOFI*	None	• EMX† • JanaCare†	• Bristol Myers Squibb*	• Vascular Interventional Advances*	None	• 2015 Defendant; Venous thromboembolism*
John Bisognano	Content Reviewer	University of Rochester Medical Center—Cardiologist	• CVRx	None	None	• CVRx* • NIH*	None	None	None
Biykem Bozkurt	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	Baylor College of Medicine—Medical Care Line Executive, Cardiology Chief, Gordon Cain Chair, Professor of Medicine, DeBakey	None	None	None	• Novartis Corporation	None	None	None

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Reviewer	Representation	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Salary
David Calhoun	Content Reviewer	University of Alabama, Birmingham School of Medicine—Professor, Department of Cardiovascular Disease	- Novartis - Valencia Technologies*	None	None	- MEDTRONIC* - ReCor Medical*	None	None	None
Joaquin E. Cigarroa	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	Oregon Health and Science University—Clinical Professor of Medicine	None	None	None	NIH	- ACC/AHA Taskforce on Clinical Practice Guidelines† - AHA, Board of Directors, Western Affiliate† - American Stroke Association, Cryptogenic Stroke Initiative Advisory Committee† - Catheterization and Cardiovascular Intervention† - SCAI Quality Interventional Council†	- Defendant, CAD, 2011† - Defendant, sudden death/ CAD, 2010†	None
William Cushman	Content Reviewer	Memphis VA Medical Center—Chief, Preventive Medicine Section; University of Tennessee College of Medicine—Professor, Medicine, Preventive Medicine, and Physiology	None	None	None	Lilly	- Novartis Corporation† - Takeda†	None	None
Anita Deswal	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	Baylor College of Medicine—Associate Professor of Medicine	None	None	None	NIH*	- Aurora Health Care Inc. - American Heart Association† - AHA Committee on Heart Failure and Transplantation – Chair† - Heart Failure Society of America†	None	None
Dave Dixon	Content Reviewer—Cardiovascular Team	Virginia Commonwealth University School of Pharmacy—Associate Professor	None	None	None	None	None	None	None
Ross Feldman	Content Reviewer	Winnipeg Regional Health Authority—Medical Director, Cardiac Sciences Program; University of Manitoba—Professor of Medicine	- GSK* - Servier* - Valeant Pharmaceuticals International*	None	None	None	None	None	None
Keith Ferdinand	Content Reviewer	Tulane University School of Medicine—Professor of Clinical Medicine	- Amgen Inc.* - Boehringer Ingelheim* - Eli Lilly* - Sanofi-Aventis* - Novartis - Quantum Genomics - Sanofi-Aventis*	None	None	None	Novartis	None	None
Stephan Fihn	Content Reviewer	University of Washington—Professor of Medicine, Health Services; Division Head, General Internal Medicine; Director, Office of Analytics and Business Intelligence for the Veterans Health Administration; VA Puget Sound Health Care System—General Internist	None	None	None	None	University of Washington	None	None

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Reviewer	Representation	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Salary
Lawrence Fine	Content Reviewer	National Heart, Lung and Blood Institute—Chief, Clinical Applications and Prevention Branch, Division of Prevention and Population Sciences	None	None	None	None	• NIH*	None	None
John Flack	Content Reviewer	Southern Illinois University School of Medicine—Chair and Professor Department of Internal Medicine; Chief, Hypertension Specialty Services	• Regeneron* • NuSirt	None	None	• Bayer Healthcare Pharmaceuticals† • GSK†	• American Journal of Hypertension* • CardioRenal Medicine† • International Journal of Hypertension† • Southern Illinois University Department of Medicine*	None	None
Joseph Flynn	Content Reviewer	Seattle Children's Hospital—Chief of the Division of Nephrology; University of Washington School of Medicine—Professor of Pediatrics	• Ultragenyx, Inc. (DSMB)	None	None	None	• UpToDate, Springer*	None	None
Federico Gentile	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	Centro Cardiologico	None	None	None	None	None	None	None
Joel Handler	Content Reviewer	Kaiser Permanente—Physician; National Kaiser Permanente Hypertension—Clinical Leader	None	None	None	None	None	None	None
Hani Jneid	Content Reviewer—ACC/AHA Task Force on Clinical Data Standards	Baylor College of Medicine—Associate Professor of Medicine, MEDVAMC	None	None	None	None	None	None	None
José A. Joglar	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	UT Southwestern Medical Center—Professor of Internal Medicine; Cardiovascular Clinical Research Center—Director	None	None	None	None	None	None	None
Amit Khera	Content Reviewer	University of Texas Southwestern Medical Center—Assistant Professor of Medicine	None	None	None	None	None	None	None
Glenn N. Levine	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	Baylor College of Medicine—Professor of Medicine; Director, Cardiac Care Unit	None	None	None	None	None	• Defendant, catheterization laboratory procedure, 2016 • Defendant, interpretation of ECG of a patient, 2014 • Defendant, interpretation of angiogram (non-ACS), 2014 • Defendant, out-of-hospital death, 2016	None

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Reviewer	Representation	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Salary
Giuseppe Mancia	Content Reviewer	University of Milan-Bicocca—Professor of Medicine; Chairman, Department of Clinical Medicine, Prevention and Applied Biotechnologies	- Boehringer Ingelheim* - CVRx - Ferrer - MEDTRONIC - Menarini International* - Recordati - Servier International* - Actavis	None	None	None	Novartis*	None	None
Andrew Miller	Content Reviewer—Geriatric Cardiology Section	Cardiovascular Associates—Cardiologist	None	None	None	- Novartis Corporation† - Pfizer Inc†	- Bristol-Myers Squibb Company - Janssen Pharmaceuticals, Inc. - NIH	None	None
Pamela Morris	Content Reviewer—Prevention Council, Chair	Seinsheimer Cardiovascular Health Program—Director; Women's Heart Care Medical University of South Carolina—Co-Director	- Amgen Inc. - AstraZeneca - Sanofi Regeneron	None	None	Amgen Inc.	None	None	None
Martin Myers	Content Reviewer	Sunnybrook Health Sciences Centre—Affiliate Scientist; University of Toronto—Professor, Cardiology	Ideal Life Inc†	None	None	None	None	None	None
Rick Nishimura	Content Reviewer	Mayo Clinic College of Medicine—Judd and Mary Morris Leighton Professor of Medicine; Mayo Clinic—Division of Cardiovascular Diseases	None	None	None	None	None	None	None
Patrick T. O'Gara	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	Harvard Medical School—Professor of Medicine; Brigham and Women's Hospital—Director, Strategic Planning, Cardiovascular Division	None	None	None	None	- MEDTRONIC - NIH†	None	None
Suzanne Oparil	Content Reviewer	University of Alabama at Birmingham—Distinguished Professor of Medicine; Professor of Cell, Developmental and Integrative Biology, Division of Cardiology	- Actelion - Lundbeck - Novo Nordisk, Inc.	None	None	- AstraZeneca (Duke University)* - Bayer Healthcare Pharmaceuticals, Inc.* - Novartis* - NIH†	- NIH/NHLBI, - Takeda WHF/ESH/EPH	None	None
Carl Pepine	Content Reviewer—CV Disease in Women Committee	Shands Hospital at University of Florida—Professor of Medicine, Chief of Cardiovascular Medicine	None	None	None	- Capricor, Inc. - NIH - Cytos Therapeutics, Inc. - Sanofi-Aventis - InVention Health Clinical. LLC	None	None	None
Mahboob Rahman	Content Reviewer	Case Western Reserve University School of Medicine—Professor of Medicine	None	None	None	None	None	None	None
Vankata Ram	Content Reviewer	UT Southwestern Medical Center; Apollo Institute for Blood Pressure Clinics	None	None	None	None	None	None	None
Barbara Riegel	Content Reviewer—ACC/AHA Task Force on Clinical Practice Guidelines	University of Pennsylvania School of Nursing- Professor	None	None	None	- Co-Investigator-mentor† - Co-investigator NIH - NIH grant - PCORI	Novartis Corp†	None	None

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Reviewer	Representation	Employment	Consultant	Speakers Bureau	Ownership/ Partnership/ Principal	Personal Research	Institutional, Organizational, or Other Financial Benefit	Expert Witness	Salary
Edward Roccella	Content Reviewer	National Heart, Lung, and Blood Institute—Coordinator, National High Blood Pressure Education Program	Medical University of South Carolina	None	None	None	- American Society of Hypertension† - Consortium for Southeast Hypertension Control† - Consortium Southeast Hypertension Control - Inter American Society of Hypertension†	None	None
Ernesto Schiffrin	Content Reviewer	Jewish General Hospital—Physician-in-Chief, Chief of the Department of Medicine and Director of the Cardiovascular Prevention Centre; McGill University—Professor, Department of Medicine, Division of Experimental Medicine	- Novartis - Servier	Novartis	None	- Servier* - Canadian Institutes for Health Research†	CME Medical Grand Rounds	None	None
Raymond Townsend	Content Reviewer	University of Pennsylvania School of Medicine—Professor of Medicine; Director, Hypertension Section, Department of Internal Medicine/Renal; Institute for Translational Medicine and Therapeutics—Member	MEDTRONIC	None	None	NIH*	- ASN - UpToDate	None	None
Michael Weber	Content Reviewer	SUNY Downstate College of Medicine—Professor of Medicine	- Ablative Solutions* - Allergan, Inc - Astellas Pharma US* - Boston Scientific* - Eli Lilly and Company - MEDTRONIC* - Novartis - Recor	- Menarini* - Merck & Co., Inc.*	None	None	None	None	None

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*Significant relationship.

†No financial benefit.

AHRQ indicates Agency for Healthcare Research and Quality; AAPA, American Academy of Physician Assistants; ACC, American College of Cardiology; ACPM, American College of Preventive Medicine; AGS, American Geriatrics Society; AHA, American Heart Association; APHA, American Pharmacists Association; ASH, American Society of Hypertension; ASPC, American Society for Preventive Cardiology; ABC, Association of Black Cardiologists; BOG, Board of Governors; CME, continuing medical education; DSMB, Data and Safety Monitoring Board; FDA, US Food and Drug Administration; NHLBI, National Heart, Lung, and Blood Institute; NIH, National Institutes of Health; NMA, National Medical Association; PCNA, Preventive Cardiovascular Nurses Association; PCORI, Patient-Centered Outcomes Research Institute; SCAI, Society for Cardiovascular Angiography and Interventions; SUNY, State University of New York; TFGP, Task Force on Practice Guidelines; and UT, University of Texas.